2-Year Master's Program in Mathematics (NEP-2020)
ABSTRACT ALGEBRA

MA/M.Sc. Mathematics (1st Semester) Total Credits: **04**
Course Code: **MMTHCAA125 (15 hours per credit)** Total Marks: **100**
Continuous Assessment: Marks 28, Theory: Marks 72 **Time Duration: 2½ hrs**

Course Learning Outcomes(CLO's): After the completion of this course students shall be able to:

- **CLO1.** Apply the structure theorems for cyclic groups and finite abelian groups.
- **CLO2.** Analyze group homomorphisms, automorphisms, and group actions.
- **CLO3.** Explore group classifications using Sylow theorems and composition series.
- **CLO4.** Gain a working knowledge of polynomial rings and factorization in rings.

UNIT-I

Semigroups and criterion for a semigroup to be a group, Cyclic groups and their generators (finite and infinite), Structure theorem for cyclic groups, Endomorphisms, automorphisms, inner and outer automorphisms, Cauchy's and Sylow's theorems for abelian groups, Groups of symmetries, alternating groups, simple groups, Classification and properties of groups of order six, Simplicity of the alternating group A_n.

UNIT-II

Finite Groups and Group Actions: Conjugacy classes and class equations for groups of orders $p, 2p, p^2, p^3$. Applications of class equations, Sylow theorems for finite groups. Direct product of groups and finite abelian group classification, Normal and subnormal series, Composition series and the Jordan-Hölder theorem, Zassenhaus lemma and Schreier's refinement theorem, Solvable groups: examples and related theorems.

UNIT-III

Rings and Integral Domains: Review of rings and basic definitions, Field of quotients for an integral domain, Embedding of integral domains, Euclidean rings with examples such as $Z[\sqrt{-1}], Z[\sqrt{2}]$, Principal Ideal Rings (PIRs), Unique Factorization Domains (UFDs), Euclidean Domains (EDs), GCD and LCM in rings, Factorization theorem and relationships among EDs, UFDs, and PIRs.

UNIT-IV

Polynomial Rings and Factorization: Polynomial rings and the division algorithm, Irreducible polynomials and primitive polynomials, Rational field and integer monic polynomials, Gauss's Lemma and Eisenstein's Irreducibility Criterion, Cyclotomic polynomials, Contraction of polynomials, Polynomial rings and some basic results of commutative ring theory.

ABSTRACT ALGEBRA (MMTHCAA125)

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Contents

Chapter 1

Groups

1.1 PERMUTATIONS

This chapter extends our study of groups by exploring the fundamental relationships between them and their internal composition. We will introduce **homomorphisms** as structure-preserving maps, leading to the **Structure Theorem for Cyclic Groups**, which completely classifies these basic groups. Key insights into the finite group structure will be provided by **Cauchy's Theorem**, which guarantees the existence of elements of prime order. Finally, we will delve into **simple groups**—the building blocks of all finite groups—and prove the **simplicity of the alternating groups A_n for $n \geq 5$**, revealing a crucial family of these fundamental structures. We begin by defining permutations that lead to permutation groups.

Definition 1.1.1. A **permutation** of a set X is a bijection from X to itself.

We use a two-rowed notation to denote the function corresponding to a rearrangement; if $\alpha(j)$ is the image of jth item on the list, then

$$\alpha = \begin{pmatrix} 1 & 2 & \cdots & j & \cdots & n \\ \alpha(1) & \alpha(2) & \cdots & \alpha(j) & \cdots & \alpha(n) \end{pmatrix}.$$

Definition 1.1.2. 1 Let $i_1, i_2, \ldots, i_r$ be distinct integers in $\{1, 2, \ldots, n\}$. If $\alpha \in S_n$ fixes the other integers (if any) and if

$$\alpha(i_1) = i_2,\ \alpha(i_2) = i_3,\ \ldots,\ \alpha(i_{r-1}) = i_r,\ \alpha(i_r) = i_1,$$

then α is called an **r-cycle**. We also say that α is a cycle of **length** r, and we denote it by

$$\alpha = (i_1\ i_2\ \ldots\ i_r).$$

A 2-cycle interchanges i_1 and i_2 and fixes everything else; 2-cycles are also called **transpositions**. A 1-cycle is the identity, for it fixes every i; thus, all 1-cycles are equal: $(i) = (1)$ for all i.

Let us now give an *algorithm* to factor a permutation into a product of cycles. For example, take

$$\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 6 & 4 & 7 & 2 & 5 & 1 & 8 & 9 & 3 \end{pmatrix}.$$

Begin by writing "(1." Now $\alpha : 1 \mapsto 6$, so write "(1 6." Next, $\alpha : 6 \mapsto 1$, and so the parentheses close: α begins "(1 6)." The first number not having appeared is 2, and so we write "(1 6)(2." Now $\alpha : 2 \mapsto 4$, so we write "(1 6)(2 4." Since $\alpha : 4 \mapsto 2$, the parentheses close once again, and we write "(1 6)(2 4)." The smallest remaining number is 3; now $3 \mapsto 7, 7 \mapsto 8, 8 \mapsto 9$, and $9 \mapsto 3$; this gives the 4-cycle $(3\ 7\ 8\ 9)$. Finally, $\alpha(5) = 5$; we claim that

$$\alpha = (1\ 6)(2\ 4)(3\ 7\ 8\ 9)(5).$$

Since multiplication in S_n is the composition of functions, our claim is that

$$\alpha(i) = [(1\ 6)(2\ 4)(3\ 7\ 8\ 9)(5)](i)$$

for every i between 1 and 9 [after all, two functions f and g are equal if and only if $f(i) = g(i)$ for every i in their domain]. The right side is the composite $\beta\gamma\delta$, where $\beta = (1\ 6)$, $\gamma = (2\ 4)$, and $\delta = (3\ 7\ 8\ 9)$ [we may ignore the 1-cycle (5) when we are evaluating, for it is the identity function]. Now $\alpha(1) = 6$; let us evaluate the composite on the right when $i = 1$.

$$\begin{aligned}\beta\gamma\delta(1) &= \beta(\gamma(\delta(1)))\\ &= \beta(\gamma(1)) \quad \text{since } \delta = (3\ 7\ 8\ 9) \text{ fixes } 1\\ &= \beta(1) \quad \text{since } \gamma = (2\ 4) \text{ fixes } 1\\ &= 6 \quad \text{since } \beta = (1\ 6).\end{aligned}$$

Similarly, $\alpha(i) = \beta\gamma\delta(i)$ for every i, proving the claim.

We multiply permutations from right to left, because multiplication here is a composite of functions; that is, to evaluate $\alpha\beta(1)$, we compute $\alpha(\beta(1))$. In the factorisation of a permutation into cycles, given by the preceding algorithm, we note that the family of cycles is *disjoint* in the following sense.

Definition 1.1.3. Two permutations $\alpha, \beta \in S_n$ are **disjoint** if every i moved by one is fixed by the other: If $\alpha(i) \neq i$, then $\beta(i) = i$, and if $\beta(j) \neq j$, then $\alpha(j) = j$. A family $\beta_1, \ldots, \beta_t$ of permutations is *disjoint* if each pair of them is disjoint.

Lemma 1.1.4. *Disjoint permutations $\alpha, \beta \in S_n$ commute.*

Proof. It suffices to prove that if $1 \leq i \leq n$, then $\alpha\beta(i) = \beta\alpha(i)$. If β moves i, say, $\beta(i) = j \neq i$, then β also moves j [otherwise, $\beta(j) = j$ and $\beta(i) = j$ contradicts β's being an injection]; since α and β are disjoint, $\alpha(i) = i$ and $\alpha(j) = j$. Hence $\beta\alpha(i) = j = \alpha\beta(i)$. The same conclusion holds if α moves i. Finally, it is clear that $\alpha\beta(i) = \beta\alpha(i)$ if both α and β fix i. ■

Proposition 1.1.5. Every permutation $\alpha \in S_n$ is either a cycle or a product of disjoint cycles.

Proof. The proof is by induction on the number k of points moved by α. The base step $k = 0$ is true, for now α is the identity, which is a 1-cycle.

If $k > 0$, let i_1 be a point moved by α. Define $i_2 = \alpha(i_1)$, $i_3 = \alpha(i_2), \ldots, i_{r+1} = \alpha(i_r)$, where r is the smallest integer for which $i_{r+1} \in \{i_1, i_2, \ldots, i_r\}$ (since there are only n possible values, the list $i_1, i_2, i_3, \ldots, i_k, \ldots$ must eventually have a repetition). We claim that $\alpha(i_r) = i_1$. Otherwise, $\alpha(i_r) = i_j$ for some $j \geq 2$; but $\alpha(i_{j-1}) = i_j$, and this contradicts the hypothesis that α is an injection. Let σ be the r-cycle $(i_1\ i_2\ \ldots\ i_r)$. If $r = n$, then $\alpha = \sigma$. If $r < n$, then σ fixes each point in Y, where Y consists of the remaining $n - r$ points, while $\alpha(Y) = Y$. Define α' to be the permutation with $\alpha'(i) = \alpha(i)$ for $i \in Y$ that fixes all $i \notin Y$, and note that

$$\alpha = \sigma\alpha'.$$

The inductive hypothesis gives $\alpha' = \beta_1 \cdots \beta_t$, where $\beta_1, \ldots, \beta_t$ are disjoint cycles. Since σ and α' are disjoint, $\alpha = \sigma\beta_1 \cdots \beta_t$ is a product of disjoint cycles. ■

Usually, we suppress the 1-cycles in this factorisation [for 1-cycles equal the identity (1)]. However, a factorisation of α in which we display one 1-cycle for each i fixed by α, if any, will arise several times.

Definition 1.1.6. A **complete factorisation** of a permutation α is a factorisation of α into disjoint cycles that contains exactly one 1-cycle (i) for every i fixed by α.

For example, the complete factorisation of the 3-cycle $\alpha = (1\ 3\ 5)$ in S_5 is

$$\alpha = (1\ 3\ 5)(2)(4).$$

There is a relation between an r-cycle $\beta = (i_1\ i_2\ \ldots\ i_r)$ and its **powers** β^k, where β^k denotes the composite of β with itself k times. Note that

$$i_2 = \beta(i_1), \quad i_3 = \beta(i_2) = \beta(\beta(i_1)), \quad i_4 = \beta(i_3) = \beta(\beta^2(i_1)) = \beta^3(i_1),$$

and, more generally,

$$i_{k+1} = \beta^k(i_1)$$

for all $k < r$.

Theorem 1.1.7. *Let $\alpha \in S_n$ and let $\alpha = \beta_1 \cdots \beta_t$ be a complete factorisation into disjoint cycles. This factorisation is unique except for the order in which the cycles occur.*

Since every complete factorisation of α has exactly one 1-cycle for each i fixed by α, it suffices to consider (not complete) factorisations into disjoint cycles of length ≥ 2. Let $\alpha = \gamma_1 \cdots \gamma_s$ be a second such factorisation of α into disjoint cycles.

The theorem is proved by induction on ℓ, the larger of t and s. The inductive step begins by noting that if β_t moves i_1, then $\beta_t^k(i_1) = \alpha^k(i_1)$ for all $k \geq 1$. Some γ_j must also move i_1 and, since disjoint cycles commute, we may assume that γ_s moves i_1. It follows that $\beta_t = \gamma_s$; right multiplying by β_t^{-1} gives

$$\beta_1 \cdots \beta_{t-1} = \gamma_1 \cdots \gamma_{s-1}.$$

Every permutation is a bijection; how do we find its inverse? In the pictorial representation of a cycle β as a clockwise rotation of a circle, the inverse β^{-1} is just a counterclockwise rotation. The proof of the next proposition is straightforward.

Proposition 1.1.8. (i) *The inverse of the cycle $\alpha = (i_1\, i_2\, \ldots\, i_r)$ is the cycle $(i_r\, i_{r-1} \ldots\, i_1)$:*

$$(i_1\, i_2\, \ldots\, i_r)^{-1} = (i_r\, i_{r-1} \ldots\, i_1).$$

(ii) *If $\gamma \in S_n$ and $\gamma = \beta_1 \cdots \beta_k$, then*

$$\gamma^{-1} = \beta_k^{-1} \cdots \beta_1^{-1}.$$

Definition 1.1.9. Two permutations $\alpha, \beta \in S_n$ have the **same cycle structure** if their complete factorisations have the same number of r-cycles for each r.

Here is a computational aid. We illustrate its statement in the following example before stating the general result.

Example 1.1.10. If $\gamma = (1\ 3)(2\ 4\ 7)(5)(6)$ and $\alpha = (2\ 5\ 6)(1\ 4\ 3)$, then

$$\alpha\gamma\alpha^{-1} = (4\ 1)(5\ 3\ 7)(6)(2) = (\alpha 1\ \alpha 3)(\alpha 2\ \alpha 4\ \alpha 7)(\alpha 5)(\alpha 6).$$

Lemma 1.1.11. *If $\gamma, \alpha \in S_n$, then $\alpha\gamma\alpha^{-1}$ has the same cycle structure as γ. In more detail, if the complete factorisation of γ is*

$$\gamma = \beta_1\beta_2 \cdots (i_1\, i_2\, \ldots) \cdots \beta_t,$$

then $\alpha\gamma\alpha^{-1}$ is the permutation that is obtained from γ by applying α to the symbols in the cycles of γ.

Proof. The idea of the proof is that $\gamma\alpha^{-1}$: $\gamma(i_1) \mapsto i_1 \mapsto i_2 \mapsto \gamma(i_2)$. Let σ denote the permutation defined in the statement.

If γ fixes i, then σ fixes $\alpha(i)$, for the definition of σ says that $\alpha(i)$ lives in a 1-cycle in the factorisation of σ. On the other hand, $\alpha\gamma\alpha^{-1}$ also fixes $\alpha(i)$:

$$\alpha\gamma\alpha^{-1}(\alpha(i)) = \alpha\gamma(i) = \alpha(i),$$

because γ fixes i.

Assume that γ moves a symbol i_1, say, $\gamma(i_1) = i_2$, so that one of the cycles in the complete factorisation of γ is

$$(i_1\, i_2\, \ldots).$$

By the definition of σ, one of its cycles is

$$(k\ \ell\ \ldots),$$

where $\alpha(i_1) = k$ and $\alpha(i_2) = \ell$; hence, $\sigma : k \mapsto \ell$. But $\alpha\gamma\alpha^{-1} : k \mapsto i_1 \mapsto i_2 \mapsto \ell$, and so $\alpha\gamma\alpha^{-1}(k) = \sigma(k)$. Therefore, σ and $\alpha\gamma\alpha^{-1}$ agree on all symbols of the form $k = \alpha(i_1)$. Since α is surjective, every k is of this form, and so $\sigma = \alpha\gamma\alpha^{-1}$. ■

Example 1.1.12. In this example, we illustrate that the converse of Lemma 1.1.11 is true; the next theorem will prove it in general. In S_5, place the complete factorisation of a 3-cycle β over that of a 3-cycle γ, and define α to be the downward function. For example, if

$$\beta = (1\ 2\ 3)(4)(5), \qquad \gamma = (5\ 2\ 4)(1)(3),$$

then

$$\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 2 & 4 & 1 & 3 \end{pmatrix},$$

and so $\alpha = (1\,5\,3\,4)$. Now $\alpha \in S_5$ and

$$\gamma = (\alpha 1\; \alpha 2\; \alpha 3),$$

so that $\gamma = \alpha\beta\alpha^{-1}$, by Lemma 1.1.11. Note that rewriting the cycles of β, for example, as $\beta = (1\,2\,3)(5)(4)$, gives another choice for α.

Theorem 1.1.13. *Permutations γ and σ in S_n have the same cycle structure if and only if there exists $\alpha \in S_n$ with $\sigma = \alpha\gamma\alpha^{-1}$.*

Proof. Sufficiency was just proved in Lemma 1.1.11. For the converse, place one complete factorisation over the other so that each cycle below is under a cycle above of the same length:

$$\gamma = \delta_1\delta_2 \cdots (i_1\; i_2\; \ldots) \cdots \delta_t \qquad \alpha\gamma\alpha^{-1} = \eta_1\eta_2 \cdots (k\; \ell\; \ldots) \cdots \eta_t.$$

Now define α to be the "downward" function, as in the example; hence, $\alpha(i_1) = k$, $\alpha(i_2) = \ell$, and so forth. Note that α is a permutation, for there are no repetitions of symbols in the factorisation of γ (the cycles η are disjoint). It now follows from the lemma that $\sigma = \alpha\gamma\alpha^{-1}$. ∎

There is another useful factorisation of a permutation.

Proposition 1.1.14. If $n \geq 2$, then every $\alpha \in S_n$ is a product of transpositions.

Proof. In light of Proposition 1.1.5, it suffices to factor an r-cycle β into a product of transpositions, and this is done as follows:

$$\beta = (1\; 2\; \cdots\; r) = (1\; r)(1\; r-1) \cdots (1\; 3)(1\; 2).$$

∎

Definition 1.1.15. A permutation $\alpha \in S_n$ is **even** if it can be factored into a product of an even number of transpositions; otherwise, α is *odd*. The **parity** of a permutation is whether it is even or odd.

It is easy to see that the permutations $(1\,2\,3)$ and (1) are even permutations, for there are factorisations

$$(1\,2\,3) = (1\,3)(1\,2) \quad \text{and} \quad (1) = (1\,2)(1\,2)$$

having two transpositions. On the other hand, we do not yet have any examples of odd permutations! If α is a product of an odd number of transpositions, perhaps it also has some other factorisation into an even number of transpositions. The definition of odd permutation α, after all, says that there is no factorisation of α into an even number of transpositions.

Definition 1.1.16. If $\alpha \in S_n$ and $\alpha = \beta_1 \cdots \beta_t$ is a complete factorisation into disjoint cycles, then the **signum** of α is defined by

$$\operatorname{sgn}(\alpha) = (-1)^{n-t}.$$

Theorem 1.1.7 shows that sgn is a (well-defined) function, for the number t is uniquely determined by α. Notice that $\operatorname{sgn}(\varepsilon) = 1$ for every 1-cycle ε because $t = n$. If τ is a transposition, then it moves two numbers, and it fixes each of the $n-2$ other numbers; therefore, $t = (n-2) + 1 = n - 1$, and so

$$\operatorname{sgn}(\tau) = (-1)^{n-(n-1)} = -1.$$

Theorem 1.1.17. *For all $\alpha, \beta \in S_n$,*

$$\operatorname{sgn}(\alpha\beta) = \operatorname{sgn}(\alpha)\operatorname{sgn}(\beta).$$

Proof. If $k, \ell \geq 0$ and the letters a, b, c_i, d_j are all distinct, then

$$(a\, b)(a\, c_1\, \ldots\, c_k\, b\, d_1\, \ldots\, d_\ell) = (a\, c_1\, \ldots\, c_k)(b\, d_1\, \ldots\, d_\ell);$$

multiplying this equation on the left by $(a\, b)$ gives

$$(a\, b)(a\, c_1\, \ldots\, c_k)(b\, d_1\, \ldots\, d_\ell) = (a\, c_1\, \ldots\, c_k\, b\, d_1\, \ldots\, d_\ell).$$

These equations are used to prove that $\operatorname{sgn}(\tau\alpha) = -\operatorname{sgn}(\alpha)$ for every $\alpha \in S_n$, where τ is the transposition $(a\, b)$. If $\alpha \in S_n$ has a factorisation $\alpha = \tau_1 \cdots \tau_m$, where each τ_i is a transposition, we now prove, by induction on m, that

$$\operatorname{sgn}(\alpha\beta) = \operatorname{sgn}(\alpha)\operatorname{sgn}(\beta)$$

for every $\beta \in S_n$. ■

Theorem 1.1.18. *(i) Let $\alpha \in S_n$; if $\operatorname{sgn}(\alpha) = 1$, then α is even, and if $\operatorname{sgn}(\alpha) = -1$, then α is odd.*

(ii) A permutation α is odd if and only if it is a product of an odd number of transpositions.

Proof. (i) If $\alpha = \tau_1 \cdots \tau_q$ is a factorisation of α into transpositions, then Theorem 1.1.17 gives

$$\operatorname{sgn}(\alpha) = \operatorname{sgn}(\tau_1) \cdots \operatorname{sgn}(\tau_q) = (-1)^q.$$

Thus, if $\operatorname{sgn}(\alpha) = 1$, then q must always be even, and if $\operatorname{sgn}(\alpha) = -1$, then q must always be odd.

(ii) If α is odd, then α is not even, and so $\operatorname{sgn}(\alpha) \neq 1$; that is, $\operatorname{sgn}(\alpha) = -1$. Now $\alpha = \tau_1 \cdots \tau_q$, where the τ_i are transpositions, so that

$$\operatorname{sgn}(\alpha) = -1 = (-1)^q;$$

hence, q is odd (we have proved more; every factorisation of α into transpositions has an odd number of factors). Conversely, if $\alpha = \tau_1 \cdots \tau_q$ is a product of transpositions with q odd, then $\operatorname{sgn}(\alpha) = -1$; therefore, α is not even and, hence, α is odd. ■

Corollary 1.1.19. *Let $\alpha, \beta \in S_n$. If α and β have the same parity, then $\alpha\beta$ is even, while if α and β have distinct parity, then $\alpha\beta$ is odd.*

Proposition 1.1.20. Let $\alpha \in S_n$.

(i) If α is an r-cycle, then α has order r.

(ii) If $\alpha = \beta_1 \cdots \beta_t$ is a product of disjoint r_i-cycles β_i, then α has order $\operatorname{lcm}\{r_1, \ldots, r_t\}$.

(iii) If p is a prime, then α has order p if and only if it is a p-cycle or a product of disjoint p-cycles.

Proof. (i) Let $\alpha = (a_1\, a_2\, \ldots\, a_r)$ be an r-cycle in the symmetric group S_n. This means that:

$$\alpha(a_i) = a_{i+1} \quad \text{for } 1 \leq i < r, \quad \text{and } \alpha(a_r) = a_1.$$

Let us apply α^k to a_1:

$$\alpha^k(a_1) = a_{k+1} \quad \text{for } 1 \leq k < r,$$

with indices taken modulo r. In particular, $\alpha^r(a_1) = a_1$, and similarly, $\alpha^r(a_i) = a_i$ for all $i = 1, 2, \ldots, r$. Moreover, since α fixes all elements not in the cycle, α^r fixes every element of the domain. Hence,

$$\alpha^r = (1).$$

Now suppose that $\alpha^k = \text{id}$ for some $1 \leq k < r$. Then in particular,

$$\alpha^k(a_1) = a_1.$$

But since α cycles through the elements $a_1, a_2, \ldots, a_r$ in order, $\alpha^k(a_1) = a_{k+1} \pmod r$. So the condition $\alpha^k(a_1) = a_1$ implies $a_{k+1} = a_1$, i.e., $k \equiv 0 \pmod r$. But $1 \leq k < r$, so this is not possible. Hence, no such $k < r$ exists with $\alpha^k = (1)$.

Therefore, the smallest positive integer k such that $\alpha^k = \text{id}$ is $k = r$. That is, the order of α is r.

(ii) Since the cycles $\beta_1, \beta_2, \ldots, \beta_t$ are disjoint, they commute. Therefore, we can write

$$\alpha^k = (\beta_1\beta_2\cdots\beta_t)^k = \beta_1^k\beta_2^k\cdots\beta_t^k.$$

Let r_i be the order of the cycle β_i and Let $m = \operatorname{lcm}(r_1, r_2, \ldots, r_t)$. Then for each i, we have $r_i \mid m$, so $\beta_i^m = (1)$. Thus,

$$\alpha^m = \beta_1^m \cdots \beta_t^m = (1)$$

Now we show that m is the smallest such positive integer. Suppose $\alpha^k = (1)$ for some positive integer k. Then:

$$(1) = \alpha^k = \beta_1^k \cdots \beta_t^k.$$

But since the cycles are disjoint, the action of each β_i^k is independent. So for this product to be the identity, each β_i^k must itself be the identity. Hence,

$$\beta_i^k = (1) \quad \text{for all } i,$$

which implies $r_i \mid k$ for all i, so $m = \operatorname{lcm}(r_1, \ldots, r_t) \mid k$.

Thus, m is the least positive integer for which $\alpha^m =$ id, and hence the order of α is $\operatorname{lcm}(r_1, \ldots, r_t)$.

(iii) Write α as a product of disjoint cycles and use (ii).

■

Definition 1.1.21 (Binary operation). Let G be a set. A binary operation, $*$ on G is a function that assigns each ordered pair of elements of G an element of G. That is,

$$* : G \times G \to G$$

Definition 1.1.22 (Algebraic structure). An **algebraic structure** (or Groupoid) consists of a non-empty set A together with one or more binary operations defined on it. That is, an algebraic structure is a pair $(A, *)$ where

$$* : A \times A \to A.$$

Example 1.1.23. The pair $(\mathbb{N}, +)$ is an algebraic structure, since the sum of two positive integers is again a positive integer.

Definition 1.1.24 (Semigroup)**.** A **semigroup** is a groupoid $(S, \cdot)$ in which the binary operation is *associative*. That is, for all $a, b, c \in S$,

$$(a \cdot b) \cdot c = a \cdot (b \cdot c).$$

Thus, a semigroup is a non-empty set with an associative binary operation.

Example 1.1.25. i. Natural numbers under addition: $(\mathbb{N}, +)$

ii. Square matrices under multiplication: $(M_n(\mathbb{R})$

iii. Functions under composition:

$$(\text{Map}(X, X), \circ) \quad \text{where Map}(X, X) \text{ is the set of all functions } f : X \to X.$$

Function composition is associative.

iv. Set of positive integers under minimum: $(\mathbb{Z}^+, \min)$

$$\min(\min(a, b), c) = \min(a, \min(b, c)) \quad \text{is associative.}$$

Definition 1.1.26 (Monoid). A **monoid** is an algebraic structure $(M, \cdot)$ consisting of a non-empty set M and a binary operation $* : M \times M \to M$ such that:

1. **Associativity:** For all $a, b, c \in M$, we have

$$(a * b) * c = a * (b * c).$$

2. **Identity element:** There exists an element $e \in M$ such that for all $a \in M$,

$$e * a = a = a * e.$$

Example 1.1.27. .

(i) $(\mathbb{N}_0, +)$: The set of natural numbers including 0 under addition.

(ii) $(\mathbb{N}, \times)$: The set of natural numbers under multiplication.

(iii) $(\mathrm{End}(V), \circ)$: The set of all endomorphisms (linear maps from V to V) of a vector space V under composition.

Definition 1.1.28 (Group). Let G be a set together with a binary operation (usually called multiplication)that assigns to each ordered pair(a, b) of elements of G an element in G denoted by ab. We say G is a group under this operation if the following three properties are satisfied.

1. (Associativity). The operation is associative; that is, $(ab)c = a(bc)$ for all $a, b, c \in G$.
2. (Identity). There is an element e[1](called the identity) in G such that $ea = e = ae$ for all $a \in G$.
3. (Inverses). For each element a in G, there is an element b in G (called an inverse of a) such that $ab = e = ba$

Example 1.1.29. The set of integers $\mathbb{Z}$, the set of rational numbers $\mathbb{Q}$ (for quotient), and the set of real numbers $\mathbb{R}$ are all groups under ordinary addition. In each case, the identity is 0 and the inverse of a is $-a$.

Example 1.1.30. The set $\mathbb{Q}^+$ of positive rationals is a group under ordinary multiplication. The inverse of any a is $\frac{1}{a} = a^{-1}$.

Example 1.1.31. A rectangular array of the form $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is called a 2×2 matrix. The set of all 2×2 matrices with real entries is a group under componentwise addition.

Example 1.1.32. The set $\mathbb{Z}_n = \{0, 1, ..., n-1\}$ for $n \geq 1$ is a group under addition modulo n. For any $j > 0$ in $\mathbb{Z}_n$, the inverse of j is $n - j$. This group is usually referred to as the group of integers modulo n.

Example 1.1.33. For all integers $n \geq 1$, the set ζ_n of complex nth roots of unity

$$\zeta_n = \left\{ \cos\frac{2k\pi}{n} + i\sin\frac{2k\pi}{n} \,\middle|\, k = 0, 1, \ldots, n-1 \right\}$$

is a group under multiplication.

Example 1.1.34. The set $\{1, 2, ..., n-1\}$ is a group under multiplication modulo n if and only if n is prime.

Example 1.1.35 (Four-group). The four permutations

$$\mathbf{V} = \{(1),\ (1\,2)(3\,4),\ (1\,3)(2\,4),\ (1\,4)(2\,3)\} \subset S_4$$

form a group, because $\mathbf{V}$ is a subgroup of $S_4 : (1) \in \mathbf{V}$; $\alpha^2 = (1)$ for each $\alpha \in \mathbf{V}$, and so $\alpha^{-1} = \alpha \in \mathbf{V}$; the product of any two distinct permutations in $\mathbf{V} \setminus \{(1)\}$ is the third one. The group $\mathbf{V}$ is called the **four-group** (or some times the **Klein's Four group**).

Theorem 1.1.36 (Cancellation laws). *Let G be a group then for all $a, b, c \in G$:*

$$ab = ac \Rightarrow b = c \quad \textit{(left cancellation law)}$$

$$ba = ca \Rightarrow b = c \quad \textit{(right cancellation law)}$$

[1]There are several notations of the identity element but we will prefer 1 in the subsequent sections.

Proof. We are given that $ab = ac$. This would imply that $a^{-1}(ab) = a^{-1}(ac)$. This gives $(a^{-1}a)b = (a^{-1}a)c$ or $eb = ec$, where e is the identity of the group. Therefore, the left cancellation law holds in G

Similarly,

$$ba = ca \Rightarrow (ba)a^{-1} = (ca)a^{-1} \Rightarrow b = c.$$

■

Remark 1.1.37. In a semi-group, the cancellation laws may not hold as is shown by the following example.

Example 1.1.38. The set S of all 2×2 matrices over integers is a semi-group under multiplication. Now

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \quad C = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

are three elements of S such that $AB = AC$ but $B \neq C$.

Remark 1.1.39. There are semi-groups which are not groups but they satisfy cancellation laws. This is shown as follows.

Example 1.1.40. The set $\mathbb{N}$ is an abelian semi-group under addition. It is well-known that for any three positive integers a, b, c,

$$a + b = a + c \Rightarrow b = c.$$

But this system is not a group.

Theorem 1.1.41. *Any finite semi-group in which both cancellation laws hold, is a group.*

Proof. Let S be a finite semi-group in which both cancellation laws hold. Let

$$S = \{a_1, a_2, a_3, \dots, a_n\} \tag{1.1}$$

Consider any $a_i \in S$, then

$$a_1a_i, \ a_2a_i, \ a_3a_i, \ \dots, \ a_na_i \tag{1.2}$$

are all distinct, since for any i and j $(1 \leq i, j \leq n)$:

$$a_ia_j = a_ja_i \Rightarrow a_i = a_j \quad \text{(right cancellation law)}.$$

Hence, the sets in (1.1) and (1.2) are identical, except possibly in some different order. So given any $a_t \in S$ there exists $a_j \in S$ such that:

$$a_t = a_ja_i \tag{1.3}$$

In particular, there exists some $a_k \in S$ such that $a_i = a_ka_i$. Then

$$a_ia_i = a_i(a_ka_i) = (a_ia_k)a_i.$$

So by right cancellation law in S, $a_i = a_ia_k$ for all i. This produces

$$a_i = a_ka_i = a_ia_k,$$

that is, a_k is the identity element of S.

Let us write e for a_k. In particular, there exists a_m in S such that $e = a_ma_i$. Then

$$a_m\, e = e\, a_m = (a_ma_i)a_m = a_m(a_ia_m)$$

So by the left cancellation law, $e = a_ia_m$. Therefore, we conclude

$$e = a_ia_m = a_ma_i$$

that is, a_m is the inverse of a_i.

Consequently, every element in S has its inverse in S. S is such that for all $a, b, c \in S$:

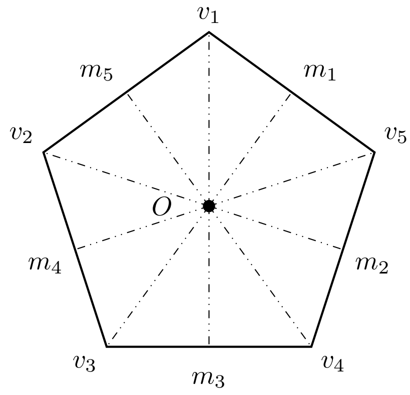


Figure 1.1: Symmetries of π_5

i. $a(bc) = (ab)c$. (S is a semi-group).

ii. There exists $e \in S$ such that $ae = a = ea$ for all $a \in S$.

iii. Given $a \in S$, $\exists\, a^{-1} \in S$ such that $aa^{-1} = e = a^{-1}a$.

Hence S is a group. ■

Remark 1.1.42. Observe that in Example 1.1.40, $\langle \mathbb{N}, + \rangle$ is an infinite semi-group with cancellation laws, but $\langle \mathbb{N}, + \rangle$ is not a group. This means that the preceding theorem cannot be generalized to infinite semi-groups.

Definition 1.1.43 (Dihedral Group). If π_n is a regular polygon with n vertices $v_1, v_2, \dots, v_n$ and center O, then the symmetry group $\Sigma(\pi_n)$ is called the **dihedral group** with $2n$ elements, and it is denoted by D_n[2].

1.2 Cyclic Groups

Definition 1.2.1. If G is a group and $a \in G$, write

$$\langle a \rangle = \{a^n \mid n \in \mathbb{Z}\};$$

a is called the **cyclic subgroup** of G generated by a. A group G is called **cyclic** if there exists $a \in G$ with $G = \langle a \rangle$, in which case a is called a generator of G.

Example 1.2.2. Example 1.1.33 shows, for every $n \geq 1$, that the multiplicative group ζ_n of all nth roots of unity is a cyclic group with the primitive nth root of unity $\zeta = e^{\frac{2\pi i}{n}}$ as a generator.

Example 1.2.3. The group $\mathbb{Z}_n$ (integers modulo n) is indeed a group under addition modulo n with $[1]$ be a generator.

Theorem 1.2.4. *If $G = \langle a \rangle$ is a cyclic group of order n, then a^k is a generator of G if and only if $\gcd(k, n) = 1$. Moreover, if $\operatorname{gen}(G) = \{\text{all generators of } G\}$, then $|\operatorname{gen}(G)| = \phi(n)$, where ϕ is the Euler ϕ-function.*

Proof. If a^k generates G, then $a \in \langle a^k \rangle$, so that $a = a^{kt}$ for some $t \in \mathbb{Z}$. Hence, $a^{kt-1} = e$; then, $n|(kt-1)$, so there is $v \in \mathbb{Z}$ with $nv = kt - 1$. Therefore, 1 is a linear combination of k and m, and so $\gcd(k, n) = 1$.

Conversely, if $\gcd(k, n) = 1$, then $nt + ku = 1$ for $t, u \in \mathbb{Z}$; hence

$$a = a^{nt+ku} = a^{nt}a^{ku} = a^{ku} \in \langle a^k \rangle.$$

Therefore, every power of a also lies in $\langle a^k \rangle$ and $G = \langle a^k \rangle$

Also, note that $\phi(n) = \{k \leq n \mid \gcd(k, n) = 1\}|$. Since $G = \{e, a, \dots, a^{n-1}\}$, by first part of the theorem, we see that $|\operatorname{gen}(G)| = \phi(n)$. ■

[2]Some authors denote D_n by D_{2n}

Theorem 1.2.5. *Let G be a group, and let a belong to G. If a has infinite order, then $a^i = a^j$ if and only if $i = j$. If a has finite order, say, n, then $\langle a\rangle = \{e, a, a^2, \ldots, a^{n-1}\}$ and $a^i = a^j$ if and only if n divides $i - j$.*

Proof. If a has infinite order, there is no nonzero n such that a^n is the identity. Since $a^i = a^j$ implies $a^{i-j} = e$, we must have $i - j = 0$, and the first statement of the theorem is proved. Now assume that $\mathrm{ord}(a) = n$. We will prove that $\langle a\rangle = \{e, a, a^2, \ldots, a^{n-1}\}$. Certainly, the elements $e, a, a^2, \ldots, a^{n-1}$ are in $\langle a\rangle$. Suppose that a^k is an arbitrary member of $\langle a\rangle$. By the division algorithm, there exist integers q and r such that $k = qn + r$ with $0 \leq r < n$. Then $a^k = a^{qn+r} = a^{qn}a^r = (a^n)^q a^r = a^r$, so that $a^k \in \{e, a, a^2, \ldots, a^{n-1}\}$. This proves that $\langle a\rangle = \{e, a, a^2, \ldots, a^{n-1}\}$. Next, we assume that $a^i = a^j$, that is $a^{i-j} = e$, so that n divides $i - j$

Conversely, suppose that $n|i - j$, then $i - j = nq$ for some integer q. Now $a^{i-j} = a^{nq} = a^{nq} = e$. This implies $a^i = a^j$ ■

Corollary 1.2.6. *Let G be a group and let a be an element of order n in G. If $a^k = e$, then n divides k.*

Proof. Since $a^k = e = a^0$, by Theorem 1.2.5, we see that n divides $k - 0$. ■

Corollary 1.2.7. *If a and b belong to a finite group and $ab = ba$, then $\mathrm{ord}(ab)$ divides $\mathrm{ord}(a)\mathrm{ord}(b)$.*

Proof. Let $\mathrm{ord}(a) = m$ and $\mathrm{ord}(b) = n$. Then $(ab)^{mn} = (a^m)^n(b^n)^m = e$. So,by Corollary 1.2.6 of Theorem 1.2.5, we have that $\mathrm{ord}(ab)$ divides mn. ■

Theorem 1.2.8. *Let a be an element of order n in a group and let k be a positive integer. Then $\langle a^k\rangle = \langle a^{\gcd(n,k)}\rangle$ and $\mathrm{ord}(a^k) = \frac{n}{\gcd(n,k)}$.*

Proof. Let $d = \gcd(n, k)$ and let $k = dr$. Since $a^k = (a^d)^r$, we have by closure that $\langle a^k\rangle \subseteq \langle a^d\rangle$. Note that d being the $\gcd$ of n and k, there exists integers s and t such that $d = ns + kt$. So, $a^d = a^{ns+kt} = a^{ns}a^{kt} = (a^n)^s(a^k)^t = (a^k)^t \in \langle a^k\rangle$. This proves $\langle a^d\rangle \subseteq \langle a^k\rangle$. Hence, $\langle a^k\rangle = \langle a^{\gcd(n,k)}\rangle$.

To prove the second part of the theorem, we show first that $\mathrm{ord}(a^d) = \frac{n}{d}$ for any divisor d of n. Clearly, $(a^d)^{\frac{n}{d}} = a^n = e$, so that $\mathrm{ord}(a^d) \leq \frac{n}{d}$. On the other hand, if i is a positive integer less than $\frac{n}{d}$, then $(a^d)^i \neq e$ by definition of $\mathrm{ord}(a)$. We now apply this fact with $d = \gcd(n, k)$ to obtain $\mathrm{ord}(a^k) = |\langle a^k\rangle| = |\langle a^{\gcd(n,k)}\rangle| = \mathrm{ord}(\mathrm{a}^{\gcd(\mathrm{n,k})}) = \frac{n}{\gcd(n,k)}$. ■

Corollary 1.2.9. *Let $\mathrm{ord}(a) = n$. Then $\langle a^i\rangle = \langle a^j\rangle$ if and only if $\gcd(n, i) = \gcd(n, j)$, and $\mathrm{ord}(a^i) = \mathrm{ord}(a^j)$ if and only if $\gcd(n, i) = \gcd(n, j)$.*

Proof. Theorem 1.2.8 shows that $\langle a^i\rangle = \langle a^{\gcd(n,i)}\rangle$, and $\langle a^j\rangle = \langle a^{\gcd(n,j)}\rangle$s o that the proof reduces to proving that $\langle a^{\gcd(n,i)}\rangle = \langle a^{\gcd(n,j)}\rangle$ if and only if $\gcd(n, i) = \gcd(n, j)$. Certainly, $\gcd(n, i) = \gcd(n, j)$ implies that $\langle a^{\gcd(n,i)}\rangle = \langle a^{\gcd(n,j)}\rangle$. On the other hand, $\langle a^{\gcd(n,i)}\rangle = \langle a^{\gcd(n,j)}\rangle$ implies that $\mathrm{ord}(a^{\gcd(n,i)}) = \mathrm{ord}(a^{\gcd(n,j)})$ so that by the second conclusion of Theorem 1.2.8, we have $\frac{n}{\gcd(n,i)} = \frac{n}{\gcd(n,j)}$, and therefore$\gcd(n, i) = \gcd(n, j)$.

The second part of the corollary follows from the first part of the corollary and Theorem 1.2.8. ■

The next two corollaries are important special cases of the preceding corollary.

Corollary 1.2.10. *Let $\mathrm{ord}(a) = n$. Then $\langle a\rangle = \langle a^j\rangle$ if and only if $\gcd(n, j) = 1$, and $\mathrm{ord}(a) = |\langle a^j\rangle|$ if and only if $\gcd(n, j) = 1$.*

Corollary 1.2.11. *An integer k in $\mathbb{Z}_n$ is a generator of $\mathbb{Z}_n$ if and only if $\gcd(n, k) = 1$.*

Theorem 1.2.12 (Fundamental Theorem of Cyclic Groups). *Every subgroup of a cyclic group is cyclic. Moreover, if $|\langle a\rangle| = n$, then the order of any subgroup of $\langle a\rangle$ is a divisor of n; and, for each positive divisor k of n, the group $\langle a\rangle$ has exactly one subgroup of order k namely, $\langle a^{\frac{n}{k}}\rangle$.*

Proof. Let $G = \langle a\rangle$ and suppose that H is a subgroup of G. We must show that H is cyclic. If it consists of the identity alone, then clearly H is cyclic. So we may assume that $H \neq \{e\}$. We now claim that H contains an element of the form a^t, where t is positive. Since $G = \langle a\rangle$, every element of H has the form a^t; and when a^t belongs to H with $t < 0$, then a^{-t} belongs to H also and $-t$ is positive. Thus, our claim is verified. Now let m be the least positive integer such that $a^m \in H$. By closure, $\langle a^m\rangle \subseteq H$. We next claim that $H = \langle a^m\rangle$. To prove this claim, it suffices to let b be an arbitrary member of H and show that b is in $\langle a^m\rangle$. Since $b \in G = \langle a\rangle$, we have $b = a^k$ for some k.

Now, apply the division algorithm to k and m to obtain integers q and r such that $k = mq + r$ where $0 \leq r < m$. Then
$$a^k = a^{mq+r} = a^{mq}a^r,$$
so that $a^r = a^{-mq}a^k$. Since $a^k = b \in H$ and $a^{-mq} = (a^m)^{-q}$ is in H also, $a^r \in H$. But, m is the *least* positive integer such that $a^m \in H$, and $0 \leq r < m$, so r must be 0. Therefore, $b = a^k = a^{mq} = (a^m)^q \in \langle a^m \rangle$. This proves the assertion of the theorem that every subgroup of a cyclic group is cyclic.

To prove the next portion of the theorem, suppose that $|\langle a \rangle| = n$ and H is any subgroup of $\langle a \rangle$. We have already shown that $H = \langle a^m \rangle$, where m is the least positive integer such that $a^m \in H$. Using $e = b = a^n$ as in the preceding paragraph, we have $n = mq$.

Finally, let k be any positive divisor of n. We will show that $\langle a^{\frac{n}{k}} \rangle$ is the one and only subgroup of $\langle a \rangle$ of order k. From Theorem 1.2.8, we see that $\langle a^{\frac{n}{k}} \rangle$ has order $\frac{n}{\gcd\left(n, \frac{n}{k}\right)} = \frac{n}{\left(\frac{n}{k}\right)} = k$. Now let H be any subgroup of $\langle a \rangle$ of order k. We have already shown above that $H = \langle a^m \rangle$, where m is a divisor of n. Then $m = \gcd(n, m)$ and $k = \operatorname{ord}(a^m) = \operatorname{ord}(a^{\gcd(n,m)} = \frac{n}{\gcd(n,m)} = \frac{n}{m}$. Thus, $m = \frac{n}{k}$ and $H = \langle a^{\frac{n}{k}} \rangle$. ■

Corollary 1.2.13. *For each positive divisor k of n, the set $\langle \frac{n}{k} \rangle$ is the unique subgroup of $\mathbb{Z}_n$ of order k; moreover, these are the only subgroups of $\mathbb{Z}_n$.*

Theorem 1.2.14. *If d is a positive divisor of n, the number of elements of order d in a cyclic group of order n is $\phi(d)$.*

Proof. By Theorem 1.2.12, the group has exactly one subgroup of order d, call it $\langle a \rangle$. Then every element of order d also generates the subgroup $\langle a \rangle$, by Corollary 1.2.9 of Theorem 1.2.8, an element a^k generates $\langle a \rangle$ if and only if $\gcd(k, d) = 1$. The number of such elements is precisely $\phi(d)$. ■

Corollary 1.2.15. *In a finite group, the number of elements of order d is a multiple of $\phi(d)$.*

Proof. If a finite group has no elements of order d, the statement is true, since $\phi(d)$ divides 0. Now suppose that $a \in G$ and $\operatorname{ord}(a) = d$. By Theorem 1.2.14, we know that $\langle a \rangle$ has $\phi(d)$ elements of order d. If all elements of order d in G are in $\langle a \rangle$, we are done. So, suppose that there is an element b in G of order d that is not in $\langle a \rangle$. Then, $\langle b \rangle$ also has $\phi(d)$ elements of order d. This means that we have found $2\phi(d)$ elements of order d in G provided that $\langle a \rangle$ and $\langle b \rangle$ have no elements of order d in common. If there is an element c of order d that belongs to both $\langle a \rangle$ and $\langle b \rangle$, then we have $\langle a \rangle = \langle b \rangle = \langle c \rangle$ so that $b \in \langle a \rangle$, which is a contradiction. Continuing in this fashion, we see that the number of elements of order d in a finite group is a multiple of $\phi(d)$. ■

1.3 HOMOMORPHISMS

Definition 1.3.1. If $(G, *)$ and $(H, \circ)$ are groups, then a function $f : G \to H$ is a **homomorphism**[3] if
$$f(x * y) = f(x) \circ f(y)$$
for all $x, y \in G$. If f is also a bijection, then f is called an **isomorphism**. Two groups G and H are called **isomorphic**, denoted by $G \cong H$, if there exists an isomorphism $f : G \to H$ between them.

Lemma 1.3.2. *Let $f : G \to H$ be a homomorphism of groups.*

(i) *$f(e) = e'$, where e and e' are the identities of G and H, respectively.*

(ii) $f(x^{-1}) = f(x)^{-1}$

(iii) *$f(x^n) = f(x)^n$ for all $n \in \mathbb{Z}$*

Proof. (i) $e \cdot e = e$ implies $f(e)f(e) = f(e)$.

(ii) $e = xx^{-1}$ implies $e' = f(1) = f(x)f(x^{-1})$, so $f(x^{-1}) = f(x)^{-1}$.

[3] The word homomorphism comes from the Greek homo meaning "same" and morph meaning "shape" or "form." Thus, a homomorphism carries a group to another group (its image) of similar form. The word isomorphism involves the Greek iso meaning "equal," and isomorphic groups have identical form.

(iii) Use induction to show that $f(x^n) = f(x)^n$ for all $n \geq 0$. Then observe that $x^{-n} = (x^{-1})^n$, and use part (ii).

■

Definition 1.3.3 (Kernel and Image). If $f : G \to H$ is a homomorphism, define

$$\textbf{kernel } f = \{x \in G : f(x) = 1\}$$

and

$$\textbf{image } f = \{h \in H : h = f(x) \text{ for some } x \in G\}.$$

We usually abbreviate kernel f to $\ker f$ and image f to $\operatorname{im} f$.

Proposition 1.3.4. Let $f : G \to H$ be a homomorphism.

(i) $\ker f$ is a subgroup of G and $\operatorname{im} f$ is a subgroup of H.

(ii) If $x \in \ker f$ and if $a \in G$, then $axa^{-1} \in \ker f$.

(iii) f is an injection if and only if $\ker f = \{e\}$.

Proof. (i) Recall that $\ker f = \{g \in G \mid f(g) = e_H\}$, where e_H is the identity element of H.

1. $\ker f$ is nonempty because $f(e_G) = e_H$, so $e_G \in \ker f$.
2. Let $a, b \in \ker f$. Then $f(a) = e_H$ and $f(b) = e_H$. We have:

$$f(ab^{-1}) = f(a)f(b^{-1}) = f(a)(f(b))^{-1} = e_H(e_H)^{-1} = e_He_H = e_H$$

Therefore, $ab^{-1} \in \ker f$.

By the subgroup criterion, $\ker f$ is a subgroup of G.
Recall that $\operatorname{im} f = \{f(g) \mid g \in G\}$.

1. $\operatorname{im} f$ is nonempty because $f(e_G) = e_H$, so $e_H \in \operatorname{im} f$.
2. Let $x, y \in \operatorname{im} f$. Then there exist $a, b \in G$ such that $x = f(a)$ and $y = f(b)$. We have:

$$xy^{-1} = f(a)(f(b))^{-1} = f(a)f(b^{-1}) = f(ab^{-1})$$

Since $ab^{-1} \in G$, we have $xy^{-1} = f(ab^{-1}) \in \operatorname{im} f$.

By the subgroup criterion, $\operatorname{im} f$ is a subgroup of H.

(ii) $f(axa^{-1}) = f(a)f(x)f(a)^{-1} = f(a)e'f(a)^{-1} = e'$.

(iii) $f(a) = f(b)$ if and only if $f(b^{-1}a) = e'$.

■

Theorem 1.3.5 (Structure Theorem for Cyclic Groups). *Let G be a cyclic group. Then either:*

1. *G is infinite and isomorphic to the additive group $\mathbb{Z}$, or*
2. *G is finite of order n and isomorphic to the additive group $\mathbb{Z}/n\mathbb{Z}$.*

Proof. Since G is cyclic, there exists an element $g \in G$ such that $G = \langle g \rangle$. Define a map $\varphi : \mathbb{Z} \to G$ by

$$\varphi(k) = g^k \quad \text{for all } k \in \mathbb{Z}.$$

We first show that φ is a group homomorphism. For any $k, l \in \mathbb{Z}$,

$$\varphi(k + l) = g^{k+l} = g^k g^l = \varphi(k)\varphi(l),$$

so φ preserves the group operation. Moreover, since every element of G is of the form g^k for some integer k, φ is surjective.

By the First Isomorphism Theorem,

$$\mathbb{Z}/\ker(\varphi) \cong G.$$

The kernel $\ker(\varphi)$ is a subgroup of $\mathbb{Z}$. It is well-known that every subgroup of $\mathbb{Z}$ is of the form $m\mathbb{Z}$ for some nonnegative integer m. Thus, $\ker(\varphi) = m\mathbb{Z}$ for some $m \geq 0$.

We now consider two cases:

1. If $m = 0$, then $\ker(\varphi) = \{0\}$, so
$$\mathbb{Z}/\{0\} \cong \mathbb{Z} \cong G.$$
In this case, G is infinite.

2. If $m > 0$, then $\ker(\varphi) = m\mathbb{Z}$, so
$$G \cong \mathbb{Z}/m\mathbb{Z}.$$
The order of $\mathbb{Z}/m\mathbb{Z}$ is m. To show that $m = n$ (where n is the order of G), note that since φ is surjective and $\ker(\varphi) = m\mathbb{Z}$, the group G has exactly m distinct elements. Specifically, for any $k \in \mathbb{Z}$, $g^k = g^r$ where $0 \le r < m$ is the remainder when k is divided by m. Thus, $G = \{g^0, g^1, \dots, g^{m-1}\}$, and these elements are distinct because if $g^a = g^b$ for $0 \le a < b < m$, then $g^{b-a} = e$, contradicting the minimality of m (since $0 < b - a < m$). Therefore, $|G| = m$, so $m = n$.

■

Example 1.3.6. We show that the quotient group G/K is precisely $\mathbb{Z}_m$ when G is the additive group $\mathbb{Z}$ and $K = \langle m \rangle$, the (cyclic) subgroup of all the multiples of a positive integer m. Since $\mathbb{Z}$ is abelian, $\langle m \rangle$ is necessarily a normal subgroup. The sets $\mathbb{Z}/\langle m \rangle$ and $\mathbb{Z}_m$ coincide because they are comprised of the same elements: The coset $a + \langle m \rangle$ is the congruence class $[a]$:
$$a + \langle m \rangle = \{a + km : k \in \mathbb{Z}\} = [a].$$
The operations also coincide: Addition in $\mathbb{Z}/\langle m \rangle$ is given by
$$(a + \langle m \rangle) + (b + \langle m \rangle) = (a + b) + \langle m \rangle;$$
since $a + \langle m \rangle = [a]$, this last equation is just $[a] + [b] = [a + b]$, which is the sum in $\mathbb{Z}_m$. Therefore, $\mathbb{Z}_m$ is equal to the quotient group $\mathbb{Z}/\langle m \rangle$.

Definition 1.3.7. Let G be a group. An **endomorphism** of G is a homomorphism $\varphi : G \to G$ from G to itself.

Example 1.3.8. 1. The identity map $\mathrm{id}_G : G \to G$ defined by $\mathrm{id}_G(g) = g$ for all $g \in G$ is an endomorphism.

2. The trivial map $\varphi : G \to G$ defined by $\varphi(g) = e$ for all $g \in G$ is an endomorphism.

3. For any integer n, the map $\varphi_n : \mathbb{Z} \to \mathbb{Z}$ defined by $\varphi_n(k) = nk$ is an endomorphism of the additive group $\mathbb{Z}$.

4. For a cyclic group $G = \langle g \rangle$ of order m, the map $\varphi_k : G \to G$ defined by $\varphi_k(g^i) = g^{ki}$ is an endomorphism for any integer k.

Remark 1.3.9. The set of all endomorphisms of a group G, denoted $\mathrm{End}(G)$, forms a monoid under composition, with identity element id_G.

Definition 1.3.10. An **automorphism** of a group G is an isomorphism $\varphi : G \to G$ from G to itself.

Example 1.3.11. 1. The identity map id_G is an automorphism.

2. For the additive group $\mathbb{R}$, the map $\varphi(x) = -x$ is an automorphism.

3. For the multiplicative group $\mathbb{R}^*$, the map $\varphi(x) = x^{-1}$ is an automorphism.

4. For the symmetric group S_n, conjugation by any element $\sigma \in S_n$ (i.e., the map $\tau \mapsto \sigma\tau\sigma^{-1}$) is an automorphism.

Theorem 1.3.12. *The set of all automorphisms of a group G, denoted $\mathrm{Aut}(G)$, forms a group under composition, called the **automorphism group** of G.*

Proof. We verify the group axioms:

1. **Closure**: If $\varphi, \psi \in \mathrm{Aut}(G)$, then $\varphi \circ \psi$ is an automorphism since the composition of isomorphisms is an isomorphism.

2. **Associativity**: Function composition is associative.
3. **Identity**: id_G serves as the identity element.
4. **Inverses**: If $\varphi \in \mathrm{Aut}(G)$, then its inverse function φ^{-1} is also an automorphism.

■

Definition 1.3.13. For a fixed element $g \in G$, define the map $\iota_g : G \to G$ by

$$\iota_g(x) = gxg^{-1}.$$

This map is called **conjugation by** g and is an automorphism of G. Automorphisms of this form are called **inner automorphisms**.

Theorem 1.3.14. *The set of inner automorphisms, denoted Inn(G), forms a normal subgroup of Aut(G).*

Proof. First, note that $\mathrm{Inn}(G)$ is nonempty because the identity automorphism id_G is equal to ι_e, where e is the identity element of G.

Now, let $\iota_g, \iota_h \in \mathrm{Inn}(G)$. We need to show that $\iota_g \circ \iota_h^{-1} \in \mathrm{Inn}(G)$.

Note that the inverse of ι_h is $\iota_{h^{-1}}$, since:

$$(\iota_h \circ \iota_{h^{-1}})(x) = \iota_h(h^{-1}xh) = h(h^{-1}xh)h^{-1} = x = \mathrm{id}_G(x)$$

and similarly $\iota_{h^{-1}} \circ \iota_h = \mathrm{id}_G$.

Therefore:

$$\iota_g \circ \iota_h^{-1} = \iota_g \circ \iota_{h^{-1}} = \iota_{gh^{-1}}$$

since for any $x \in G$:

$$(\iota_g \circ \iota_{h^{-1}})(x) = \iota_g(\iota_{h^{-1}}(x)) = \iota_g(h^{-1}xh) = g(h^{-1}xh)g^{-1} = (gh^{-1})x(gh^{-1})^{-1} = \iota_{gh^{-1}}(x)$$

Since $gh^{-1} \in G$, we have $\iota_{gh^{-1}} \in \mathrm{Inn}(G)$.

Thus, by the subgroup test, $\mathrm{Inn}(G)$ is a subgroup of $\mathrm{Aut}(G)$. To show normality, let $\varphi \in \mathrm{Aut}(G)$ and $\iota_g \in \mathrm{Inn}(G)$. Then

$$(\varphi \circ \iota_g \circ \varphi^{-1})(x) = \varphi(g\varphi^{-1}(x)g^{-1}) = \varphi(g)x\varphi(g)^{-1} = \iota_{\varphi(g)}(x),$$

so $\varphi \circ \iota_g \circ \varphi^{-1} \in \mathrm{Inn}(G)$. ■

Definition 1.3.15. The quotient group

$$\mathrm{Out}(G) = \mathrm{Aut}(G)/\mathrm{Inn}(G)$$

is called the **outer automorphism group** of G. Elements of Out(G) are cosets of Inn(G) in Aut(G), and automorphisms representing nontrivial cosets are called **outer automorphisms**.

Example 1.3.16.
1. For an abelian group G, Inn(G) is trivial, so Aut(G) = Out(G).
2. For the symmetric group S_n with $n \neq 6$, Out(S_n) is trivial. For $n = 6$, Out(S_6) has order 2.
3. For the alternating group A_n with $n \geq 4$, $n \neq 6$, Out(A_n) has order 2.
4. For the dihedral group D_4 of order 8, Out(D_4) has order 2.

Remark 1.3.17. Not every automorphism is inner. For example, the automorphism of the additive group $\mathbb{Z}$ defined by $\varphi(x) = -x$ is not inner (since $\mathbb{Z}$ is abelian, all inner automorphisms are trivial).

Theorem 1.3.18. *The map $\Phi : G \to$ Aut(G) defined by $\Phi(g) = \iota_g$ is a homomorphism with kernel $Z(G)$, the center of G. Thus,*

$$G/Z(G) \cong \mathit{Inn}(G).$$

Proof. For $g, h \in G$, we have

$$\Phi(gh) = \iota_{gh} = \iota_g \circ \iota_h = \Phi(g) \circ \Phi(h),$$

so Φ is a homomorphism. The kernel consists of elements $g \in G$ such that $\iota_g = \mathrm{id}_G$, i.e., $gxg^{-1} = x$ for all $x \in G$. This is exactly the center $Z(G)$. The result follows from the First Isomorphism Theorem. ■

Theorem 1.3.19. *If G is a complete group (i.e., $Z(G) = \{e\}$ and $Aut(G) = Inn(G)$), then $Aut(G) \cong G$.*

Example 1.3.20. The symmetric group S_n for $n \neq 2, 6$ is complete, so $\mathrm{Aut}(S_n) \cong S_n$.

Groups of permutations led us to abstract groups; the next result, due to A. Cayley, shows that abstract groups are not so far removed from permutations.

Theorem 1.3.21 (Cayley). *Every group G is isomorphic to a subgroup of the symmetric group S_G. In particular, if $|G| = n$, then G is isomorphic to a subgroup of S_n.*

Proof. For each $a \in G$, define "translation" $\tau_a : G \to G$ by $\tau_a(x) = ax$ for every $x \in G$ (if $a \neq e$, then τ_a is not a homomorphism). For $a, b \in G$, $(\tau_a \circ \tau_b)(x) = \tau_a(\tau_b(x)) = \tau_a(bx) = a(bx) = (ab)x$, by associativity, so that

$$\tau_a \tau_b = \tau_{ab}.$$

It follows that each τ_a is a bijection, for its inverse is $\tau_{a^{-1}}$:

$$\tau_a \tau_{a^{-1}} = \tau_{aa^{-1}} = \tau_1 = 1_G = \tau_{a^{-1}a},$$

and so $\tau_a \in S_G$.

Define $\varphi : G \to S_G$ by $\varphi(a) = \tau_a$. Note that,

$$\varphi(a)\varphi(b) = \tau_a \tau_b = \tau_{ab} = \varphi(ab),$$

so that φ is a homomorphism. Finally, φ is an injection. If $\varphi(a) = \varphi(b)$, then $\tau_a = \tau_b$, and hence $\tau_a(x) = \tau_b(x)$ for all $x \in G$; in particular, when $x = 1$, this gives $a = b$, as desired.

The last statement follows from the fact that if X is a set with $|X| = n$, then $S_X \cong S_n$. ■

Theorem 1.3.22 (Representation of Cosets). *Let G be a group, and let H be a subgroup of G having finite index n. Then there exists a homomorphism $\varphi : G \to S_n$ with $\ker \varphi \leq H$.*

Proof. Even though H may not be a normal subgroup, we still denote the family of all the cosets of H in G by G/H.

For each $a \in G$, define "translation" $\tau_a : G/H \to G/H$ by $\tau_a(xH) = axH$ for every $x \in G$. For $a, b \in G$,

$$(\tau_a \circ \tau_b)(xH) = \tau_a(\tau_b(xH)) = \tau_a(bxH) = a(bxH) = (ab)xH,$$

by associativity, so that $\tau_a \tau_b = \tau_{ab}$. It follows that each τ_a is a bijection, for its inverse is $\tau_{a^{-1}}$:

$$\tau_a \tau_{a^{-1}} = \tau_{aa^{-1}} = \tau_1 = 1_{G/H} = \tau_{a^{-1}} \tau_a,$$

and so $\tau_a \in S_{G/H}$. Define $\varphi : G \to S_{G/H}$ by $\varphi(a) = \tau_a$. Rewriting,

$$\varphi(a)\varphi(b) = \tau_a \tau_b = \tau_{ab} = \varphi(ab),$$

so that φ is a homomorphism. Finally, if $a \in \ker \varphi$, then $\varphi(a) = 1_{G/H}$, so that $\tau_a(xH) = xH$ for all $x \in G$; in particular, when $x = 1$, this gives $aH = H$. The result follows, for $|G/H| = n$, and so $S_{G/H} \cong S_n$. ■

Note that, when $H = \{1\}$, this is the Cayley theorem.

We are now going to classify all groups of order up to 7. By Theorem 1.3.5, every group of prime order p is isomorphic to $\mathbb{Z}_p$, and so, to isomorphism, there is just one group of order p. Of the possible orders through 7, four of them, 2, 3, 5, and 7, are primes, and so we need look only at orders 4 and 6.

Proposition 1.3.23. Every group G of order 4 is isomorphic to either $\mathbb{Z}_4$ or the four-group $\mathbf{V}$. Moreover, $\mathbb{Z}_4$ and $\mathbf{V}$ are not isomorphic.

Proof. By Lagrange's theorem, every element in G, other than 1, has order either 2 or 4. If there is an element of order 4, then G is cyclic. Otherwise, $x^2 = 1$ for all $x \in G$, so that G is abelian.

If distinct elements x and y in G are chosen, neither being 1, then we quickly check that $xy \notin \{1, x, y\}$; hence,

$$G = \{1, x, y, xy\}.$$

It is easy to see that the bijection $f : G \to \mathbf{V}$, defined by $f(1) = 1$, $f(x) = (1\ 2)(3\ 4)$, $f(y) = (1\ 3)(2\ 4)$, and $f(xy) = (1\ 4)(2\ 3)$, is an isomorphism.

Also, note that $\mathbb{Z}_4$ being cyclic and $\mathbf{V}$ being non-cyclic, we see that $\mathbb{Z}_4 \ncong \mathbf{V}$. ■

Proposition 1.3.24. If G is a group of order 6, then G is isomorphic to either $\mathbb{Z}_6$ or S_3. Moreover, $\mathbb{Z}_6$ and S_3 are not isomorphic.[4]

Proof. By Lagrange's theorem, the only possible orders of nonidentity elements are 2, 3, and 6. Of course, $G \cong \mathbb{Z}_6$ if G has an element of order 6. Notice that G must contain an element of order 2, say, t. We now consider the cases G abelian and G nonabelian separately.

Case 1. G is abelian: If there is a second element of order 2, say, a, then it is easy to see, using $at = ta$, that $H = \{1, a, t, at\}$ is a subgroup of G. This contradicts Lagrange's theorem, because 4 is not a divisor of 6. It follows that G must contain an element b of order 3. But then tb has order 6. Therefore, G is cyclic if it is abelian.

Case 2. G is not abelian: If G has no elements of order 3, then $x^2 = 1$ for all $x \in G$, and G is abelian. Therefore, G contains an element s of order 3 as well as the element t of order 2.

Now $|\langle s\rangle| = 3$, so that $[G : \langle s\rangle] = |G|/|\langle s\rangle| = 6/3 = 2$, and so $\langle s\rangle$ is a normal subgroup of G. Since $t = t^{-1}$, we have $tst = tst^{-1} \in \langle s\rangle$; hence, $tst = s^i$ for $i = 0, 1$ or 2. Now $i \neq 0$, for $tst = s^0 = 1$ implies $s = 1$. If $i = 1$, then s and t commute, and this gives st of order 6, as in Case 1 (which forces G to be cyclic, hence abelian, contrary to our present hypothesis). Therefore, $tst = s^2 = s^{-1}$.

We now use Theorem 1.3.22 to construct an isomorphism $G \to S_3$. Let $H = \langle t\rangle$, and consider the homomorphism $\varphi : G \to S_{G/\langle t\rangle}$ given by

$$\varphi(g) : x\langle t\rangle \mapsto gx\langle t\rangle.$$

By the theorem, $\ker\varphi \leq \langle t\rangle$, so that either $\ker\varphi = \{1\}$ (and φ is injective), or $\ker\varphi = \langle t\rangle$. Now $G/\langle t\rangle = \{\langle t\rangle, s\langle t\rangle, s^2\langle t\rangle\}$, and, in two-rowed notation,

$$\varphi(t) = \begin{pmatrix} \langle t\rangle & s\langle t\rangle & s^2\langle t\rangle \\ t\langle t\rangle & ts\langle t\rangle & ts^2\langle t\rangle \end{pmatrix}.$$

If $\varphi(t)$ is the identity permutation, then $ts\langle t\rangle = s\langle t\rangle$, so that $s^{-1}ts \in \langle t\rangle = \{1, t\}$, by Lemma 2.40. But now $s^{-1}ts = t$ (it cannot be 1), hence $ts = st$, contradicting t and s not commuting. Therefore, $t \notin \ker\varphi$, and $\varphi : G \to S_{G/\langle t\rangle} \cong S_3$ is an injective homomorphism. Since both G and S_3 have order 6, φ must be a bijection, and so $G \cong S_3$.

It is clear that $\mathbb{Z}_6$ and S_3 are not isomorphic, for one is abelian and the other is not. ■

[4]Cayley states this proposition in an article he wrote in 1854. However, in 1878, in the American Journal of Mathematics, he wrote, "The general problem is to find all groups of a given order n; ... if $n = 6$, there are three groups; a group

$$1, \alpha, \alpha^2, \alpha^3, \alpha^4, \alpha^5 \quad (\alpha^6 = 1),$$

and two more groups

$$1, \beta, \beta^2, \alpha, \alpha\beta, \alpha\beta^2 \quad (\alpha^2 = 1, \beta^3 = 1),$$

viz., in the first of these $\alpha\beta = \beta\alpha$ while in the other of them, we have $\alpha\beta = \beta^2\alpha, \alpha\beta^2 = \beta\alpha$." Cayley's list is $\mathbb{Z}_6, \mathbb{Z}_2 \times \mathbb{Z}_3$, and S_3. Of course, $\mathbb{Z}_2 \times \mathbb{Z}_3 \cong \mathbb{Z}_6$; even Homer nods.

Chapter 2

Group Actions

2.1 Conjugacy Classes

A fundamental concept in the study of groups is that of a group acting on a set. This leads to powerful tools for understanding group structure, including the notion of conjugacy classes.

Definition 2.1.1 (Group Action)**.** Let G be a group and X be a set. A *left group action* of G on X is a map $\cdot : G \times X \to X$, denoted by $(g, x) \mapsto g \cdot x$, satisfying the following two axioms:

1. $e \cdot x = x$ for all $x \in X$ (where e is the identity element of G).
2. $(g_1 g_2) \cdot x = g_1 \cdot (g_2 \cdot x)$ for all $g_1, g_2 \in G$ and $x \in X$.

Similarly, a *right group action* is a map $\cdot : X \times G \to X$, denoted by $(x, g) \mapsto x \cdot g$, satisfying:

1. $x \cdot e = x$ for all $x \in X$.
2. $x \cdot (g_1 g_2) = (x \cdot g_1) \cdot g_2$ for all $g_1, g_2 \in G$ and $x \in X$.

Example 2.1.2. Consider the symmetric group S_n acting on the set $\{1, 2, \ldots, n\}$. The action is defined by $\sigma \cdot i = \sigma(i)$ for $\sigma \in S_n$ and $i \in \{1, 2, \ldots, n\}$.

Example 2.1.3. Let G be any group. Consider the action of G on itself by left multiplication. For $g, x \in G$, define $g \cdot x = gx$. This is a left action.

Example 2.1.4. Let G be any group. Consider the action of G on itself by right multiplication. For $g, x \in G$, define $x \cdot g = xg$. This is a right action.

One of the most important group actions is conjugation.

Definition 2.1.5 (Action by Conjugation)**.** Let G be a group. The group G acts on itself by *conjugation*. This action is defined for $g, x \in G$ as $g \cdot x = gxg^{-1}$.

Let's verify this is indeed a group action:

Proof.
1. Identity: $e \cdot x = exe^{-1} = x$ since $e^{-1} = e$. This holds for all $x \in G$.
2. Associativity: For $g_1, g_2, x \in G$: $(g_1 g_2) \cdot x = (g_1 g_2) x (g_1 g_2)^{-1} = g_1 g_2 x g_2^{-1} g_1^{-1}$ And $g_1 \cdot (g_2 \cdot x) = g_1 \cdot (g_2 x g_2^{-1}) = g_1 (g_2 x g_2^{-1}) g_1^{-1} = g_1 g_2 x g_2^{-1} g_1^{-1}$. Since these are equal, the associativity axiom is satisfied.

Therefore, conjugation is a group action. ∎

When a group acts on a set, the set is partitioned into orbits. For the action of conjugation, these orbits are called conjugacy classes.

Definition 2.1.6 (Conjugacy Class)**.** Let G be a group. For an element $x \in G$, the *conjugacy class* of x in G, denoted by C_x or $\mathrm{cl}(x)$, is the set of all elements $y \in G$ such that $y = gxg^{-1}$ for some $g \in G$.

$$C_x = \{gxg^{-1} \mid g \in G\}$$

Proposition 2.1.7. Conjugacy is an equivalence relation on a group G. That is, for elements $x, y, z \in G$:

1. Reflexivity: $x \sim x$ (where $x \sim y$ means y is conjugate to x).
2. Symmetry: If $x \sim y$, then $y \sim x$.
3. Transitivity: If $x \sim y$ and $y \sim z$, then $x \sim z$.

Proof. 1. Reflexivity: For any $x \in G$, $x = exe^{-1}$. So x is conjugate to itself.

2. Symmetry: If $x \sim y$, then there exists $g \in G$ such that $y = gxg^{-1}$. Multiplying by g^{-1} on the left and g on the right, we get $g^{-1}yg = x$. Since $(g^{-1})^{-1} = g$, we can write $x = g^{-1}y(g^{-1})^{-1}$. Thus, $y \sim x$.

3. Transitivity: If $x \sim y$ and $y \sim z$, then there exist $g_1, g_2 \in G$ such that $y = g_1xg_1^{-1}$ and $z = g_2yg_2^{-1}$. Substituting the expression for y into the equation for z: $z = g_2(g_1xg_1^{-1})g_2^{-1} = (g_2g_1)x(g_1^{-1}g_2^{-1}) = (g_2g_1)x(g_2g_1)^{-1}$. Let $h = g_2g_1 \in G$. Then $z = hxh^{-1}$. Thus, $x \sim z$.

Therefore, conjugacy is an equivalence relation. ■

Since conjugacy is an equivalence relation, it partitions the group G into disjoint conjugacy classes.

Example 2.1.8. Consider the symmetric group S_3. Its elements are: $e = (1)(2)(3)$, $(12) = (12)(3)$, $(13) = (13)(2)$, $(23) = (23)(1)$, (123), and (132)

Let's find the conjugacy classes in S_3.

Solution. We know that two permutations in S_n are conjugate if and only if they have the same cycle structure.

1. The identity element e is always in its own conjugacy class: $C_e = \{e\}$. This is because $geg^{-1} = e$ for all $g \in G$.
2. Elements of order 2 (transpositions): $(12), (13), (23)$. These all have cycle structure (2-cycle)(1-cycle). They form one conjugacy class: $C_{(12)} = \{(12), (13), (23)\}$.
3. Elements of order 3 (3-cycles): $(123), (132)$. These all have cycle structure (3-cycle). They form another conjugacy class: $C_{(123)} = \{(123), (132)\}$.

The conjugacy classes of S_3 are $\{e\}$, $\{(12), (13), (23)\}$, and $\{(123), (132)\}$. ■

Example 2.1.9. Consider the group of quaternions $Q_8 = \{1, -1, i, -i, j, -j, k, -k\}$.

Solution. We list the conjugacy classes:

1. $C_1 = \{1\}$ (always the case for the identity).
2. $C_{-1} = \{-1\}$ (always the case for elements in the center of the group, and -1 commutes with all elements in Q_8).
3. For i: $1i1^{-1} = i$ $(-1)i(-1)^{-1} = (-1)i(-1) = i$ $iii^{-1} = i$ $jij^{-1} = ji(-j) = (-k)(-j) = kj = -i$ $kik^{-1} = ki(-k) = j(-k) = -jk = -i$ So $C_i = \{i, -i\}$.
4. Similarly, $C_j = \{j, -j\}$.
5. And $C_k = \{k, -k\}$.

The conjugacy classes of Q_8 are $\{1\}$, $\{-1\}$, $\{i, -i\}$, $\{j, -j\}$, $\{k, -k\}$. ■

The size of conjugacy classes is related to centralizers.

Definition 2.1.10 (Centralizer)**.** Let G be a group and $x \in G$. The *centralizer* of x in G, denoted by $C_G(x)$, is the set of elements in G that commute with x:

$$C_G(x) = \{g \in G \mid gx = xg\}$$

Proposition 2.1.11. $C_G(x)$ is a subgroup of G.

Proof. 1. Identity: $ex = xe = x$, so $e \in C_G(x)$.

2. Closure: If $g_1, g_2 \in C_G(x)$, then $g_1x = xg_1$ and $g_2x = xg_2$. Consider $(g_1g_2)x = g_1(g_2x) = g_1(xg_2) = (g_1x)g_2 = (xg_1)g_2 = x(g_1g_2)$. So $g_1g_2 \in C_G(x)$.

3. Inverse: If $g \in C_G(x)$, then $gx = xg$. Multiply by g^{-1} on the left and right: $g^{-1}(gx)g^{-1} = g^{-1}(xg)g^{-1}$. $(g^{-1}g)xg^{-1} = g^{-1}x(gg^{-1})\ exg^{-1} = g^{-1}xe\ xg^{-1} = g^{-1}x$. So $g^{-1} \in C_G(x)$.

Therefore, $C_G(x)$ is a subgroup of G. ■

Definition 2.1.12 (Stabilizer of a Point)**.** Let G be a group of permutations of a set S. For each i in S, let $\text{stab}_G(i) = \{\phi \in G \mid \phi(i) = i\}$. We call $\text{stab}_G(i)$ the **stabilizer of** i **in** G.

The student should verify that $\text{stab}_G(i)$ is a subgroup of G. (See Exercise 31 in Chapter 5.)

Definition 2.1.13 (Orbit of a Point)**.** Let G be a group of permutations of a set S. For each s in S, let $\text{orb}_G(s) = \{\phi(s) \mid \phi \in G\}$. The set $\text{orb}_G(s)$ is a subset of S called the **orbit of** s **under** G. We use $|\text{orb}_G(s)|$ to denote the number of elements in $\text{orb}_G(s)$.

Example 6 should clarify these two definitions.

Example 2.1.14. Let

$$G = \{(1), (132)(465)(78), (132)(465), (123)(456), (123)(456)(78), (78)\}.$$

Then,

$$\begin{aligned} \text{orb}_G(1) &= \{1, 3, 2\}, & \text{stab}_G(1) &= \{(1), (78)\}, \\ \text{orb}_G(2) &= \{2, 1, 3\}, & \text{stab}_G(2) &= \{(1), (78)\}, \\ \text{orb}_G(4) &= \{4, 6, 5\}, & \text{stab}_G(4) &= \{(1), (78)\}, \\ \text{orb}_G(7) &= \{7, 8\}, & \text{stab}_G(7) &= \{(1), (132)(465), (123)(456)\}. \end{aligned}$$

2.1.1 The Orbit Stabilizer Theorem

Theorem 2.1.15 (Orbit-Stabilizer Theorem)**.** *Let G be a finite group of permutations of a set S. Then, for any i from S, $|G| = |orb_G(i)||stab_G(i)|$.*

Proof. By Lagrange's Theorem, $|G|/|\text{stab}_G(i)|$ is the number of distinct left cosets of $\text{stab}_G(i)$ in G. Thus, it suffices to establish a one-to-one correspondence between the left cosets of $\text{stab}_G(i)$ and the elements in the orbit of i. To do this, we define a correspondence T by mapping the coset $\phi\,\text{stab}_G(i)$ to $\phi(i)$ under T. To show that T is a well-defined function, we must show that $\alpha\,\text{stab}_G(i) = \beta\,\text{stab}_G(i)$ implies $\alpha(i) = \beta(i)$. But $\alpha\,\text{stab}_G(i) = \beta\,\text{stab}_G(i)$ implies $\alpha^{-1}\beta \in \text{stab}_G(i)$, so that $(\alpha^{-1}\beta)(i) = i$ and, therefore, $\beta(i) = \alpha(i)$. Reversing the argument from the first step shows that T is also one-to-one. We conclude the proof by showing that T is onto. Let $j \in \text{orb}_G(i)$. Let $\alpha(i) = j$ for some $\alpha \in G$ and clearly $T(\alpha\,\text{stab}_G(i)) = \alpha(i) = j$, so that T is onto. ■

Example 2.1.16. Use the class equation to confirm the conjugacy classes for S_3.

Solution. We have $|S_3| = 6$. The center of S_3 is $Z(S_3) = \{e\}$ (since only the identity commutes with all elements). So $|Z(S_3)| = 1$. The conjugacy classes we found were $\{e\}$, $\{(12), (13), (23)\}$, and $\{(123), (132)\}$. Let $x_1 = e$, $x_2 = (12)$, $x_3 = (123)$. $|C_e| = 1$. $C_{S_3}(e) = S_3$, so $|C_e| = |S_3|/|S_3| = 6/6 = 1$.

For $x_2 = (12)$: $C_{S_3}((12)) = \{\sigma \in S_3 \mid \sigma(12) = (12)\sigma\}$. The elements are e and (12). Check: $(12)(12) = e = (12)(12)$. $(13)(12) = (132)$, $(12)(13) = (123)$. Not equal. $(23)(12) = (123)$, $(12)(23) = (132)$. Not equal. So $C_{S_3}((12)) = \{e, (12)\}$, and $|C_{S_3}((12))| = 2$. Thus, $|C_{(12)}| = |S_3|/|C_{S_3}((12))| = 6/2 = 3$. This matches $C_{(12)} = \{(12), (13), (23)\}$.

For $x_3 = (123)$: $C_{S_3}((123)) = \{\sigma \in S_3 \mid \sigma(123) = (123)\sigma\}$. The elements are e and (123) and (132). Check: $(123)(123) = (132) = (123)(123)$. So $C_{S_3}((123)) = \{e, (123), (132)\}$, and $|C_{S_3}((123))| = 3$. Thus, $|C_{(123)}| = |S_3|/|C_{S_3}((123))| = 6/3 = 2$. This matches $C_{(123)} = \{(123), (132)\}$.

The class equation for S_3 is: $|S_3| = |C_e| + |C_{(12)}| + |C_{(123)}|$, that is, $6 = 1 + 3 + 2$. ■

Since the conjugacy classes partition the group G, we can sum their sizes to get the order of the group. This leads to the class equation.

2.1.2 The Class Equation

Theorem 2.1.17 (The Class Equation)**.** *Let G be a finite group. Then*

$$|G| = |Z(G)| + \sum_{i=1}^{k} [G : C(a_i)]$$

where $a_1, a_2, \ldots, a_k$ are representatives from each distinct conjugacy class that contains more than one element.

Proof. Note that $a \in Z(G)$ iff $\mathrm{cl(a)} = \{\mathrm{a}\}$. Let $a_1, a_2, \ldots, a_k$ be representatives from the distinct conjugacy classes with more than one element. We can write the partition as:

$$|G| = |Z(G)| + \sum_{i=1}^{k} |\mathrm{Cl}(a_i)|$$

By the Orbit-Stabilizer Theorem:

$$|\mathrm{Cl}(a_i)| = [G : C(a_i)] = \frac{|G|}{|C(a_i)|}$$

Substituting this into the equation gives the class equation:

$$|G| = |Z(G)| + \sum_{i=1}^{k} [G : C(a_i)]$$

■

A crucial consequence of the class equation is the following lemma, which is essential for analyzing groups of prime-power order.

Lemma 2.1.18. *Let G be a finite group of order p^n for some prime p and $n \geq 1$ (a p-group). Then its center is non-trivial, i.e., $|Z(G)| > 1$.*

Proof. Consider the class equation: $|G| = |Z(G)| + \sum_i [G : C(a_i)]$. For any $a_i \notin Z(G)$, its centralizer $C(a_i)$ is a proper subgroup of G. By Lagrange's Theorem, $|C(a_i)| = p^k$ for some $k < n$. Therefore, the size of its conjugacy class is $[G : C(a_i)] = |G|/|C(a_i)| = p^n/p^k = p^{n-k}$, which is a multiple of p since $n - k \geq 1$. The class equation can be rewritten as:

$$p^n = |Z(G)| + (\text{a sum of multiples of } p)$$

This implies that $|Z(G)| = p^n - (\text{a sum of multiples of } p)$. Therefore, p must divide $|Z(G)|$. Since the identity element is always in the center, $|Z(G)| \geq 1$. Because p divides $|Z(G)|$, we must have $|Z(G)| \geq p > 1$. ■

Theorem 2.1.19 (G/Z Theorem)**.** *For any group G, $G/Z(G)$ is cyclic inplies G is abelian.*

Proof. Let $G/Z(G)$ be generated by the coset $gZ(G)$. Then any element $x \in G$ can be written as $x = g^k z$ for some integer k and some $z \in Z(G)$. Let $x_1 = g^{k_1} z_1$ and $x_2 = g^{k_2} z_2$. $x_1 x_2 = (g^{k_1} z_1)(g^{k_2} z_2) = g^{k_1} g^{k_2} z_1 z_2 = g^{k_1+k_2} z_2 z_1 = (g^{k_2} z_2)(g^{k_1} z_1) = x_2 x_1$. Thus G is abelian. ■

Groups of Order p

Let $|G| = p$. By Lagrange's Theorem, the only subgroups of G are $\{e\}$ and G itself. Let $a \in G$ be a non-identity element. The subgroup $\langle a \rangle$ cannot be $\{e\}$, so it must be G. This means G is cyclic. Every cyclic group is abelian, so G is abelian. If G is abelian, then $Z(G) = G$, and every element is its own conjugacy class.

- **Class Equation:** $|G| = |Z(G)|$, so $p = p$. In terms of classes, this is written as:

$$p = \underbrace{1 + 1 + \cdots + 1}_{p \text{ times}}$$

Groups of Order p^2

Let $|G| = p^2$. By the lemma above, since G is a p-group, its center must be non-trivial. So $|Z(G)| > 1$. By Lagrange's Theorem, $|Z(G)|$ must divide $|G| = p^2$. The possible orders for the center are p and p^2.

- **Case 1:** $|Z(G)| = p^2$**.** Then $Z(G) = G$, which means G is an abelian group.
- **Case 2:** $|Z(G)| = p$**.** Consider the quotient group $G/Z(G)$. Its order is $|G/Z(G)| = |G|/|Z(G)| = p^2/p = p$. Any group of prime order is cyclic.

This means that Case 2 leads to a contradiction: if $|Z(G)| = p$, then G must be abelian, which would imply $|Z(G)| = p^2$. Therefore, Case 2 is impossible. The only possibility is that $|Z(G)| = p^2$. This proves that **every group of order** p^2 **is abelian**.

- **Class Equation:** Since every group of order p^2 is abelian, the class equation is always:

$$p^2 = \underbrace{1 + 1 + \cdots + 1}_{p^2 \text{ times}}$$

Groups of Order $2p$ (for an odd prime p)

Let $|G| = 2p$. By Cauchy's Theorem, G has an element a of order p and an element b of order 2. Let $H = \langle a \rangle$. Since $[G : H] = 2$, H is a normal subgroup. The order of the center, $|Z(G)|$, must divide $2p$, so $|Z(G)| \in \{1, 2, p, 2p\}$.

- If $|Z(G)| = 2p$, G is abelian (and cyclic, isomorphic to $\mathbb{Z}_{2p}$).
- If $|Z(G)| = p$, then $|G/Z(G)| = 2$, which is cyclic. This implies G is abelian, a contradiction.
- If $|Z(G)| = 2$, then $|G/Z(G)| = p$, which is cyclic. This implies G is abelian, a contradiction.

So, either G is abelian ($|Z(G)| = 2p$) or it has a trivial center ($|Z(G)| = 1$).

Case 1: G is abelian. The class equation is $2p = \underbrace{1 + 1 + \cdots + 1}_{2p \text{ times}}$.

Case 2: G is non-abelian. An example is the dihedral group D_p. Here, $|Z(G)| = 1$. The class equation is $2p = 1 + \sum [G : C(a_i)]$. The sizes of the non-trivial classes, $[G : C(a_i)]$, must divide $2p$ and be greater than 1. Possible sizes are 2 and p. For the dihedral group D_p (symmetries of a p-gon), the classes are:

1. The identity element: $\{e\}$. Size 1.
2. The p reflections: These form a single conjugacy class of size p.
3. The rotations: The $p - 1$ non-identity rotations are paired up, (r^k, r^{-k}), into $(p - 1)/2$ classes, each of size 2.

- **Class Equation (Non-abelian case):**

$$2p = 1 + p + \underbrace{2 + 2 + \cdots + 2}_{(p-1)/2 \text{ times}}$$

Groups of Order p^3

If G is abelian then $|G| = p^3$. Since this is a p-group, we know $|Z(G)| > 1$. The possible orders are p, p^2, p^3.

- If $|Z(G)| = p^3$, G is abelian. The class equation is $p^3 = \underbrace{1 + \cdots + 1}_{p^3 \text{ times}}$.
- If $|Z(G)| = p^2$, then $|G/Z(G)| = p$, which is cyclic, implying G is abelian. This is a contradiction, so this case is impossible.

Case: G is non-abelian. We have $|Z(G)| = p$. The class equation is $|G| = |Z(G)| + \sum[G : C(a_i)]$, so $p^3 = p + \sum |\mathrm{Cl}(a_i)|$. For any element $a \notin Z(G)$, $C(a)$ is a subgroup that properly contains $Z(G)$ and is properly contained in G. Thus, the only possible order for $C(a)$ is p^2. This means that every conjugacy class for an element not in the center has size:

$$|\mathrm{Cl}(a)| = [G : C(a)] = \frac{p^3}{p^2} = p$$

The total number of elements outside the center is $p^3 - p$. Since each of these elements is in a conjugacy class of size p, there must be $(p^3 - p)/p = p^2 - 1$ such classes.

- **Class Equation (Non-abelian case):**

$$p^3 = p + \underbrace{p + p + \cdots + p}_{p^2-1 \text{ times}}$$

Theorem 2.1.20. *Let G be a finite group of order $|G| = p^n$, where p is a prime number and $n \geq 1$. Then the center of G, denoted $Z(G)$, is non-trivial. That is, $|Z(G)| > 1$.*

Proof. The class equation for the group G is:

$$|G| = |Z(G)| + \sum_{i=1}^{k}[G : C(a_i)]$$

where the sum is over representatives a_i of the distinct conjugacy classes not contained in the center $Z(G)$.

Given $|G| = p^n$, we can write:

$$p^n = |Z(G)| + \sum_{i=1}^{k} \frac{|G|}{|C(a_i)|}$$

For any $a_i \notin Z(G)$, its centralizer $C(a_i)$ is a proper subgroup of G. By Lagrange's Theorem, $|C(a_i)| = p^{k_i}$ for some integer $k_i < n$. Therefore, the size of each corresponding conjugacy class is

$$[G : C(a_i)] = \frac{p^n}{p^{k_i}} = p^{n-k_i}$$

Since $k_i < n$, the exponent $n - k_i \geq 1$, which means p divides $[G : C(a_i)]$ for all i.

Rearranging the class equation gives:

$$|Z(G)| = p^n - \sum_{i=1}^{k} p^{n-k_i}$$

The right side is a difference of terms, each of which is divisible by p. Thus, p must divide $|Z(G)|$.

Since the identity element is always in the center, we know $|Z(G)| \geq 1$. Because p divides $|Z(G)|$, it is not possible for $|Z(G)|$ to be 1. Therefore, $|Z(G)| \geq p$, which proves that the center is non-trivial. ■

Theorem 2.1.21. *Every group of order p^2 is abelian.*

Proof. Let $|G| = p^2$. From the previous theorem, we know $|Z(G)| > 1$. By Lagrange's theorem, $|Z(G)|$ must divide $|G| = p^2$, so the possibilities are $|Z(G)| = p$ or $|Z(G)| = p^2$.

- If $|Z(G)| = p^2$, then $G = Z(G)$ and G is abelian.
- If $|Z(G)| = p$, consider the quotient group $G/Z(G)$. Its order is $|G/Z(G)| = p^2/p = p$. Any group of prime order is cyclic.

There is a standard lemma stating that if $G/Z(G)$ is cyclic, then G must be abelian. This leads to a contradiction: if we assume $|Z(G)| = p$, we conclude that G is abelian, which would mean $|Z(G)| = p^2$. Therefore, the case $|Z(G)| = p$ is impossible. The only possibility is $|Z(G)| = p^2$, so every group of order p^2 is abelian. ■

2.1.3 Cauchy's Theorem for Finite Groups

Theorem 2.1.22 (Cauchy's Theorem for Finite Groups). *Let G be a finite group and let p be a prime number dividing the order of G. Then G contains an element of order p.*

Proof. We proceed by induction on $|G|$. The base case $|G| = 1$ is trivial.

Now assume $|G| > 1$ and that the theorem holds for all groups of order less than $|G|$. Let p be a prime dividing $|G|$.

Case 1: G is abelian. Since G is abelian, every subgroup is normal. Choose a non-identity element $a \in G$, and let $o(a) = m$.

- If $p \mid m$, then $a^{m/p}$ has order p.
- If $p \nmid m$, consider the subgroup $H = \langle a \rangle$. Since H is normal, the quotient group G/H has order $|G|/m$. Since $p \mid |G|$ and $p \nmid m$, we have $p \mid |G/H|$. By the induction hypothesis, G/H contains an element of order p, say bH. Then the order of b in G is a multiple of p, say kp, so b^k has order p.

Case 2: G is not abelian. Since G is not abelian, the center $Z(G)$ is a proper subgroup. Consider the class equation:

$$|G| = |Z(G)| + \sum [G : N(x_i)],$$

where the sum is over representatives of conjugacy classes not contained in $Z(G)$.

- If $p \mid |Z(G)|$, then since $|Z(G)| < |G|$, by the induction hypothesis, $Z(G)$ contains an element of order p, which is also in G.
- If $p \nmid |Z(G)|$, then since $p \mid |G|$, there exists some $x_i \notin Z(G)$ such that $p \nmid [G : N(x_i)]$. For such x_i, we have

  $$|G| = \frac{|G|}{o[N(x_i)]} o[N(x_i)].$$

  Since $p \nmid [G : N(x_i)]$ and $p \mid |G|$, it follows that $p \mid |N(x_i)|$. Also, since $x_i \notin Z(G)$, $N(x_i)$ is a proper subgroup of G. By the induction hypothesis, $N(x_i)$ contains an element of order p, which is also in G.

In all cases, G contains an element of order p, completing the proof. ■

2.2 THE SYLOW THEOREMS

The Sylow theorems are a cornerstone of finite group theory. They provide a partial converse to Lagrange's theorem by guaranteeing the existence of subgroups of specific prime-power orders.

Definition 2.2.1 (Sylow p-subgroup). Let G be a group of order $|G| = p^k m$, where p is a prime and $\gcd(p, m) = 1$. A subgroup of order p^k is called a **Sylow p-subgroup** of G. Let n_p denote the number of Sylow p-subgroups of G.

Theorem 2.2.2 (Existence of Subgroups of Prime-Power Order (Sylow's First Theorem, 1872)). *Let G be a finite group and let p be a prime. If p^k divides $|G|$, then G has at least one subgroup of order p^k.*

Proof. We proceed by induction on $|G|$. If $|G| = 1$, Theorem is trivially true. Now assume that the statement is true for all groups of order less than $|G|$. If G has a proper subgroup H such that p^k divides $|H|$, then, by our inductive assumption, H has a subgroup of order p^k and we are done. Thus, we may henceforth assume that p^k does not divide the order of any proper subgroup of G. Next, consider the class equation for G in the form

$$|G| = |Z(G)| + \Sigma |G : C(a)|,$$

where we sum over a representative of each conjugacy class $\text{cl}(a)$, where $a \notin Z(G)$. Since p^k divides $|G| = |G : C(a)||C(a)|$ and p^k does not divide[1] $|C(a)|$, therefore p must divide $|G : C(a)|$ for all

[1] Factors of p^k are $1, p, p^2, \ldots, p^k$. We are only guaranteed that $p^k \nmid |C(a)|$, but p or p^2 or $\cdots$ or p^{k-1} may divide $|C(a)|$, hence it guarantees that p divides $|Z(G)|$ but p^k need not divide $|Z(G)|$.

$a \notin Z(G)$. It then follows from the class equation that p divides $|Z(G)|$. The Cauchy's theorem then guarantees that $Z(G)$ contains an element of order p, say x. Since x is in the center of G, $\langle x \rangle$ is a normal subgroup of G, and we may form the factor group $G/\langle x \rangle$. Now observe that p^{k-1} divides $|G/\langle x \rangle|$. Thus, by the induction hypothesis, $G/\langle x \rangle$ has a subgroup of order p^{k-1}, this subgroup has the form $H/\langle x \rangle$, where H is a subgroup of G. Finally, note that $|H/\langle x \rangle| = p^{k-1}$ and $|\langle x \rangle| = p$ imply that $|H| = p^k$, and this completes the proof. ■

Remark 2.2.3. If $|G| = p^k m$, where k is the largest integer with the property that $p^k \mid o(G)$, then G has subgroups of orders $p^1, p^2, \ldots, p^k$.

Corollary 2.2.4. *If G is a finite group and p is a prime such that $p \mid o(G)$, then G has an element of order p.*

Definition 2.2.5. Let H and K be subgroups of a group G. We say that H and K are conjugate in G if there is an element g in G such that $H = gKg^{-1}$.

Theorem 2.2.6 (Sylow's Second Theorem). *If H is a subgroup of a finite group G and $|H|$ is a power of a prime p, then H is contained in some Sylow p-subgroup of G.*

Proof. Let K be a Sylow p-subgroup of G and let $C = \{K_1, K_2, \ldots, K_n\}$ be the set of all conjugates of K in G, where $K = K_1$. Since conjugation is an automorphism, each element of C is a Sylow p-subgroup of G. Let S_C denote the group of all permutations of C. For each $g \in G$, define $\phi_g \colon C \to C$ by

$$\phi_g(K_i) = gK_ig^{-1}$$

It is easy to show that each $\phi_g \in S_C$.

Now define a mapping $T \colon G \to S_C$ by $T(g) = \phi_g$. Since

$$\phi_{gh}(K_i) = (gh)K_i(gh)^{-1} = g(hK_ih^{-1})g^{-1} = g\phi_h(K_i)g^{-1} = \phi_g(\phi_h(K_i)) = (\phi_g \circ \phi_h)(K_i),$$

we have $\phi_{gh} = \phi_g \circ \phi_h$, and therefore T is a homomorphism from G to S_C.

Next consider $T(H)$, the image of H under T. Since $|H|$ is a power of p, so is $|T(H)|$, since $|T(H)|$ divides $|H|$. Thus, by the Orbit-Stabilizer Theorem 2.3.13 for each i, $|\mathcal{O}_{T(H)}(K_i)|$ divides $|T(H)|$, so that $|\mathcal{O}_{T(H)}(K_i)|$ is a power of p.

Note that $|\mathcal{O}_{T(H)}(K_i)| = 1$ means that $\phi_g(K_i) = gK_ig^{-1} = K_i$ for all $g \in H$; that is, $|\mathcal{O}_{T(H)}(K_i)| = 1$ if and only if $H \leq N(K_i)$. But the only elements of $N(K_i)$ that have orders that are powers of p are those of K_i. Thus, $|\mathcal{O}_{T(H)}(K_i)| = 1$ if and only if $H \leq K_i$.

So, to complete the proof, all we need to do is show that for some i, $|\mathcal{O}_{T(H)}(K_i)| = 1$. From Proposition 2.3.16, we have $|C| = [G : N(K)]$. And since $[G : K] = [G : N(K)][N(K) : K]$ is not divisible by p, neither is $|C|$. Because the orbits partition C, $|C|$ is the sum of powers of p. If no orbit has size 1, then p divides each summand and, therefore, p divides $|C|$, which is a contradiction. Thus, there is an orbit of size 1, and the proof is complete. ■

Theorem 2.2.7 (Sylow's Third Theorem). *Let p be a prime and let G be a group of order $p^e m$, where p does not divide m. If n_p is the the number of Sylow p-subgroups of G then $n \equiv 1 \mod p$ and $n_p \mid m$. Furthermore, any two Sylow p-subgroups of G are conjugate.*

Proof. Let K be any Sylow p-subgroup of G and let $C = \{K_1, K_2, \ldots, K_n\}$, with $K = K_1$, be the set of all conjugates of K in G. We first prove that $n \mod p = 1$.

Let S_C and T be as in the proof of Theorem 2.2.6. This time we consider $T(K)$, the image of K under T. As before, we have $|\mathcal{O}_{T(K)}(K_i)|$ is a power of p for each i and $|\mathcal{O}_{T(K)}(K_i)| = 1$ if and only if $K \leq K_i$. Thus, $|\mathcal{O}_{T(K)}(K_1)| = 1$ and $|\mathcal{O}_{T(K)}(K_i)|$ is a power of p greater than 1 for all $i \neq 1$. Since the orbits partition C, it follows that, $n_p = |C| \equiv 1 \mod p$.

Next we show that every Sylow p-subgroup of G belongs to C. To do this, suppose that H is a Sylow p-subgroup of G that is not in C. Let S_C and T be as in the proof of Theorem 2.2.6 and this time consider $T(H)$. As in the previous paragraph, $|C|$ is the sum of the orbits' sizes under the action of $T(H)$. However, no orbit has size 1, since H is not in C. Thus, $|C|$ is a sum of terms each divisible by p, so that, $n_p = |C| \equiv 0 \mod p$. This contradiction proves that H belongs to C, and that n_p is the number of Sylow p-subgroups of G.

Finally, that n_p divides m follows directly from the fact that $n_p = [G : N(K)]$ and n_p is relatively prime to p. ■

Corollary 2.2.8. *A Sylow p-subgroup of a finite group G is a normal subgroup of G if and only if it is the only Sylow p-subgroup of G.*

SYLOW'S SECOND THEOREM - SECOND PROOF

Double Cosets: Let H and K be two (not necessarily distinct) subgroups of a group G and let $x \in G$. The set HxK given by $\{hxk \mid h \in H, k \in K\}$ is called a double coset.

Lemma 2.2.9. *Two double cosets HxK and HyK are either disjoint or identical.*

Proof. Let $z \in HxK \cap HyK$. Then for $z \in HxK$, $z = h_1xk_1$ and for $z \in HyK$, we have, $z = h_2yk_2$; $h_1, h_2 \in H$ and $k_1, k_2 \in K$. This implies

$$\begin{aligned} HzK &= H(h_1xk_1)K = (Hh_1)x(k_1K) = HxK \\ \text{and } HzK &= H(h_2yk_2)K = (Hh_2)y(k_2K) = HyK \end{aligned}$$

Then $HxK = HzK = Hyk$. ∎

Lemma 2.2.10. *For subgroups H and K of a finite group G, $o(HxK) = \frac{o(H)o(K)}{o(x^{-1}Hx \cap K)}$.*

Proof. Define $\Phi : HxK \to x^{-1}HxK$ by $\Phi(hxk) = x^{-1}hxk$ for all $h \in H, k \in K$. Then Φ is a bijection, because $\Phi(hxk) = \Phi(h'xk')$ iff $x^{-1}hxk = x^{-1}h'xk'$ iff $hxk = h'xk'$, so that $o(HxK) = o(x^{-1}HxK)$. And we have, $o(x^{-1}HxK) = \frac{o(x^{-1}Hx)o(K)}{o(x^{-1}Hx \cap K)}$. But $o(x^{-1}Hx) = o(H)$. So

$$o(HxK) = \frac{o(H)o(K)}{o(x^{-1}Hx \cap K)}.$$

∎

Theorem 2.2.11 (Frobenius). *If G is a finite group, then*

$$o(G) = \sum \frac{o(H)o(K)}{o(x^{-1}Hx \cap K)}$$

where summation on right hand side is taken over elements x chosen one from each disjoint double cosets HxK.

Proof. For each $a \in G$, $a \in HaK$.

This at once implies that $G = \bigcup\limits_{x \in G} HxK$. Writing disjoint union, we get

$$\begin{aligned} o(G) &= \sum o(HxK) \\ &= \sum \frac{o(H)o(K)}{o(x^{-1}Hx \cap K)} \end{aligned}$$

∎

Theorem 2.2.12 (Sylow's Second Theorem). *All Sylow p-subgroups of a finite group G are conjugate.*

Proof. Let $o(G) = p^k m$, $k \geq 1$ and $p \nmid m$. Suppose there are two Sylow p-subgroups A and B of G which are not conjugate. Assuming that A and B generate r distinct double cosets.

Let $G = \bigcup\limits_{i=1}^{r} Ax_iB, x_1 = e$. Then $o(G) = \sum\limits_{i=1}^{r} o(Ax_iB)$. But $o(Ax_iB) = \dfrac{o(A)o(B)}{o(x_i^{-1}Ax_i \cap B)} = \dfrac{p^{2k}}{p^t} = p^{2k-t}$, where $o(x_i^{-1}Ax_i \cap B) = p^t$.

Since B is not conjugate to A, $x_i^{-1}Ax_i \cap B \subsetneq B \Rightarrow t < k$. So $2k - t \gg k$, that is, $2k - t \geq k + 1$ $\Rightarrow p^{m+1} \mid o(Ax_iB)$ for all $i = 1, 2, \ldots, r$. Consequently $p^{m+1} \mid o(G)$. This is absurd as $p \nmid m$. Hence, all Sylow p-subgroups are conjugate. ∎

Theorem 2.2.13 (Sylow's Third Theorem). *Number of Sylow p-subgroups of a finite group G is equal to $n_p = 1 + kp$ where k is an integer ≥ 0 such that $n_p \mid o(G)$.*

Proof. Let $o(G) = p^\alpha m$, $\alpha \geq 1$ and $p \nmid m$. If P is a Sylow p-subgroup of G then $o(P) = p^\alpha$. We know that

Now,

$$o(G) = \sum_{x \in G} o(PxP) = \sum_{x \in N(P)} PxP + \sum_{x \notin N(P)} PxP \tag{2.1}$$

$$= \sum_{x \in N(P)} PxP + \sum_{x \notin N(P)} \frac{o(P)o(P)}{o(x^{-1}Px \cap P)} \tag{2.2}$$

Now, if $x \in N(P)$, then $Px = xP$ gives $PxP = xPP = xP = Px$ and in this case, $\bigcup_{x \in N(P)} Px = N(P)$ and in the case, when $x \notin N(p)$, then $p^{k+1} \mid o(PxP)$ for each x because $x^{-1}Px \cap P$ is a proper subgroup of P. Thus, from equation (2.2), we have

$$o(G) = o[N(P)] + p^{\alpha+1}t$$

This gives

$$\frac{o(G)}{o(N(P))} = 1 + \frac{p^{\alpha+1}t}{o(N(P))} = 1 + p \cdot \frac{p^\alpha t}{o(N(P))}$$

As $N(P)$ is a subgroup of G, therefore, $p^s \mid o[N(P)]$ implies $s \leq \alpha$. Hence, $\frac{p^\alpha t}{o(N(P))}$ is an integer. Therefore $\frac{o(G)}{o(N(P))} = 1 + kp$.

But $G/N(P)$ is the number of distinct conjugates of P in G. So the number of conjugates of P is equal to an integer of the form $1 + kp$, such that $1 + kp \mid o(G)$. By Sylow's Second theorem all Sylow p-subgroups are conjugate. Consequently, the number of Sylow p-subgroups is $1 + kp$. ■

Lemma 2.2.14. $\bigcup_{x \in N(P)} Px = N(P)$

$\Rightarrow \bigcup_{x \in N(P)} Px \subseteq N(P)$**:** Let $y \in \bigcup_{x \in N(P)} Px$. Then $y = px$ for some $p \in P$ and $x \in N(P)$. Since $x \in N(P)$, $xPx^{-1} = P$. Then $yPy^{-1} = (px)P(px)^{-1} = p(xPx^{-1})p^{-1} = pPp^{-1} = P$. Thus $y \in N(P)$.

$\Leftarrow N(P) \subseteq \bigcup_{x \in N(P)} Px$**:** Let $g \in N(P)$. Then g is an element of the right coset Pg. Since $g \in N(P)$, Pg is one of the cosets included in the union. Therefore, $g \in Pg \subseteq \bigcup_{x \in N(P)} Px$.

Combining these, we get $\bigcup_{x \in N(P)} Px = N(P)$.

Theorem 2.2.15. *A Sylow p-subgroup P of a finite group G is normal in G if and only if it is unique.*

Proof. ($\Rightarrow$) Assume $P \triangleleft G$. Let Q be any Sylow p-subgroup. By Sylow's Second Theorem, $Q = gPg^{-1}$ for some $g \in G$. Since $P \triangleleft G$, $gPg^{-1} = P$. Thus, $Q = P$. Therefore, P is unique.

($\Leftarrow$) Assume P is the unique Sylow p-subgroup of G. For any $g \in G$, gPg^{-1} is also a subgroup of the same order as P, i.e., $o(gPg^{-1}) = o(P)$. This means gPg^{-1} is also a Sylow p-subgroup. By uniqueness, $gPg^{-1} = P$. Therefore, $P \triangleleft G$. ■

Example 2.2.16. Consider the Sylow 2-subgroups of S_3. They are $\{(1), (12)\}$, $\{(1), (23)\}$, and $\{(1), (13)\}$. According to Sylow's Third Theorem, we should be able to obtain the latter two of these from the first by conjugation. Indeed,

$$(13)\{(1), (12)\}(13)^{-1} = \{(1), (23)\},$$
$$(23)\{(1), (12)\}(23)^{-1} = \{(1), (13)\}.$$

The main application of the Sylow theorems is to determine the structure of a group or to prove that a group of a certain order cannot be simple. A group is **simple** if its only normal subgroups are the trivial group and the group itself.

Corollary 2.2.17. *If $n_p = 1$ for some prime p dividing $|G|$, then G is not simple (unless $|G| = p$).*

Solution. If $n_p = 1$, let P be the unique Sylow p-subgroup. By the Second Sylow Theorem, all Sylow p-subgroups are conjugate. Since there is only one, P must be conjugate only to itself, i.e., $gPg^{-1} = P$ for all $g \in G$. This is the definition of a normal subgroup. If $|G|$ is not prime, P is a non-trivial proper normal subgroup, so G is not simple. ■

Example 2.2.18 (Groups of order 15)**.** Show that any group G of order 15 is cyclic.

Solution. Let $|G| = 15 = 3 \cdot 5$. We examine the number of Sylow subgroups.

- **Sylow 5-subgroups (**$p = 5$**):** The number of Sylow 5-subgroups, n_5, must satisfy:
  - $n_5 \equiv 1 \pmod 5$ (so $n_5 \in \{1, 6, 11, \dots\}$)
  - n_5 must divide $m = 3$ (so $n_5 \in \{1, 3\}$)

  The only common value is $n_5 = 1$. Let P_5 be this unique Sylow 5-subgroup. Since it is unique, it is normal in G. $|P_5| = 5$.

- **Sylow 3-subgroups (**$p = 3$**):** The number of Sylow 3-subgroups, n_3, must satisfy:
  - $n_3 \equiv 1 \pmod 3$ (so $n_3 \in \{1, 4, 7, \dots\}$)
  - n_3 must divide $m = 5$ (so $n_3 \in \{1, 5\}$)

  The only common value is $n_3 = 1$. Let P_3 be this unique Sylow 3-subgroup. It is normal in G. $|P_3| = 3$.

Since P_3 and P_5 are normal subgroups with trivial intersection ($|P_3 \cap P_5|$ must divide both 3 and 5, so it is 1), and $|P_3||P_5| = 15 = |G|$, it follows that $G \cong P_3 \times P_5$. As groups of prime order, $P_3 \cong \mathbb{Z}_3$ and $P_5 \cong \mathbb{Z}_5$. Therefore, $G \cong \mathbb{Z}_3 \times \mathbb{Z}_5$. Since $\gcd(3, 5) = 1$, this direct product is isomorphic to $\mathbb{Z}_{15}$. Thus, any group of order 15 is cyclic. ■

Example 2.2.19 (Groups of order 30)**.** Show that any group G of order 30 is not simple.

Solution. Let $|G| = 30 = 2 \cdot 3 \cdot 5$. We use an argument by contradiction. Assume G is simple. This would imply that $n_p > 1$ for all primes p dividing $|G|$.

- **Sylow 5-subgroups (**n_5**):** $n_5 \equiv 1 \pmod 5$ and n_5 divides 6. The possibilities are $n_5 = 1$ or $n_5 = 6$. Since we assume G is simple, we must have $n_5 = 6$.

- **Sylow 3-subgroups (**n_3**):** $n_3 \equiv 1 \pmod 3$ and n_3 divides 10. The possibilities are $n_3 = 1$ or $n_3 = 10$. Since G is simple, we must have $n_3 = 10$.

Let's count the number of elements these subgroups would require.

- The 6 Sylow 5-subgroups are distinct. Since they are of prime order, their intersection is just the identity. So, each provides $5 - 1 = 4$ unique elements of order 5. Total elements of order 5: $6 \times 4 = 24$.

- The 10 Sylow 3-subgroups are distinct. Each provides $3 - 1 = 2$ unique elements of order 3. Total elements of order 3: $10 \times 2 = 20$.

The total number of elements accounted for so far is $24 + 20 = 44$. This is impossible, as the group G only has 30 elements. Our assumption that G is simple must be false. Therefore, at least one of n_3 or n_5 must be 1, which guarantees the existence of a non-trivial proper normal subgroup. ■

Example 2.2.20 (Groups of order pq)**.** Let p and q be primes with $p < q$. If p does not divide $q - 1$, then any group of order pq is cyclic.

Solution. Let $|G| = pq$.

- **Sylow q-subgroups (**n_q**):** $n_q \equiv 1 \pmod q$ and n_q divides p. Since p is prime, the only divisors are 1 and p. As $p < q$, p cannot be congruent to $1 \pmod q$. Thus, the only possibility is $n_q = 1$. This gives a normal subgroup Q of order q.

- **Sylow p-subgroups (n_p):** $n_p \equiv 1 \pmod p$ and n_p divides q. The divisors of q are 1 and q. So $n_p = 1$ or $n_p = q$. If $n_p = q$, then $q \equiv 1 \pmod p$, which means p divides $q-1$. But the theorem's hypothesis is that p does *not* divide $q-1$. Therefore, n_p cannot be q. The only possibility is $n_p = 1$. This gives a normal subgroup P of order p.

Since both the Sylow p-subgroup P and the Sylow q-subgroup Q are normal, we can conclude (as in the order 15 example) that $G \cong P \times Q \cong \mathbb{Z}_p \times \mathbb{Z}_q \cong \mathbb{Z}_{pq}$. The group is cyclic. ■

Example 2.2.21 (Sylow 3-subgroups of S_4)**.** Let $G = S_4$, so $|G| = 24 = 2^3 \cdot 3^1$. Sylow 3-subgroups have order 3. By Sylow's Third Theorem:

- $n_3 \equiv 1 \pmod 3 \implies n_3 \in \{1, 4, 7, \dots\}$
- $n_3 \mid 24 \implies n_3 \in \{1, 2, 3, 4, 6, 8, 12, 24\}$

The common values are $1, 4$. The elements of order 3 in S_4 are the 3-cycles: $(123), (132), (124), (142), (134), (14$ There are 8 such elements. Each Sylow 3-subgroup (order 3) contains 2 elements of order 3. Thus, $2n_3 = 8 \implies n_3 = \mathbf{4}$.

Example 2.2.22 (Sylow 2-subgroups of S_3)**.** Let $G = S_3$, so $|G| = 6 = 2^1 \cdot 3^1$. Sylow 2-subgroups have order 2. By Sylow's Third Theorem:

- $n_2 \equiv 1 \pmod 2 \implies n_2 \in \{1, 3, 5, \dots\}$
- $n_2 \mid 6 \implies n_2 \in \{1, 2, 3, 6\}$

The common values are $1, 3$. The elements of order 2 in S_3 are the transpositions: $(12), (13), (23)$. There are 3 such elements. Each Sylow 2-subgroup (order 2) contains 1 element of order 2. Thus, $1n_2 = 3 \implies n_2 = \mathbf{3}$.

Example 2.2.23 (Sylow 7-subgroups of a group of order 147)**.** Let G be a group with $|G| = 147 = 3 \cdot 7^2$. Sylow 7-subgroups have order $7^2 = 49$. By Sylow's Third Theorem:

- $n_7 \equiv 1 \pmod 7 \implies n_7 \in \{1, 8, 15, \dots\}$
- $n_7 \mid 147 \implies n_7 \in \{1, 3, 7, 21, 49, 147\}$

The only common value is 1. Thus, $n_7 = \mathbf{1}$. This means the Sylow 7-subgroup is unique and normal.

2.3 Group Actions

Definition 2.3.1. If X is a set and G is a group, then G *acts* on X if there is a function $G \times X \to X$, denoted by $(g, x) \mapsto g \cdot x$, such that:

(i) $(gh)x = g(hx)$ for all $g, h \in G$ and $x \in X$;

(ii) $1x = x$ for all $x \in X$, where 1 is the identity in G.

We also call X a *G-set* if G acts on X.
If a group G acts on a set X, then fixing the first variable, say g, gives a function

$$\alpha_g : X \to X, \quad \text{namely, } \alpha_g : x \mapsto g \cdot x.$$

This function is a permutation of X, for its inverse is $\alpha_{g^{-1}}$:

$$\alpha_g \alpha_{g^{-1}} = \alpha_1 = 1_X = \alpha_{g^{-1}} \alpha_g.$$

We now show that an action of a group G on a set X is merely another way of viewing a homomorphism $G \to S_X$, where S_X is the symmetric group on X.

Theorem 2.3.2. *Every action $\alpha \colon G \times X \to X$ of a group G on a set X determines a homomorphism $G \to S_X$. Conversely, every homomorphism $G \to S_X$ makes X into a G-set.*

Proof. Given an action α, fixing the first variable, say, g, gives a function $\alpha_g \colon X \to X$, namely, $\alpha_g : x \mapsto gx$. This function is a permutation of X, for its inverse is $\alpha_{g^{-1}} \colon \alpha_g\alpha_{g^{-1}} = \alpha_1 = 1_X = \alpha_{g^{-1}}\alpha_g$. It is easy to see that $\alpha \colon G \to S_X$, defined by $\alpha \colon g \mapsto \alpha_g$, is a homomorphism.

Conversely, given any homomorphism $\varphi \colon G \to S_X$, define $\alpha \colon G \times X \to X$ by $\alpha(g,x) = \varphi(g)(x)$. It is easy to see that α is an action. ∎

We say that a G-set X is **faithful** if the homomorphism $\alpha : G \to S_X$ is an injection; that is, G can be viewed (via α) as a subgroup of S_X.

The following two definitions are fundamental.

Definition 2.3.3 (Orbit). If G acts on X and $x \in X$, then the **orbit** of x, denoted by $\mathcal{O}(x)$, is the subset of X
$$\mathcal{O}(x) = \{gx : g \in G\} \subseteq X;$$
The **orbit space**, denoted by X/G, is the set of all the orbits. We say that G acts **transitively** on a set X if $|X/G| = 1$, that is if there is exactly one orbit.

Definition 2.3.4 (Stabilizer). the **stabilizer** of x, denoted by G_x, is the subgroup
$$G_x = \{g \in G : gx = x\} \leq G.$$

Example 2.3.5. We show that G acts on itself by conjugation. For each $g \in G$, define $\alpha_g : G \to G$ to be conjugation
$$\alpha_g(x) = gxg^{-1}.$$
To verify axiom (i), note that for each $x \in G$,
$$(\alpha_g\alpha_h)(x) = \alpha_g(\alpha_h(x)) = \alpha_g(hxh^{-1}) = g(hxh^{-1})g^{-1} = (gh)x(gh)^{-1} = \alpha_{gh}(x).$$
Therefore, $\alpha_g\alpha_h = \alpha_{gh}$. To prove axiom (ii), note that $\alpha_1(x) = 1x1^{-1} = x$ for each $x \in G$, and so $\alpha_1 = 1_G$.

Let us find some orbits and stabilizers.

Example 2.3.6. Cayley's Theorem says that G acts on itself by translations: $\tau_g : a \mapsto ga$. If $a \in G$, then the orbit $\mathcal{O}(a) = G$, for if $b \in G$, then $b = (ba^{-1})a = \tau_{ba^{-1}}(a)$. The stabilizer G_a of $a \in G$ is $\{1\}$, for if $a = \tau_g(a) = ga$, then $g = 1$.

More generally, G acts transitively on G/H (the family of (left) cosets of a (not necessarily normal) subgroup H) by translations $\tau_g : aH \mapsto gaH$. The orbit $\mathcal{O}(aH) = G/H$, for if $bH \in G/H$, then $\tau_{ba^{-1}} : aH \mapsto bH$. The stabilizer G_{aH} of the coset aH is aHa^{-1}, for $gaH = aH$ if and only if $a^{-1}ga \in H$ if and only if $g \in aHa^{-1}$.

Example 2.3.7. Let $X = \{1, 2, \ldots, n\}$, let $\alpha \in S_n$, and regard the cyclic group $G = \langle\alpha\rangle$ as acting on X. If $i \in X$, then
$$\mathcal{O}(i) = \{\alpha^k(i) : k \in \mathbb{Z}\}.$$

Definition 2.3.8 (Conjugacy class). When a group G acts on itself by conjugation, then the orbit $\mathcal{O}(x)$ is
$$\{y \in G : y = axa^{-1} \text{ for some } a \in G\};$$
in this case, $\mathcal{O}(x)$ is called the **conjugacy class** of x, and it is commonly denoted by x^G.

Definition 2.3.9 (Centralizer). If $x \in G$, then the stabilizer G_x of x is
$$C_G(x) = \{g \in G : gxg^{-1} = x\}.$$
This subgroup of G, consisting of all $g \in G$ that commute with x, is called the **centralizer** of x in G.

Every group G acts on the set X of all its subgroups, by conjugation: If $a \in G$, then a acts by $H \mapsto aHa^{-1}$, where $H \leq G$.

Definition 2.3.10 (Conjugate). If H is a subgroup of a group G, then a **conjugate** of H is a subgroup of G of the form
$$aHa^{-1} = \{aha^{-1} : h \in H\},$$
where $a \in G$.

Since conjugation $h \mapsto aha^{-1}$ is an injection $H \to G$ with image aHa^{-1}, it follows that conjugate subgroups of G are isomorphic. For example, in S_3, all cyclic subgroups of order 2 are conjugate (for their generators are conjugate).

The orbit of a subgroup H consists of all its conjugates; notice that H is the only element in its orbit if and only if $H \trianglelefteq G$; that is, $aHa^{-1} = H$ for all $a \in G$.

Definition 2.3.11 (Normalizer). The stabilizer of H is

$$N_G(H) = \{g \in G : gHg^{-1} = H\}.$$

This subgroup of G is called the **normalizer** of H in G.

Proposition 2.3.12. If G acts on a set X, then X is the disjoint union of the orbits. If X is finite, then

$$|X| = \sum_i |\mathcal{O}(x_i)|,$$

where one x_i is chosen from each orbit.

Proof. As we have mentioned earlier, the relation on X, given by $x \equiv y$ if there exists $g \in G$ with $y = gx$, is an equivalence relation whose equivalence classes are the orbits. Therefore, the orbits partition X.

The count given in the second statement is correct: Since the orbits are disjoint, no element in X is counted twice. ■

Here is the connection between orbits and stabilizers.

Theorem 2.3.13 (Orbit-Stabilizer Theorem). *If G acts on a set X and $x \in X$, then*

$$|\mathcal{O}(x)| = [G : G_x]$$

the index of the stabilizer G_x in G.

Proof. Let G/G_x denote the family of all the left cosets of G_x in G. We will exhibit a bijection $\varphi : G/G_x \to \mathcal{O}(x)$, and this will give the result, since $|G/G_x| = [G : G_x]$.

Define $\varphi : gG_x \mapsto gx$. Now φ is well-defined: If $gG_x = hG_x$, then $h = gf$ for some $f \in G_x$; that is, $fx = x$; hence, $hx = g(fx) = gx$. Now φ is an injection: if $gx = \varphi(gG_x) = \varphi(hG_x) = hx$, then $h^{-1}gx = x$; hence, $h^{-1}g \in G_x$, and $gG_x = hG_x$. Lastly, φ is a surjection: if $y \in \mathcal{O}(x)$, then $y = gx$ for some $g \in G$, and so $y = \varphi(gG_x)$. ■

Corollary 2.3.14. *If a finite group G acts on a set X, then the number of elements in any orbit is a divisor of $|G|$.*

Proof. This follows at once from Lagrange's Theorem. ■

Corollary 2.3.15. *If x lies in a finite group G, then the number of conjugates of x is the index of its centralizer:*

$$|x^G| = [G : C_G(x)],$$

and hence it is a divisor of $|G|$.

Proof. This follows from the fact that the orbit of x is its conjugacy class x^G, and the stabilizer G_x is the centralizer $C_G(x)$. ■

Proposition 2.3.16. If H is a subgroup of a finite group G, then the number of conjugates of H in G is $[G : N_G(H)]$.

Proof. The proof follows from the fact that the orbit of H is the family of all its conjugates, and the stabilizer is its normalizer $N_G(H)$. ■

Example 2.3.17 (Symmetric Group Action)**.** Let S_n be the symmetric group on n elements, and let $X = \{1, 2, \dots, n\}$. Define an action of S_n on X by

$$\sigma \cdot x = \sigma(x) \quad \text{for all } \sigma \in S_n, x \in X.$$

This is a group action because:

(a) The identity permutation id $\in S_n$ satisfies $\text{id}(x) = x$ for all $x \in X$

(b) For any $\sigma, \tau \in S_n$ and $x \in X$,

$$(\sigma\tau) \cdot x = (\sigma\tau)(x) = \sigma(\tau(x)) = \sigma \cdot (\tau \cdot x)$$

The orbits of this action are the subsets of X that can be mapped to each other by permutations. In fact, since S_n is transitive on X, there is only one orbit: X itself.

The stabilizer of an element $x \in X$ is

$$(S_n)_x = \{\sigma \in S_n \mid \sigma(x) = x\},$$

which is isomorphic to S_{n-1}. By the Orbit-Stabilizer Theorem,

$$|S_n| = |S_n \cdot x| \cdot |(S_n)_x| \Rightarrow n! = n \cdot (n-1)!.$$

Example 2.3.18 (Dihedral Group Action)**.** Let D_4 be the dihedral group of order 8 (symmetries of a square), and let $X = \{1, 2, 3, 4\}$ be the vertices of the square labeled consecutively. Define an action of D_4 on X by

$$g \cdot x = \text{the vertex to which } x \text{ is mapped by the symmetry } g.$$

This is a group action because:

(a) The identity symmetry fixes all vertices

(b) Composition of symmetries corresponds to composition of vertex mappings

Consider the rotation $r \in D_4$ by $90°$ counterclockwise. The orbits are:

$$\mathcal{O}_1 = \{1, 2, 3, 4\}$$

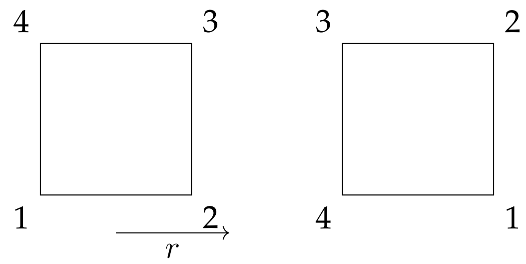
since repeated application of r cycles through all vertices.

The stabilizer of vertex 1 consists of the identity and the reflection across the diagonal passing through vertex 1:

$$D_{4_1} = \{e, s\}$$

where s is the reflection that fixes vertex 1. By the Orbit-Stabilizer Theorem,

$$|D_4| = |\mathcal{O}_1| \cdot |D_{4_1}| \Rightarrow 8 = 4 \cdot 2.$$



2.4 SIMPLE GROUPS

Abelian groups (and the quaternions) have the property that every subgroup is normal. At the opposite pole are groups having no normal subgroups other than the two obvious ones: $\{1\}$ and G.

Definition 2.4.1 (Simple group). A group $G \neq \{1\}$ is called **simple** if G has no normal subgroups other than $\{1\}$ and G itself.

Proposition 2.4.2. An abelian group G is simple if and only if it is finite and of prime order.

Proof. If G is finite of prime order p, then G has no subgroups H other than $\{1\}$ and G, otherwise Lagrange's theorem would show that $|H|$ is a divisor of p. Therefore, G is simple.

Conversely, assume that G is simple. Since G is abelian, every subgroup is normal, and so G has no subgroups other than $\{1\}$ and G. Choose $x \in G$ with $x \neq 1$. Since $\langle x \rangle$ is a subgroup, we have $\langle x \rangle = G$. If x has infinite order, then all the powers of x are distinct, and so $\langle x^2 \rangle < \langle x \rangle$ is a forbidden subgroup of $\langle x \rangle$, a contradiction. Therefore, every $x \in G$ has finite order. If x has (finite) order m and if m is composite, say $m = k\ell$, then $\langle x^k \rangle$ is a proper nontrivial subgroup of $\langle x \rangle$, a contradiction. Therefore, $G = \langle x \rangle$ has prime order. ∎

2.4.1 Simplicity of Alternating Groups

Lemma 2.4.3. *Every element of $A_{n\geq 3}$ is a product of 3 cycles.*

Proof. Note that for $\alpha \neq 1, \beta \neq 1,$, we have $(\alpha\beta) = (1\,\alpha)(1\,\beta)(1\,\alpha)$. So $x \in A_n$ can be written as $(1\,\alpha_1)(1\,\alpha_2)\ldots(1\,\alpha_r)$ where r is even and $\alpha_i \in \{2, 3, \ldots, n\}$. Further $(1\,\alpha)(1\,\beta) = (1\,\beta\,\alpha)$ and r is even implying every even permutation is a product of 3 cycles. ∎

Proposition 2.4.4. If $\sigma = (x_1 x_2 \ldots x_k)$ is a cycle in S_n and $\theta \in S_n$, then $\theta\sigma\theta^{-1} = (\theta(x_1)\theta(x_2)\ldots\theta(x_k))$.

Proof. Let $\tau = \theta\sigma\theta^{-1}$.

A. **Action on $\theta(x_i)$:** For any $i \in \{1, \ldots, k-1\}$, we have $\tau(\theta(x_i)) = \theta(\sigma(\theta^{-1}(\theta(x_i)))) = \theta(\sigma(x_i)) = \theta(x_{i+1})$. For $i = k$, $\tau(\theta(x_k)) = \theta(\sigma(x_k)) = \theta(x_1)$. This shows that τ cyclically permutes $\theta(x_1), \ldots, \theta(x_k)$.

B. **Action on $y \notin \{\theta(x_1), \ldots, \theta(x_k)\}$:** If $y \notin \{\theta(x_1), \ldots, \theta(x_k)\}$, then $\theta^{-1}(y) \notin \{x_1, \ldots, x_k\}$. Since σ fixes elements not in $\{x_1, \ldots, x_k\}$, we have $\sigma(\theta^{-1}(y)) = \theta^{-1}(y)$. Therefore, $\tau(y) = \theta(\sigma(\theta^{-1}(y))) = \theta(\theta^{-1}(y)) = y$. This shows that τ fixes all elements outside the set $\{\theta(x_1), \ldots, \theta(x_k)\}$.

Combining these, τ acts as the cycle $(\theta(x_1)\theta(x_2)\ldots\theta(x_k))$. ∎

Lemma 2.4.5. *If a normal subgroup H of A_n contains a cycle of length 3, then $H = A_n$.*

Proof. In view of Lemma 2.4.3, it is sufficient to prove that every cycle of length 3 belongs to H.

Let $(\alpha\beta\gamma)$ be any cycle of length 3 and let $(xyz) \in H$. Consider a $\theta \in S_n$ such that $\theta(x) = \alpha, \theta(y) = \beta$ and $\theta(z) = \gamma$. Then $\theta(xyz)\theta^{-1} = (\alpha\beta\gamma)$; thus if $\theta \in A_n$, then $(\alpha\beta\gamma) \in H$, as H is a normal subgroup of A_n.

So let $\theta \in S_n$ be an odd permutation. Take u, v in $\{1, 2, \ldots, n\}$ different from x, y, z (this is possible as $n \geq 5$). As θ is odd, $\theta(uv) \in A_n$. So that $\theta(uv)(xyz)[\theta(uv)]^{-1} \in H \implies \theta(uv)(xyz)(uv)\theta^{-1} \in H$, as (xyz) and (uv) are disjoint and hence $(uv)(xyz)(uv) = (xyz)(uv)^2 = (xyz)$. $\implies \theta(xyz)\theta^{-1} \in H$ $\implies (\alpha\beta\gamma) \in H$. Hence, the lemma follows. ∎

Lemma 2.4.6. *If any normal subgroup H of A_n contains a product of two disjoint transpositions, then $H = A_n$.*

Proof. Let $(xy)(uv) \in H$ where (xy) and (uv) are disjoint. Take w in $\{1, 2, 3, \ldots, n\}$ different from x, y, u, v; this is possible as $n \geq 5$.

Put $\theta = (uvw)$. Clearly $\theta \in A_n$, so that $\theta(xy)(uv)\theta^{-1} \in H$, i.e. $(uvw)(xy)(uv)(uvw)^{-1} \in H$. i.e., $(uvw)(xy)(uv)(wvu) \in H$. i.e., $(vw)(xy) \in H$. However H is a subgroup, therefore $(xy)(uv)(vw)(xy) \in H$ i.e., $(uvw) \in H$. Hence $H = A_n$ (Lemma 2.4.5). Now we are ready to prove our main result. ∎

Theorem 2.4.7 ((Galois))**.** *A_n is simple for $n \geq 5$.*

Proof. Let $H \neq \{I\}$ be a normal subgroup of A_n. We shall show that $H = A_n$. Since $H \neq \{I\}$, there exists $\alpha(\neq I) \in H$. Let $\alpha = c_1 c_2 \ldots c_k$ be the decomposition of α into disjoint cycles, each of length > 1. Since disjoint cycles commute we assume: Length$(c_1) \geq$ Length$(c_2) \geq \cdots \geq$ Length(c_k) Four cases arise.

(i) Length$(c_1) \geq 4$

(ii) Length$(c_1) = 3$ and Length$(c_2) = 3$

(iii) Length$(c_1) = 3$ and all other c_i's, $i \geq 2$ are transpositions.

(iv) All c_i's are transpositions.

(i) Without loss of generality, let $c_1 = (123\ldots k)$, $k > 3$. Consider $\theta = (123)$. Clearly $\theta \in A_n$, so that $\theta\alpha\theta^{-1} \in H$ because H is normal in A_n. Now

$$\begin{aligned}
\theta\alpha\theta^{-1} &= (123)c_1c_2\ldots c_k(321) \\
&= c_2c_3\ldots c_k(123)c_1(321) && \text{[because } c_i\text{'s are disjoint, hence they commute]} \\
&= c_2c_3\ldots c_k(123)(123\ldots k)(321) && \text{[because } c_1 = (123\ldots k)\text{]} \\
&= c_2c_3\ldots c_k(1456\ldots k23)
\end{aligned}$$

But H is a subgroup, therefore $\alpha^{-1}c_2c_3\dots c_k(1456\dots k23)\in H$, but

$$\begin{aligned}\alpha^{-1}c_2c_3\dots c_k(1456\dots k23) &= c_k^{-1}\cdots c_2^{-1}c_1^{-1}c_2c_3\cdots c_k(1456\cdots k23)\\ &= c_1^{-1}(1456\dots k23) \qquad \text{[because } c_i\text{'s are disjoint cycles, so they comm}\\ &= (123\cdots k)(1456\cdots k23)\\ &= (13k), \text{ which is a 3-cycle.}\end{aligned}$$

Consequently $H = A_n$ (Lemma 2.4.5).

(ii) Let $c_1 = (123)$ and $c_2 = (456)$. Consider $\theta = (124)$. Since $\theta \in A_n$ and H is a normal subgroup of A_n, so $\theta\alpha\theta^{-1}\in H$; and we have

$$\begin{aligned}\theta\alpha\theta^{-1} &= (124)c_1c_2\cdots c_k(421)\\ &= c_3\cdots c_k(124)c_1c_2(421)\\ &= c_3\cdots c_k(124)(123)(456)(421)\\ &= c_3\cdots c_k(156)(243)\end{aligned}$$

But H being a subgroup gives $\alpha^{-1}c_3\cdots c_k(156)(243)\in H$ and we have

$$\begin{aligned}\alpha^{-1}c_3\cdots c_k(156)(243) &= c_k^{-1}\cdots c_2^{-1}c_1^{-1}c_3\cdots c_k(156)(243)\\ &= c_k^{-1}\cdots c_3^{-1}c_3\cdots c_kc_2^{-1}c_1^{-1}(156)(243) \qquad \text{[because disjoint cycles commute]}\\ &= (654)(321)(156)(243)\\ &= (14263),\ \text{which is a } 5\text{−cycle in } H\end{aligned}$$

and we are led to case (I). Thus, $H = A_n$.

(iii) Let $c_1 = (123)$ and for $i \geq 2$ each c_i be a transposition. Then $\alpha^2 = c_1^2c_2^2\dots c_k^2 = c_1^2 = (321)\in H$. Consequently $H = A_n$ (Lemma 2.4.5).

(iv) Let $c_1 = (12)$ and $c_2 = (34)$. Consider $\theta = (123)$. Because $\theta\in A_n$ and H is normal subgroup of A_n, $\theta\alpha\theta^{-1}\in H$. i.e., $(123)c_1c_2\dots c_k(321)\in H$, and we have

$$\begin{aligned}\theta\alpha\theta^{-1} &= (123)c_1c_2\dots c_k(321)\\ &= c_3\cdots c_k(123)c_1c_2(321)\\ &= c_3\cdots c_k(123)(12)(34)(321)\\ &= c_3\cdots c_k(14)(23)\end{aligned}$$

Now H is a subgroup, therefore $\alpha^{-1}c_3c_4\dots c_k(14)(23)\in H$, and we have

$$\begin{aligned}\alpha^{-1}c_3c_4\dots c_k(14)(23) &= c_k^{-1}\cdots c_2^{-1}c_1^{-1}c_3\cdots c_k(14)(23)\\ &= c_K^{-1}\cdots c_3^{-1}c_3\cdots c_kc_2^{-1}c_1^{-1}(14)(23)\\ &= c_2^{-1}c_1^{-1}(14)(23)\\ &= (43)(21)(14)(23)\\ &= (13)(24),\end{aligned}$$

which is a product of two disjoint transpositions. This implies that $H = A_n$ (Lemma 2.4.6).

Hence A_n is a simple group for every $n \geq 5$. ■

Remark: Trivially, A_n is simple for $n = 3$. So we can say that A_n is simple for all natural numbers n except 1, 2 and 4.

Proposition 2.4.8. The alternating group A_4 is a group of order 12 having no subgroup of order 6.

Proof. First, note that $|A_4| = 12$. If A_4 contains a subgroup H of order 6, then H has index 2, and hence normal. Then $A_4/H = \{H, \alpha H\}$ where $\alpha(\neq I)\in A_4$. Then $(\alpha H)^2 = \alpha^2 H = H$, so $\alpha^2\in H$ for every $\alpha\in A_4$. If α is a 3-cycle, then so is α^2 and $(\alpha^2)^2 = \alpha$. Thus, H contains every 3-cycle. This is a contradiction, for there are 8 3-cycles in A_4. ■

2.5 Direct Product of Groups

Definition 2.5.1. (1) The direct product $G_1 \times G_2 \times \cdots \times G_n$ of the groups $G_1, G_2, \dots, G_n$ with operations $*_1, *_2, \dots, *_n$, respectively, is the set of n-tuples $(g_1, g_2, \dots, g_n)$ where $g_i \in G_i$ with operation defined componentwise:

$$(g_1, g_2, \dots, g_n) * (h_1, h_2, \dots, h_n) = (g_1 *_1 h_1, g_2 *_2 h_2, \dots, g_n *_n h_n).$$

(2) Similarly, the direct product $G_1 \times G_2 \times \dots$ of the groups $G_1, G_2, \dots$ with operations $*_1, *_2, \dots$, respectively, is the set of sequences $(g_1, g_2, \dots)$ where $g_i \in G_i$ with operation defined componentwise:

$$(g_1, g_2, \dots) * (h_1, h_2, \dots) = (g_1 *_1 h_1, g_2 *_2 h_2, \dots).$$

Although the operations may be different in each of the factors of a direct product, we shall, as usual, write all abstract groups multiplicatively, so that the operation in (1) above, for example, becomes simply

$$(g_1, g_2, \dots, g_n)(h_1, h_2, \dots, h_n) = (g_1 h_1, g_2 h_2, \dots, g_n h_n).$$

Proposition 2.5.2. Let $G_1, G_2, \dots, G_n$ be groups and let $G = G_1 \times \cdots \times G_n$ be their direct product.

1. For each fixed i the set of elements of G which have the identity of G_j in the j^{th} position for all $j \neq i$ and arbitrary elements of G_i in position i is a subgroup of G isomorphic to G_i:
$$G_i' = \{(1, 1, \dots, 1, g_i, 1, \dots, 1) \mid g_i \in G_i\} \cong G_i$$
(here g_i appears in the i^{th} position). If we identify G_i with this subgroup, then $G_i \trianglelefteq G$ and
$$G/G_i \cong G_1 \times \cdots \times G_{i-1} \times G_{i+1} \times \cdots \times G_n.$$

2. For each fixed i define $\pi_i : G \to G_i$ by
$$\pi_i((g_1, g_2, \dots, g_n)) = g_i.$$
Then π_i is a surjective homomorphism with
$$\ker \pi_i = \{(g_1, \dots, g_{i-1}, 1, g_{i+1}, \dots, g_n) \mid g_j \in G_j \text{ for all } j \neq i\}$$
(here the 1 appears in position i) $\cong G_1 \times \cdots \times G_{i-1} \times G_{i+1} \times \cdots \times G_n$.

3. Under the identifications in part (1), if $x \in G_i$ and $y \in G_j$ for some $i \neq j$, then $xy = yx$.

Proof. (1) Since the operation in G is defined componentwise, it follows easily from the subgroup criteria that $\{(1, 1, \dots, 1, g_i, 1, \dots, 1) \mid g_i \in G_i\}$ is a subgroup of G. Furthermore, the map $\phi : G_i \to \{(1, 1, \dots, 1, g_i, 1, \dots, 1) \mid g_i \in G_i\}$ defined by $\phi(g_i) = (1, \dots, g_i, \dots, 1)$ is seen to be an isomorphism of G_i with this subgroup. Identify G_i with this isomorphic copy in G. To prove the remaining parts of (1) consider the map

$$\varphi : G \to G_1 \times \cdots \times G_{i-1} \times G_{i+1} \times \cdots \times G_n$$

defined by

$$\varphi((g_1, g_2, \dots, g_n)) = (g_1, \dots, g_{i-1}, g_{i+1}, \dots, g_n)$$

(i.e., φ erases the i^{th} component of G). The map φ is a homomorphism since

$$\begin{aligned}\varphi((g_1, \dots, g_n)(h_1, \dots, h_n)) &= \varphi((g_1 h_1, \dots, g_n h_n)) \\ &= (g_1 h_1, \dots, g_{i-1} h_{i-1}, g_{i+1} h_{i+1}, \dots, g_n h_n) \\ &= (g_1, \dots, g_{i-1}, g_{i+1}, \dots, g_n)(h_1, \dots, h_{i-1}, h_{i+1}, \dots, h_n) \\ &= \varphi((g_1, \dots, g_n))\varphi((h_1, \dots, h_n)).\end{aligned}$$

Since the entries in position j are arbitrary elements of G_j for all j, φ is surjective. Furthermore,

$$\ker \varphi = \{(g_1, \dots, g_n) \mid g_j = 1 \text{ for all } j \neq i\} = G_i'.$$

This proves that G_i' is a normal subgroup of G (in particular, it again proves this copy of G_i is a subgroup) and the First Isomorphism Theorem gives the final assertion of part (1).

(2) In the argument that π_i is a surjective homomorphism and the kernel is the subgroup described is very similar to that in part (1), so the details are left to the reader. In part (3) if $x = (1, \dots, 1, x_j, 1, \dots, 1)$ and $y = (1, \dots, 1, y_i, 1, \dots, 1)$, where the indicated entries appear in positions j, respectively, then

$$xy = (1, \dots, 1, x_j, 1, \dots, 1, y_i, 1, \dots, 1) = yx$$

(where the notation is chosen so that $i < j$). This completes the proof. ■

Exercise 2.5.3. A direct product $G_1 \times G_2$ of groups G_1 and G_2 is abelian iff both G_1 and G_2 are abelian.

EXAMPLES

(1) Under the notation of Proposition 2.4.2, it follows from part (3) that if $x_i \in G_i, 1 \leq i \leq n$, then for all $k \in \mathbb{Z}$

$$(x_1 x_2 \dots x_n)^k = x_1^k x_2^k \dots x_n^k.$$

Since the order of $x_1 x_2 \dots x_n$ is the smallest positive integer k such that $x_i^k = 1$ for all i, we see that

$$|(x_1 x_2 \dots x_n)| = \text{LCM}\,[|x_1|, |x_2|, \dots, |x_n|]$$

(where this order is infinite if and only if one of the x_i's has infinite order).

(2) Let p be a prime and for $n \in \mathbb{Z}^+$ consider

$$E_{p^n} = \mathbb{Z}_p \times \mathbb{Z}_p \times \dots \times \mathbb{Z}_p \quad (n \text{ factors}).$$

Then E_{p^n} is an abelian group of order p^n with the property that $x^p = 1$ for all $x \in E_{p^n}$. This group is the elementary abelian group of order p^n

(3) For a prime p, we show that the elementary abelian group of order p^2 has exactly $p+1$ subgroups of order p (in particular, there are more than the two obvious ones). Let $E = \mathbb{Z}_p \times \mathbb{Z}_p$. Since nonidentity element of E has order p, each of these generates a cyclic subgroup of E of order p. By Lagrange's Theorem, distinct subgroups of order p intersect trivially. Thus the $p^2 - 1$ nonidentity elements of E are partitioned into subsets of size $p - 1$ (i.e., each of these subsets consists of the nonidentity elements of some subgroup of order p). There must therefore be

$$\frac{p^2 - 1}{p - 1} = \frac{(p-1)(p+1)}{p-1} = p + 1$$

subgroups of order p. When $p = 2$, E is the Klein 4-group which we have already seen has 3 subgroups of order 2

2.5.1 Internal Direct Product of Groups

Definition 2.5.4. A group G is the **internal direct product** of its subgroups $H_1, H_2, \dots, H_k$ if:

1. for $i \neq j$, $h_i h_j = h_j h_i$ for all $h_i \in H_i$ & $h_j \in H_j$, whenever, $i \neq j$.
2. each $g \in G$ has a unique representation of the form $h_1 h_2 \dots h_k$, where, $h_i \in H_i$, $1 \leq i \leq k$.

Theorem 2.5.5. *If G is the internal direct product of its subgroups $H_1, H_2, \dots, H_k$, then $G \cong H_1 \times H_2 \times \dots \times H_k$.*

Proof. A mapping $\varphi : G \to H_1 \times H_2 \times \dots \times H_k$ defined by $\varphi(g) = (h_1, h_2, \dots, h_k)$, where, $g = h_1 h_2 \dots h_k$ is a bijective homomorphism. ■

Theorem 2.5.6. *Let G be a group and let $H_1, H_2, \dots, H_n$ be n subgroups of G. Then G is an internal direct product of $H_1, H_2, \dots, H_n$ if and only if the following conditions are satisfied:*

1. *For each i, H_i is a normal subgroup of G.*
2. $H_i \cap \left(\prod_{j \neq i} H_j\right) = (e)$.

3. $G = \prod_{i=1}^{n} H_i$.

Proof. Let G be the internal direct product of its subgroups $H_1, H_2, \ldots, H_n$.

1. Let $h_k \in H_k$ and let $g = h_1h_2\ldots h_k$, $h_i \in H_i$, $1 \leq i \leq k$, be any member of G. Since $h_ih_j = h_hh_i$ for all $i \neq j$, we get $g^{-1}h_ig = h_i^{-1}h_ih_i \in H_i$. This yields that H_i is a normal subgroup of G. This proves (1).

2. Now $d \in H_i \cap \left(\prod\limits_{j\neq i} H_j\right)$. $\Rightarrow d \in H_i$ and $d \in \prod\limits_{j\neq i} H_j \Rightarrow d = d_i$ and $d = d_1d_2\ldots d_{i-1}d_{i+1}\ldots d_k$ for some $d_s \in H_s$ for all $s = 1, 2, \ldots, k \Rightarrow ee\ldots ed_ie\ldots e = d_1d_2\ldots d_{i-1}ed_{i+1}\ldots d_n \Rightarrow d_i = e$, by the uniqueness of expression as products $\Rightarrow d = e$. Hence $H_i \cap \left(\prod_{j\neq i} H_j\right) = (e)$. This establishes (2).

3. (3) follows directly from the definition of internal direct product.

Conversely, let $H_1, H_2, \ldots, H_n$ be subgroups of G satisfying the given three conditions. Consider any two distinct integers $k \neq l$, between 1 and n. By (2) $H_k \cap \left(\prod_{j\neq k} H_j\right) = (e)$.

Since $l \neq k$, $H_l \subseteq \prod_{j\neq k} H_j$. Consequently $H_k \cap H_l = (e)$. Let $x_k \in H_k$, $x_l \in H_l$. Then $x_k^{-1}x_l^{-1}x_kx_l = (x_k^{-1}x_l^{-1}x_k)x_l \in H_l$, since H_l is normal in G. Similarly $x_k^{-1}x_l^{-1}x_kx_l = x_k^{-1}(x_l^{-1}x_kx_l) \in H_k$. So that $x_k^{-1}x_l^{-1}x_kx_l \in H_k \cap H_l = (e)$. This gives $x_k^{-1}x_l^{-1}x_kx_l = e \Rightarrow x_kx_l = x_lx_k$.

Now by (3), $G = H_1H_2\ldots H_k$. Therefore given $x \in G$, $x = x_1x_2\ldots x_k$, $x_i \in H_i$ for all i. If $x = y_1y_2\ldots y_k$, then

$$\begin{aligned}
x_1x_2\ldots x_k &= y_1y_2\ldots y_k \\
\Rightarrow y_2^{-1}y_3^{-1}\ldots y_k^{-1}x_1x_2\cdots \Rightarrow x_k &= y_2^{-1}y_3^{-1}\ldots y_k^{-1}y_1y_2\ldots y_k \\
\Rightarrow x_1y_2^{-1}y_3^{-1}\ldots y_k^{-1}x_2\ldots x_k &= y_1 \\
\Rightarrow y_2^{-1}y_3^{-1}\ldots y_k^{-1}x_2\ldots x_k &= x_1^{-1}y_1
\end{aligned}$$

Therefore, $x_1^{-1}y_1 \in H_1 \cap \left(\prod\limits_{i=2}^{k} H_i\right)$, that is, $x_1^{-1}y_1 = e$, implying, $x_1 = y_1$. Hence, $x_i = y_i$ for all $1 \leq i \leq k$. This completes the proof. ∎

2.6 THE FUNDAMENTAL THEOREM OF FINITELY GENERATED ABELIAN GROUPS

Definition 2.6.1. 1. A group G is finitely generated if there is a finite subset A of G such that $G = \langle A \rangle$.

2. For each $r \in \mathbb{Z}$ with $r \geq 0$, let $\mathbb{Z}^r = \mathbb{Z} \times \mathbb{Z} \times \cdots \times \mathbb{Z}$ be the direct product of r copies of the group $\mathbb{Z}$, where $\mathbb{Z}^0 = 1$. The group $\mathbb{Z}^r$ is called the free abelian group of rank r.

Note that any finite group G is, a fortiori, finitely generated: simply take $A = G$ as a set of generators. Also, $\mathbb{Z}^r$ is finitely generated by $e_1, e_2, \ldots, e_r$, where e_i is the r-tuple with 1 in position i and zeros elsewhere. We can now state the fundamental classification theorem for (finitely generated) abelian groups.

Theorem 2.6.2 (Fundamental Theorem of Finitely Generated Abelian Groups)**.** *Let G be a finitely generated abelian group. Then*

1. *$G \cong \mathbb{Z}^r \times \mathbb{Z}_{n_1} \times \mathbb{Z}_{n_2} \times \cdots \times \mathbb{Z}_{n_k} \times \mathbb{Z}_{n_k}$ for some integers $r, n_1, n_2, \ldots, n_k$ satisfying the following conditions:*

(a) $r \geq 0$ and $n_j \geq 2$ for all j, and

(b) $n_{i+1} \mid n_i$ for $1 \leq i \leq k-1$.

(2) the expression in (1) is unique: if $G \cong \mathbb{Z}^{r'} \times \mathbb{Z}_{m_1} \times \cdots \times \mathbb{Z}_{m_s}$, where r' and $m_1, m_2, \ldots, m_s$ satisfy (a) and (b) (i.e., $r' \geq 0, m_j \geq 2$ for all j and $m_{i+1} \mid m_i$ for $1 \leq i \leq s-1$), then $r = r', s = k$ and $m_i = n_i$ for all i.

Definition 2.6.3. The integer r in Theorem 2.6.2 is called the *free rank* or Betti number of G and the integers $n_1, n_2, \dots, n_k$ are called the *invariant factors* of G. The description of G in Theorem 2.6.2(1) is called the *invariant factor decomposition* of G.

Theorem 2.6.2 asserts that the free rank and (ordered) list of invariant factors of an abelian group are uniquely determined. If G is a finite abelian group with invariant factors $n_1, n_2, \dots, n_k$, where $n_{i+1} \mid n_i$, $1 \le i \le k-1$, then G is said to be of type $(n_1, n_2, \dots, n_k)$. Theorem 2.6.2 gives an effective way of listing all finite abelian groups of a given order. Namely, to find (up to isomorphism) all abelian groups of a given order N one must find all finite sequences of integers $n_1, n_2, \dots, n_k$ such that

1. $n_j \ge 2$ for all $j \in \{1, 2, \dots, s\}$,
2. $n_{i+1} \mid n_i$, $1 \le i \le s-1$, and
3. $n_1 n_2 \dots n_s = N$.

Corollary 2.6.4. *If n is the product of distinct primes, then up to isomorphism the only abelian group of order n is the cyclic group of order n, $\mathbb{Z}_n$.*

Example 2.6.5. Suppose $N = 180 = 2^2 \cdot 3^2 \cdot 5$. As noted above we must have $2 \cdot 3 \cdot 5 \mid n_1$, so possible values of n_1 are

$$n_1 = 2^2 \cdot 3^2 \cdot 5, \quad 2^2 \cdot 3 \cdot 5, \quad 2 \cdot 3^2 \cdot 5, \quad \text{or } 2 \cdot 3 \cdot 5.$$

To each of these one must work out all possible n_i's (subject to $n_{i+1} \mid n_i$ and $n_1 n_2 \dots n_s = N$). For each resulting pair n_1, n_2 one must work out all possible n_3's etc. until all lists satisfying (1) to (3) are obtained. For instance, if $n_1 = 2 \cdot 3^2 \cdot 5$, the only number n_2 dividing n_1 with $n_1 n_2$ dividing N is $n_2 = 2$. In this case $n_1 n_2 = N$, so this list is complete: $2 \cdot 3^2 \cdot 5, 2$. The abelian group corresponding to this list is $\mathbb{Z}_{90} \times \mathbb{Z}_2$. If $n_1 = 2 \cdot 3 \cdot 5$, the only candidates for n_2 are $n_2 = 2, 3$ or 6. If $n_2 = 2$ or 3, then $n_1 n_2 \ne N$ we would necessarily have $n_3 = N/(n_1 n_2)$ (and there must be a third term in the list by property (3)). This leads to a contradiction because $n_1 n_2 n_3$ would be divisible by 2^3 or 3^3 respectively, but N is not divisible by either of these numbers. Thus the only list of invariant factors whose first term is $2 \cdot 3 \cdot 5$ is $2 \cdot 3 \cdot 5, 2 \cdot 3 \cdot 5$. The corresponding abelian group is $\mathbb{Z}_{30} \times \mathbb{Z}_6$.

Similarly, all permissible lists of invariant factors and the corresponding abelian groups of order 180 are easily seen to be the following:

Invariant Factors	**Abelian Groups**
$2^2 \cdot 3^2 \cdot 5$	$\mathbb{Z}_{180}$
$2 \cdot 3^2 \cdot 5, 2$	$\mathbb{Z}_{90} \times \mathbb{Z}_2$ (not $\mathbb{Z}_{180}$)
$2^2 \cdot 3 \cdot 5, 3$	$\mathbb{Z}_{60} \times \mathbb{Z}_3$
$2 \cdot 3 \cdot 5, 2 \cdot 3$	$\mathbb{Z}_{30} \times \mathbb{Z}_6$

Remark: Define partition of a positive integer t as the representation of t as the sum of positive integers, listed in non-increasing order. For example, the distinct partitions of 3 are 3, 2+1, 1+1+1. The number of abelian groups of order $p^{n_1} p^{n_2} \dots p^{n_k}$ is equal to $\mathcal{P}(n_1)\mathcal{P}(n_2) \cdots \mathcal{P}(n_k)$, where, $\mathcal{P}(t)$ denotes the number of distinct partitions of the integer t.

2.7 Solvable Groups

Definition 2.7.1. A group G is said to be solvable if it has a subnormal series $G = G_0 \supseteq G_1 \supseteq G_2 \supseteq \cdots \supseteq G_n = \langle e \rangle$, such that the quotients $\frac{G_i}{G_{i+1}}$ are abelian.

1. Any abelian group G is solvable. Now $G > \langle e \rangle$ is a normal series of G and its only factor group is $G/\langle e \rangle$, which being isomorphic to G, is commutative.
2. Consider the series $S_3 > A_3 > \langle I \rangle$. Its factor groups are S_3/A_3 and A_3, which have orders 2 and 3, respectively. Since any group of prime order is commutative, it follows that both the factor groups are commutative. Hence, S_3 is solvable.

3. Consider $S_4 > A_4 > V_4 > \langle I \rangle$, a subnormal series of S_4, where $V_4 = \{(I), (12)(34), (13)(24), (14)(23)\}$. Its factor groups are S_4/A_4, A_4/V_4, and V_4, which have orders 2, 3, and 4, respectively. Since every group of order 4 is abelian, it follows that all the factor groups are abelian, hence S_4 is solvable.

Theorem 5.8 Any subgroup H of a solvable group G is solvable.

Proof. Let $G = G_0 \supseteq G_1 \supseteq G_2 \supseteq \cdots \supseteq G_n = \langle e \rangle$ be a solvable series for G. We claim that

$$H = H_0 \supseteq (H \cap G_1) \supseteq (H \cap G_2) \supseteq \cdots \supseteq H \cap G_n = \langle e \rangle \quad \ldots (1)$$

is a solvable series for H. Since for $i = 0, 1, \ldots, n-1$, G_{i+1} is normal in G_i, we get $H \cap G_{i+1}$ is normal in $H \cap G_i$. Define a mapping $f : H_i \to G_i/G_{i+1}$ such that $f(x) = xG_{i+1}$, $x \in H_i$. f is a homomorphism. Further

$$x \in \text{Ker} f \iff xG_{i+1} = G_{i+1}$$
$$\iff x \in G_{i+1}$$
$$\iff x \in H \cap G_{i+1} \text{ since } x \in H$$

This yields that $\text{Ker} f = H \cap G_{i+1}$. H_i hence by the Fundamental Theorem of Homomorphism $H_i/(H_i \cap G_{i+1}) \cong f(H_i)$. As $f(H_i)$ is a subgroup of G_i/G_{i+1} and G_i/G_{i+1} is abelian, $f(H_i)$ is also abelian. Consequently $H_i/(H_i \cap G_{i+1})$ is abelian. This proves that (1) is a solvable series for H. Hence H is a solvable group. ■

Theorem 5.9 If H is a normal subgroup of a solvable group G, then G/H is also solvable.

Proof. Let $G = G_0 \supseteq G_1 \supseteq G_2 \supseteq \cdots \supseteq G_n = \langle e \rangle$ be a solvable series for G. Consider the series

$$G/H = H_0/H \supseteq G_1H/H \supseteq G_2H/H \supseteq \cdots \supseteq G_nH/H = \langle e \rangle \quad \ldots (1)$$

For each $i = 0, 1, 2, \ldots, n-1$, G_{i+1} is normal in G_i. Consider any $x \in G_iH$ then $x = gh$, for some $g \in G_i, h \in H$. Thus

$$xG_{i+1}H = ghG_{i+1}H$$

$$= ghG_{i+1}H \text{ (since for any subgroup K of G, HK = KH)}$$
$$= gG_{i+1}H \text{ since } H = H$$
$$= gG_{i+1}hH = gG_{i+1}h$$

This proves that $G_{i+1}H$ is a normal subgroup G_iH. Hence

$$\frac{G_iH/H}{G_{i+1}H/H} \cong \frac{G_iH}{G_{i+1}H} \quad \ldots (2)$$

Now define $f : G_i \to G_iH/G_{i+1}H$ such that $f(x) = xG_{i+1}H$, $x \in G_i$, then f is a homomorphism. As $G_iH = HG_i$, given $y \in G_iH$, we can write

$$y = hg \text{ for some } h \in H, g \in G_i$$

Then $Hy = Hhg = Hg = f(g)$. This shows that f is also onto. Thus $G_iH/G_{i+1}H \cong \text{Ker} f$, and hence f induces a homomorphism

$$\bar{f} : G_i/\text{G}_{i+1} \to G_iH/G_{i+1}H$$

such that $\bar{f}(gG_{i+1}) = gG_{i+1}H \; x \in G_i$. This f is also onto. Thus $G_iH/G_{i+1}H$ is a homomorphic image of the abelian group G_i/G_{i+1}. So that it must be itself abelian. Consequently each factor group of the subnormal series (1) is abelian. This proves that G/H is solvable. ■

Theorem 5.10 Let H be a normal subgroup of a group G. If both H and G/H are solvable then G is solvable.

Proof. Let $G/H = G_0/H \supseteq G_1/H \supseteq \cdots \supseteq G_r/H = \langle \bar{e} \rangle \quad \ldots (1)$ be a solvable series for G/H. Here each G_i is a subgroup of G containing H. Since G_{i+1}/H is normal in G_i/H, each G_{i+1} is normal in G_i. Further $G_i/H/G_{i+1}/H \cong G_i/G_{i+1}$. Now let

$$H = H_0 \supseteq H_1 \supseteq H_2 \supseteq \cdots \supseteq H_m = \langle e \rangle \quad \ldots (2)$$

Then $G = G_0 \supseteq G_1 \supseteq G_2 \supseteq \cdots \supseteq G_r \supseteq H_1 \supseteq H_2 \supseteq \cdots \supseteq H_m = \langle e \rangle$ is a solvable series for G. This proves that G is a solvable group. ■

Corollary 5.11 Let H, K be two groups and $G = H \times K$. Then G is solvable iff H and K are solvable.

Proof. Let G be solvable. Then any subgroup of G is solvable. As $H \cong H \times \{e\}$ and $K \cong \{e\} \times K$, we get H is solvable. Similarly K is solvable. Conversely, let H and K be solvable. Consider $H' = H \times \{e\}$. H' isomorphic to H gives that H' is solvable. Moreover, $G/H' \cong K$, gives that G/H' is solvable. Hence, by the above theorem, G is solvable. ■

Corollary 5.12 Any finite p-group is solvable.

Proof. Let G be a finite p-group (p is a prime number) then $o(G) = p^n$, for some $n \geq 0$. If $n = 1$, G is abelian and so it is solvable. So let $n > 1$, we apply induction on n. Suppose the result holds for groups of order p^m with $m < n$. Now $Z(G)$, the centre of G is not equal to $\langle e \rangle$. Consequently $o(Z(G)) = p^k$ for some $k \geq 1$. This yields that $G/Z(G)$ is a group of order p^{n-k} with $n - k < n$. Hence by the induction hypothesis $G/Z(G)$ is solvable. Also $Z(G)$ being abelian, is solvable. Hence, taking $H = Z(G)$, we see from the above theorem that G is solvable. ■

2.8 The Jordon-Hölder Theorem

Definition 2.8.1 (Normal Series). A **normal series** of a group G is a finite sequence of subgroups,

$$G = G_0 \geq G_1 \geq G_2 \geq \cdots \geq G_n = \{1\},$$

with $G_{i+1} \lhd G_i$ for all i. The **length** of the series is the number of strict inclusions or equivalently, the number of nontrivial factor groups.

2.8.1 Zassenhaus Lemma

Lemma 2.8.2 (Zassenhaus Lemma). *Given four subgroups $A \lhd A^*$ and $B \lhd B^*$ of a group G, then*

$$A(A^* \cap B) \lhd A^* \cap B^*, \quad B(B^* \cap A) \lhd B^* \cap A^*,$$

and there is an isomorphism

$$\frac{A(A^* \cap B^*)}{A(A^* \cap B)} \cong \frac{B(B^* \cap A^*)}{B(B^* \cap A)}.$$

Remark 2.8.3. The Zassenhaus lemma is sometimes called the **butterfly lemma** because of the following picture.

$$\begin{tikzcd}[arrows=dash]
A(A^* \cap B^*) \arrow[drr] \arrow[dd] & & & & B(A^* \cap B^*) \arrow[dll] \arrow[dd] \\
& & A^* \cap B^* & & \\
A(A^* \cap B) \arrow[d] \arrow[drr] & & & & B(A \cap B^*) \arrow[dll] \arrow[d] \\
A \arrow[d] & & D = (A^* \cap B)(A \cap B^*) \arrow[dll] \arrow[drr] & & B \arrow[d] \\
A \cap B^* & & & & A^* \cap B
\end{tikzcd}$$

The isomorphism is symmetric in the sense that the right side is obtained from the left by interchanging the symbols A and B.

Proof. We claim that $(A \cap B^*) \lhd (A^* \cap B^*)$: that is, if $c \in A \cap B^*$ and $x \in A^* \cap B^*$, then $xcx^{-1} \in A \cap B^*$. Now $xcx^{-1} \in A$ because $c \in A, x \in A^*$, and $A \lhd A^*$; but also $xcx^{-1} \in B^*$, because $c, x \in B^*$. Hence, $(A \cap B^*) \lhd (A^* \cap B^*)$; similarly, $(A^* \cap B) \lhd (A^* \cap B^*)$. Therefore, the subset D, defined by

$$D = (A \cap B^*)(A^* \cap B),$$

is a normal subgroup of $A^* \cap B^*$, because it is generated by two normal subgroups.

We show that there is an isomorphism from $\dfrac{A(A^* \cap B^*)}{A(A^* \cap B)}$ to $\dfrac{A^* \cap B^*}{D}$. Define $\varphi : A(A^* \cap B^*) \to (A^* \cap B^*)/D$ by $ax \mapsto xD = Dx$, where $a \in A$ and $x \in A^* \cap B^*$. Now φ is well-defined: if $ax = a'x'$, where $a' \in A$ and $x' \in A^* \cap B^*$, then $(a')^{-1}a = x'x^{-1} \in A \cap (A^* \cap B^*) = A \cap B^* \leq D$; which gives $Dx = Dx'$, that is, $\varphi(ax) = \varphi(a'x')$. Also, φ is a homomorphism: $axa'x' = a''xx'$, where $a'' = a(xa'x^{-1}) \in A$ (because $A \lhd A^*$), and so $\varphi(axa'x') = \varphi(a''xx') = xx'D = \varphi(ax)\varphi(a'x')$. For every $Dx \in \frac{A^* \cap B^*}{D}$, we choose $a \in A$ such that $ax \mapsto Dx$. Therefore, φ is surjective and that $\ker(\phi) = \{ax \mid Dx = D\} = \{ax \mid x \in D\}$, so that, $a \in A$ and $x \in D$, we have $ax \in AD = A(A \cap B^*)(A^* \cap B) = A(A^* \cap B)$ because $A \cap B^* \subseteq A$ (and $HK = K$ if $H \subseteq K$). That is, $\ker \varphi = A(A^* \cap B)$. The first isomorphism theorem gives

$$\frac{A(A^* \cap B^*)}{A(A^* \cap B)} \cong \frac{A^* \cap B^*}{D} = \frac{A^* \cap B^*}{(A \cap B^*)(A^* \cap B)}. \tag{2.3}$$

Using symmetry, (interchanging roles of A and B), we obtain

$$\frac{B(B^* \cap A^*)}{B(A^* \cap A)} \cong \frac{B^* \cap A^*}{(B \cap A^*)(B^* \cap A)}. \tag{2.4}$$

Combining equations (2.3) and (2.4), the conclusion follows. ■

Observation: Zassenhaus lemma implies the second isomorphism theorem: If S and T are subgroups of a group G with $T \lhd G$, then

$$TS/T \cong S/(S \cap T);$$

set $A^* = G, A = T, B^* = S, B = S \cap T$.

Definition 2.8.4 (Composition Series)**.** A **composition series** is a normal series all of whose nontrivial factor groups are simple. The nontrivial factor groups of a composition series are called **composition factors** of G.

A group need not have a composition series; for example, the abelian group $\mathbb{Z}$ has no composition series. However, every finite group does have a composition series.

Proposition 2.8.5. Every finite group G has a composition series.

Proof. If the proposition is false, let G be a finite group of smallest order that does not have a composition series. Now G is not simple, otherwise $G > \{1\}$ is a composition series. Hence, G has a proper normal subgroup H; we may assume that H is a maximal normal subgroup, so that G/H is a simple group. But $|H| < |G|$, so that H does have a composition series: say, $H = H_0 > H_1 > \cdots > \{1\}$, and

$$G > H_0 > H_1 > \cdots > \{1\}$$

is a composition series for G, a contradiction. ■

Here are two composition series of $G = \langle a \rangle$, a cyclic group of order 30 (note that normality of subgroups is automatic because G is abelian). The first is

$$G = \langle a \rangle \geq \langle a^2 \rangle \geq \langle a^{10} \rangle \geq \{1\};$$

the factor groups of this series are $\langle a \rangle/\langle a^2 \rangle \cong \mathbb{Z}_2$, $\langle a^2 \rangle/\langle a^{10} \rangle \cong \mathbb{Z}_5$, and $\langle a^{10} \rangle/\{1\} \cong \mathbb{Z}_3$. Another normal series is

$$G = \langle a \rangle \geq \langle a^5 \rangle \geq \langle a^{15} \rangle \geq \{1\};$$

the factor groups of this series are $\langle a\rangle/\langle a^5\rangle \cong \mathbb{Z}_5$, $\langle a^5\rangle/\langle a^{15}\rangle \cong \mathbb{Z}_3$, and $\langle a^{15}\rangle/\{1\} \cong \mathbb{Z}_2$. Notice that the same factor groups arise, although the order in which they arise is different. We will see that this phenomenon always occurs: Different composition series of the same group have the same factor groups. This is the **Jordan–Hölder theorem**, and the next definition makes its statement more precise.

Definition 2.8.6 (Equivalent)**.** Two normal series of a group G are equivalent if there is a bijection between the sets of nontrivial factor groups of each so that corresponding factor groups are isomorphic.

The Jordan–Hölder theorem says that any two composition series of a group are equivalent. It will be more efficient to prove a more general theorem, due to Schreier.

Definition 2.8.7 (Refinement)**.** A **refinement** of a normal series is a normal series $G = N_0, N_1, \ldots, N_k = \{1\}$ having the original series as a subsequence.

In other words, a refinement of a normal series is a new normal series obtained from the original by inserting more subgroups.

Notice that a composition series admits only insignificant refinements; one can merely repeat terms (if G_i/G_{i+1} is simple, then it has no proper nontrivial normal subgroups and, hence, there is no intermediate subgroup L with $G_i > L > G_{i+1}$ and $L \lhd G_i$). Therefore, any refinement of a composition series is equivalent to the original composition series.

2.8.2 Schreier Refinement Theorem

Theorem 2.8.8 (Schreier Refinement Theorem)**.** *Any two normal series*

$$G = G_0 \geq G_1 \geq \cdots \geq G_n = \{1\}$$

and

$$G = N_0 \geq N_1 \geq \cdots \geq N_k = \{1\}$$

of a group G have equivalent refinements.

Proof. We insert a copy of the second series between each pair of adjacent terms in the first series. In more detail, for each $i \geq 0$, define

$$G_{ij} = G_{i+1}(G_i \cap N_j)$$

(this is a subgroup because $G_{i+1} \lhd G_i$). Note that

$$G_{i0} = G_{i+1}(G_i \cap N_0) = G_{i+1}G_i = G_i,$$

because $N_0 = G$, and that

$$G_{ik} = G_{i+1}(G_i \cap N_k) = G_{i+1},$$

because $N_k = \{1\}$. Therefore, the series of G_{ij} is a subsequence of the series of G_i:

$$\cdots \geq G_i = G_{i0} \geq G_{i1} \geq G_{i2} \geq \cdots \geq G_{ik} = G_{i+1} \geq \cdots .$$

Similarly, there is a subsequence of the second series arising from subgroups

$$N_{pq} = N_{p+1}(N_p \cap G_q).$$

Both subsequences have nk terms. For each i, j, the Zassenhaus lemma, for the four subgroups $G_{i+1} \lhd G_i$ and $N_{j+1} \lhd N_j$, says both subsequences are normal series, hence are refinements, and there is an isomorphism

$$\frac{G_{i+1}(G_i \cap N_j)}{G_{i+1}(G_i \cap N_{j+1})} \cong \frac{N_{j+1}(N_j \cap G_i)}{N_{j+1}(N_j \cap G_{i+1})};$$

that is,

$$G_{i,j}/G_{i,j+1} \cong N_{j,i}/N_{j,i+1}.$$

The association $G_{i,j}/G_{i,j+1} \mapsto N_{j,i}/N_{j,i+1}$ is a bijection showing that the two refinements are equivalent. ■

In other words, a refinement of a normal series is a new normal series obtained from the original by inserting more subgroups.

Theorem 2.8.9 (Jordan–Hölder Theorem). *Any two composition series of a group G are equivalent. In particular, the length of a composition series, if one exists, is an invariant of G.*

Proof. Notice that a composition series admits only insignificant refinements; one can merely repeat terms (if G_i/G_{i+1} is simple, then it has no proper nontrivial normal subgroups and, hence, there is no intermediate subgroup L with $G_i > L > G_{i+1}$ and $L \lhd G_i$). Therefore, any refinement of a composition series is equivalent to the original composition series. It now follows from Schreier's theorem that any two composition series are equivalent. ■

Here is a new proof of the fundamental theorem of arithmetic.

Corollary 2.8.10. *Every integer $n \geq 2$ has a factorisation into primes, and the prime factors are uniquely determined by n.*

Proof. Since the group $\mathbb{Z}_n$ is finite, it has a composition series; let $S_1, \ldots, S_r$ be the factor groups. Now, an abelian group is simple if and only if it is of prime order. Since $n = |\mathbb{Z}_n|$ is the product of the orders of the factor groups, we have proved that n is a product of primes. Moreover, the Jordan-Hölder theorem gives the uniqueness of the (prime) orders of the factor groups. ■

Example 2.8.11.

1. Non-isomorphic groups can have the same composition factors. For example, both $\mathbb{Z}_4$ and V have composition series whose factor groups are $\mathbb{Z}_2, \mathbb{Z}_2$.

2. Let $G = \mathrm{GL}(2, \mathbb{F}_4)$ be the general linear group of all 2×2 nonsingular matrices with entries in the field $\mathbb{F}_4$ with four elements. Now $\det\colon G \to (\mathbb{F}_4)^\times$, where $(\mathbb{F}_4)^\times \cong \mathbb{Z}_3$ is the multiplicative group of nonzero elements of $\mathbb{F}_4$. Since $\ker\det = \mathrm{SL}(2, \mathbb{F}_4)$, the special linear group consisting of those matrices of determinant 1, there is a normal series
$$G = \mathrm{GL}(2, \mathbb{F}_4) \geq \mathrm{SL}(2, \mathbb{F}_4) \geq \{1\}.$$
The factor groups of this normal series are $\mathbb{Z}_3$ and $\mathrm{SL}(2, \mathbb{F}_4)$. It is true that $\mathrm{SL}(2, \mathbb{F}_4)$ is a non-abelian simple group and so this series is a composition series. We cannot yet conclude that G is not solvable, for the definition of solvability requires that there be some composition series, not necessarily this one, having factor groups of prime order. However, the Jordan-Hölder theorem says that if one composition series of G has all its factor groups of prime order, then so does every other composition series. We may now conclude that $\mathrm{GL}(2, \mathbb{F}_4)$ is not a solvable group.

2.8.3 Solvability Revisited

Proposition 2.8.12. Every quotient G/N of a solvable group G is itself a solvable group.

Proof. Let $G = G_0 \geq G_1 \geq G_2 \geq \cdots \geq G_r = \{1\}$ be a sequence of subgroups as in the definition of a solvable group. Since $N \trianglelefteq G$, we have NG_i a subgroup of G for all i, and so there is a sequence of subgroups
$$G = G_0N \geq G_1N \geq \cdots \geq G_rN \geq \{1\}.$$
This is a normal series. With obvious notation,
$$(g_in)G_{i+1}N(g_in)^{-1} = g_iG_{i+1}Ng_i^{-1} = g_iG_{i+1}g_i^{-1}N \subseteq G_{i+1}N,$$
That is, $G_{i+1}N$ is normal in G_iN. The second isomorphism theorem gives
$$\frac{G_i}{G_i \cap (G_{i+1}N)} \cong \frac{G_i(G_{i+1}N)}{G_{i+1}N} = \frac{G_iN}{G_{i+1}N},$$
the last equation holding because $G_iG_{i+1} = G_i$. Since $G_{i+1} \lhd G_i \cap G_{i+1}N$, the third isomorphism theorem gives a surjection
$$G_i/G_{i+1} \twoheadrightarrow G_i/(G_i \cap G_{i+1}N),$$
And so the composite is a surjection
$$G_i/G_{i+1} \twoheadrightarrow G_iN/G_{i+1}N.$$

As G_i/G_{i+1} is cyclic of prime order, its image is either cyclic of prime order or trivial. Therefore, G/N is a solvable group. ■

Proposition 2.8.13. Every subgroup H of a solvable group G is itself a solvable group.

Proof. Since G is solvable, there is a sequence of subgroups

$$G = G_0 \geq G_1 \geq G_2 \geq \cdots \geq G_r = \{1\}$$

with G_i normal in G_{i-1} and G_{i-1}/G_i cyclic for all i. Consider the sequence of subgroups

$$H = H \cap G_0 \geq H \cap G_1 \geq H \cap G_2 \geq \cdots \geq H \cap G_r = \{1\}.$$

This is a normal series: If $h_{i+1} \in H \cap G_{i+1}$ and $g_i \in H \cap G_i$, then $g_i h_{i+1} g_i^{-1} \in H$, for $g_i, h_{i+1} \in H$; also, $g_i h_{i+1} g_i^{-1} \in G_{i+1}$ because G_i is normal in G_i. Therefore,

$$g_i h_{i+1} g_i^{-1} \in H \cap G_{i+1},$$

and so $H \cap G_{i+1} \triangleleft H \cap G_i$. Finally, the second isomorphism theorem gives

$$\frac{(H \cap G_i)}{(H \cap G_{i+1})} = \frac{(H \cap G_i)}{[(H \cap G_i) \cap G_{i+1}]} \cong \frac{(H \cap G_i)G_{i+1}}{G_{i+1}}.$$

But the last (quotient) group is a subgroup of G_i/G_{i+1}. Since the only subgroups of a cyclic group C of prime order are C and $\{1\}$, it follows that the nontrivial factor groups

$$(H \cap G_i)/(H \cap G_{i+1})$$

are cyclic of prime order. Therefore, H is a solvable group. ■

Example 2.8.14. We have seen above that S_4 is a solvable group. However, if $n \geq 5$, the symmetric group S_n is not a solvable group. If, on the contrary, S_n were solvable, then so would each of its subgroups. But as $A_5 \leq S_5 \leq S_n$, and A_5 is not solvable because it is a nonabelian simple group.

Proposition 2.8.15. If $H \triangleleft G$ and if both H and G/H are solvable groups, then G is solvable.

Proof. Since G/H is solvable, there is a normal series

$$G/H \geq K_1^* \geq K_2^* \geq \cdots \geq K_m^* = \{1\}$$

Having factor groups of prime order. By the correspondence theorem for groups, there are subgroups K_i of G,

$$G \geq K_1 \geq K_2 \geq \cdots \geq K_m = H,$$

with $K_i/H = K_i^*$ and $K_{i+1}/H = K_{i+1}^*$ for all i. By the third isomorphism theorem,

$$K_i^*/K_{i+1}^* \cong K_i/K_{i+1}$$

for all i, and so K_i/K_{i+1} is cyclic of prime order for all i.

Since H is solvable, there is a normal series

$$H \geq H_1 \geq H_2 \geq \cdots \geq H_q = \{1\}$$

having factor groups of prime order. Splice these two series together,

$$G \geq K_1 \geq K_2 \geq \cdots \geq K_m \geq H \geq H_1 \geq H_2 \geq \cdots \geq H_q = \{1\},$$

to obtain a normal series of G having factor groups of prime order. ■

Corollary 2.8.16. *If H and K are solvable groups, then $H \times K$ is solvable.*

Proof. Since $(H \times K)/H \cong K$, the result follows at once from Proposition 2.8.15. ■

Theorem 2.8.17. *Every $p-$group is solvable.*

Proof. We use induction on n, where $o(G) = p^n$. Base case is trivially true. Assume the induction hypothesis. Finally, $Z(G)$ is non-trivial and abelian. Thus, $Z(G)$ is solvable, and because of the induction hypothesis, $G/Z(G)$ is solvable. Hence, G is solvable, by the previous theorem. ■

Chapter 3

Rings

3.1 Rings - Integral Domains

Definition 3.1.1. A non-empty set R together with two binary operations $+$ and $\cdot$, called *addition* and *multiplication* respectfully, is called a *ring* if it has the following three properties.
($\mathcal{A}_1$) $(R,+)$ is an abelian group,
($\mathcal{A}_2$) $(R,\cdot)$ is a semi-group and
($\mathcal{A}_3$) Distributive laws hold, i.e., for all $r, r_1, r_2 \in R$, $\begin{cases} r.(r_1+r_2) = r.r_1 + r.r_2, \\ (r_1+r_2).r = r_1.r + r_2.r. \end{cases}$

Definition 3.1.2. A ring in which every non-zero element is a unit is called a division ring. If such a ring is also commutative, it is called a field.

Example 3.1.3. For an example of a non-commutative division ring, let $\mathbf{1} = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$, $\mathbf{i} = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, $\mathbf{j} = \left(\begin{smallmatrix} 0 & i \\ i & 0 \end{smallmatrix}\right)$ and $\mathbf{k} = \left(\begin{smallmatrix} i & 0 \\ 0 & -i \end{smallmatrix}\right)$, where $i^2 = -1$. These elements satisfy the following relations: $\mathbf{i^2} = \mathbf{j^2} = \mathbf{k^2} = -\mathbf{1}$, $\mathbf{ij} = \mathbf{k},\ \mathbf{jk} = \mathbf{i},\ \mathbf{ki} = \mathbf{j}$ and $\mathbf{ji} = -\mathbf{k},\ \mathbf{kj} = -\mathbf{i}, \mathbf{ik} = -\mathbf{j}$.

Definition 3.1.4. Let R be a ring. A non-zero element $a \in R$ is called a left zero divisor if $ab = 0$ for some non-zero $b \in R$, and if $ba = 0$ for some non-zero $b \in R$, then a is a right zero divisor. Thus, a zero-divisor is a nonzero element a of a commutative ring R such that there is a nonzero element $b \in R$ with $ab = 0 = ba$. 0 is a trivial zero divisor of a ring. A commutative ring R which contains no zero divisors is called an **integral domain. An integral domain R in which every non-zero element has an inverse is called a *field*.**

Example 3.1.5.

1. The ring $R = \mathbb{Z}_8$ is commutative but not an integral domain.
2. The ring of quaternions is a division ring, but it is not an integral domain since it is not commutative.
3. The ring $\mathbb{Z}$ is an integral domain but not a field.
4. The set $\mathbb{Q}$ under ordinary addition and multiplication is a field.

Proposition 3.1.6. A ring R is without zero divisors if and only if $ab = ac$ implies $b = c$ for all $a, b, c \in R$ with $a \neq 0$.

Proof. If R has no zero divisors and $a \neq 0$, then $ab = ac$ gives $a(b-c) = 0$ so that $b - c = 0$, that is, $b = c$. Conversely, assume cancellation laws hold under given assumptions. If $a \neq 0$ and $ab = 0$, then $ab = a0$ which implies $b = 0$, (by cancellation law). Hence R has no zero divisors. ■

Theorem 3.1.7. *Any finite non-zero integral domain is a field.*

Proof. Let $R = \{x_1, x_2, \ldots, x_n\}$ be a finite non-zero integral domain. Clearly, $n \geq 2$ and let $x \in R$ be non-zero. Then $R = \{xx_1, xx_2, \ldots, xx_n\}$, since if $xx_k = xx_l$ for $k \neq l$, then Proposition 3.1.6 implies $x_k = x_l$, a contradiction. Therefore, $x = xx_i$ for some i, $1 \leq i \leq n$. Now, let $y \in R$. Then $xy = (xx_i)y$, hence commutativity of R and Proposition 3.1.6 gives $y = x_i y = y x_i$. That is x_i is an identity element of R. Denote x_i by e. Then $e = xx_j = x_j x$ for some j, $1 \leq j \leq n$. So x_j is the inverse of x, that is, every non-zero element of R has an inverse. Hence R is a field. ■

2nd proof. By definition, R is a commutative ring with identity $1 \neq 0$. It remains only to check that each nonzero element is a unit. Take $0 \neq a \in R$. Consider the set $\{a^i : i > 0\}$. It consists of infinitely many powers of a. But R is finite. Thus, there cannot be infinitely many distinct powers. Let us say that $a^i = a^j$ with $i > j > 0$. Then $a^j a^{i-j} = a^i = a^j$. More importantly, $a^j a^{i-j} = a^j \cdot 1$. Now, a is a non-zero element of an integral domain, and products of nonzero elements in such a domain do not become zero. Thus, $a^j \neq 0$. By the cancellation law, $a^{i-j} = 1$. If $i - j = 1$, then $a = 1$,which is surely a unit. Otherwise, $aa^{i-j-1} = 1$. Since $i - j - 1 \in \mathbb{N}$, $a^{i-j-1} \in R$, and we have an inverse for a.

Corollary 3.1.8. *For every prime p, $\mathbb{Z}_p$, the ring of integers modulo p, is a field.*

Theorem 3.1.9. *A finite non-zero ring R without zero divisors is a division ring.*

Proof. Let $R = \{x_1, x_2, \ldots, x_n\}$ be a finite non-zero integral domain. Clearly, $n \geq 2$ and let $x \in R$ be non-zero. Then $R = \{xx_1, xx_2, \ldots, xx_n\}$, since if $xx_k = xx_l$ for $k \neq l$, then Proposition 3.1.6 implies $x_k = x_l$, a contradiction. Therefore, $x = xx_i$ for some i, $1 \leq i \leq n$. Now, let $y \in R$. Then $xy = (xx_i)y$, Proposition 3.1.6 gives $y = x_i y$. This gives, $yx_i y = y^2$, that is, $(yx_i - y)y = 0 = 0y$, hence $yx_i = y$, by Proposition 3.1.6. Thus, $y = x_i y = yx_i$, which proves x_i is an identity element of R. Denote x_i by e. Then $e = xx_j$ for some j, $1 \leq j \leq n$. Then, $xx_j x - ex = 0 = x0$, which gives, $x(x_j x - e) = x0$, hence, $x_j x = e$. Therefore, $e = xx_j = x_j x$, that is, x_j is the inverse of x. Hence, R is a division ring. ■

Corollary 3.1.10. *A finite non-zero commutative ring R without zero divisors is a field.*

Theorem 3.1.11. *Let $n \geq 2$ be a positive integer. Then the following are equivalent:*

(i) *$\mathbb{Z}_n$ is an integral domain;*

(ii) *$\mathbb{Z}_n$ is a field; and*

(iii) *n is prime.*

Proof. Since every field is an integral domain and every finite integral domain is a field thus, (i) and (ii) are equivalent. We need only show that they are equivalent to (3). If n is composite, then write $n = kl$, where k and l are positive integers smaller than n. Then k and l are not 0 in $\mathbb{Z}_n$, and yet $kl = 0$ in $\mathbb{Z}_n$. Thus, $\mathbb{Z}_n$ is not an integral domain. On the other hand, suppose that n is prime. Surely $\mathbb{Z}_n$ is a commutative ring with identity $1 \neq 0$. Suppose we have integers i and j such that $ij = 0$ in $\mathbb{Z}_n$. Then $n \mid ij$. Thus, $n \mid i$ or $n \mid j$. That is, $i = 0$ or $j = 0$ in$\mathbb{Z}_n$. Thus, $\mathbb{Z}_n$ is an integral domain. ■

Theorem 3.1.12 (Wedderburn's Theorem). *Every finite division ring is a field.*

The proof is beyond the scope of this book. The proof can be found in [**?**].

Definition 3.1.13. The characteristic of a ring R is defined to be the least positive integer n such that $nr = 0$ for all $r \in R$. If no such integer exists, then the characteristic of R is defined to be 0.

Theorem 3.1.14. *Let R be a ring with unity 1. If 1 has infinite order under addition, then the characteristic of R is 0. If 1 has order n under addition, then the characteristic of R is n.*

Proof. If 1 has infinite order, then there is no positive integer n such that $n \cdot 1 = 0$, so R has characteristic 0. Now suppose that 1 has additive order n. Then $n \cdot 1 = 0$, and n is the least positive integer with this property. So, for any x in R, we have

$$\begin{aligned} n \cdot x &= x + x + \cdots + x \ (n \text{ summands}) \\ &= 1x + 1x + \cdots + 1x \ (n \text{ summands}) \\ &= (1 + 1 + \cdots + 1)x \ (n \text{ summands}) \\ &= (n \cdot 1)x = 0x = 0. \end{aligned}$$

Thus, R has characteristic n. ■

In the case of an integral domain, the possibilities for the characteristic are severely limited.

Theorem 3.1.15. *The characteristic of an integral domain is either prime or zero.*

Proof. Let D be an integral domain and suppose that the characteristic of D is n with $n \neq 0$. If n is not prime, then $n = ab$, where$1 < a < n$ and $1 < b < n$. Since $0 = n1 = (ab)1 = (a1)(b1)$ and there are no zero divisors in D, either $a1 = 0$ or $b1 = 0$. Hence, the characteristic of D must be less than n, which is a contradiction. Therefore, n must be prime. ∎

Example 3.1.16 (Characteristic of $\mathbb{Z}$, an Infinite Ring)**.** Find the characteristic of the ring of integers $(\mathbb{Z}, +, \cdot)$.

The ring $\mathbb{Z}$ has unity 1. We need to determine if there is a positive integer n such that $n \cdot 1 = 0$. $n \cdot 1 = \underbrace{1 + 1 + \cdots + 1}_{n \text{ times}} = n$. For $n \cdot 1 = 0$ in $\mathbb{Z}$, we would need $n = 0$. However, the definition of characteristic (when it's non-zero) requires n to be a positive integer. There is no positive integer n such that $n = 0$. Therefore, by definition, $\text{char}(\mathbb{Z}) = 0$.

Example 3.1.17 (Characteristic of $\mathbb{Q}$ and $\mathbb{R}$, Infinite Rings)**.** Find the characteristic of the field of rational numbers $(\mathbb{Q}, +, \cdot)$ and the field of real numbers $(\mathbb{R}, +, \cdot)$.

Example 3.1.18 (Characteristic of $\mathbb{Z}_n$, a Finite Ring)**.** Find the characteristic of the ring of integers modulo n, $(\mathbb{Z}_n, +, \cdot)$, where n is a positive integer greater than 1.

Example 3.1.19 (Characteristic of a Product Ring $R_1 \times R_2$)**.** Let $R_1 = \mathbb{Z}_2$ and $R_2 = \mathbb{Z}_3$. Consider the product ring $R = R_1 \times R_2 = \mathbb{Z}_2 \times \mathbb{Z}_3$. Find $\text{char}(R)$.

The ring $R = \mathbb{Z}_2 \times \mathbb{Z}_3$ has elements $([a]_2, [b]_3)$ where $[a]_2 \in \mathbb{Z}_2$ and $[b]_3 \in \mathbb{Z}_3$. Addition and multiplication are component-wise. The zero element is $0_R = ([0]_2, [0]_3)$. The unity element is $1_R = ([1]_2, [1]_3)$. We need to find the smallest positive integer n such that $n \cdot 1_R = 0_R$. $n \cdot 1_R = n \cdot ([1]_2, [1]_3) = (n \cdot [1]_2, n \cdot [1]_3) = ([n]_2, [n]_3)$. For $([n]_2, [n]_3) = ([0]_2, [0]_3)$, we need:

1. $[n]_2 = [0]_2 \implies n \equiv 0 \pmod{2} \implies 2 \mid n$.
2. $[n]_3 = [0]_3 \implies n \equiv 0 \pmod{3} \implies 3 \mid n$.

So n must be a common multiple of 2 and 3. We are looking for the smallest positive such n. This is the least common multiple of 2 and 3. $\text{lcm}(2, 3) = 6$. Let's check $n = 6$: $6 \cdot 1_R = ([6]_2, [6]_3) = ([0]_2, [0]_3) = 0_R$. Any positive integer smaller than 6 will not satisfy both conditions simultaneously. For example, if $n = 2$, $([2]_2, [2]_3) = ([0]_2, [2]_3) \neq 0_R$. If $n = 3$, $([3]_2, [3]_3) = ([1]_2, [0]_3) \neq 0_R$. Thus, $\text{char}(\mathbb{Z}_2 \times \mathbb{Z}_3) = 6$.

In general, if $R = R_1 \times R_2 \times \cdots \times R_k$ is a direct product of rings with unity, and $\text{char}(R_i) = n_i > 0$, then $\text{char}(R) = \text{lcm}(n_1, n_2, \ldots, n_k)$. If any R_i has characteristic 0, then R has characteristic 0 (unless it's a product with a zero ring, which is a trivial case).

Example 3.1.20 (A Ring Without Unity)**.** Consider the ring $2\mathbb{Z}$ (the even integers) under usual addition and multiplication. Find its characteristic.

The ring $2\mathbb{Z} = \{\ldots, -4, -2, 0, 2, 4, \ldots\}$ does not have a multiplicative identity. We use the general definition: $\text{char}(2\mathbb{Z})$ is the smallest positive integer n such that $n \cdot a = 0$ for all $a \in 2\mathbb{Z}$. Suppose such a positive integer n exists. Let $a = 2 \in 2\mathbb{Z}$. Then $n \cdot a = n \cdot 2 = 2n$. For $2n = 0$ (in $\mathbb{Z}$), we must have $n = 0$. Since n must be a positive integer, there is no such n. Therefore, $\text{char}(2\mathbb{Z}) = 0$.

Theorem 3.1.21 (Freshman's Dream). *Let R be a commutative ring of prime characteristic p. Then for any $a, b \in R$, we have $(a + b)^p = a^p + b^p$.*

Proof. Apply the Binomial Theorem. We have

$$(a + b)^p = \sum_{k=0}^{p} \binom{p}{k} a^{p-k} b^k \qquad (1)$$

For $1 < k < p$, $\binom{p}{k} = \dfrac{p!}{(p-k)!k!}$ so that the numerator is divisible by p. However, p does not divide any terms in the denominator. Since R has characteristic p, thus, each term except the first and the last term becomes zero in equation (1). ∎

3.2 Subrings and Ideals

Definition 3.2.1. A subset S of a ring R is a subring of R if S is itself a ring with the operations of R.

Theorem 3.2.2 (Subring Test). *A non-empty subset S of a ring $(R, +, \cdot)$ is a subring of R if and only if for all $a, b \in S$:*

1. *$a - b \in S$ (closed under subtraction)*
2. *$ab \in S$ (closed under multiplication)*

Definition 3.2.3. A subring A of a ring R is called a (two-sided) ideal of R if for every $r \in R$ and every $a \in A$ both ra and ar are in A.

So, a subring A of a ring R is an ideal of R if A "absorbs" elements from R, that is, if $rA = \{ra : a \in A\} \subseteq A$ and $Ar = \{ar : a \in A\} \subseteq A$ for all $r \in R$. An ideal A of R is called a proper ideal of R if A is a proper subset of R.

Theorem 3.2.4. *A nonempty subset A of a ring R is an ideal of R if*

(i) $a - b \in A$ whenever $a, b \in A$.

(ii) ra and ar are in A whenever $a \in A$ and $r \in R$.

Definition 3.2.5. Let R be a commutative ring with unity and let $a_1, a_2, \ldots, a_n \in R$. Then $I = \langle a_1, a_2, \ldots, a_n\rangle = \{r_1a_1 + r_2a_2 + \cdots + r_na_n : r_i \in R\}$ is an ideal of R called the ideal generated by $a_1, a_2, \ldots, a_n$.
Let R be a commutative ring with unity and let $a \in R$. The set $\langle a\rangle = \{ra : r \in R\}$ is an ideal of R called the *principal ideal* generated by a.

Theorem 3.2.6. *Let R be a ring with identity 1_R. If an ideal I of R contains a unit, then $I = R$.*

Corollary 3.2.7. *Let F be a field. Then the only ideals of F are $\{0\}$ and F.*

Theorem 3.2.8. *If I and J are ideals of a ring R, then so is $I + J$.*

Theorem 3.2.9. *Let I and J be ideals in a ring R. Then IJ is also an ideal.*

Definition 3.2.10. Let R be a ring. An element $a \in R$ is said to be nilpotent if $a^k = 0$ for some positive integer k. An ideal N of a ring R is called a **nil ideal** if each element of N is nilpotent.

Theorem 3.2.11 (Existence of Factor Rings). *Let R be a ring and let A be a subring of R. The set of cosets $\{r + A \mid r \in R\}$ is a ring under the operations $(s + A) + (t + A) = s + t + A$ and $(s + A)(t + A) = st + A$ if and only if A is an ideal of R.*

Definition 3.2.12. Let R be a ring. Then a polynomial with coefficients in R is a formal expression

$$a_0 + a_1x + a_2x_2 + \cdots + a_nx_n,$$

where $a_i \in R$ and n is a non-negative integer. Suppose that $b_0 + b_1x + \cdots + b_mx_m$ is also a polynomial with coefficients in R. Without loss of generality, let us say that $n \neq m$. Then these polynomials are equal if and only if $a_i = b_i$ for all $i \leq n$ and $b_i = 0$ for all $i > n$. The set of all polynomials with coefficients in R is denoted $R[x]$.
More details regarding polynomial rings can be found in Section 3.4.

Theorem 3.2.13. *Let R be a ring and I an ideal. Then*

(i) if R is commutative, then so is R/I;

(ii) if R has an identity, then so does R/I; and

(iii) if u is a unit of R, then $u + I$ is a unit of R/I.

Proof. (i) If $a, b \in R$, then $(a + I)(b + I) = ab + I = ba + I = (b + I)(a + I)$. (ii) If $a \in R$, then $(1 + I)(a + I) = a + I = (a + I)(1 + I)$. Hence, $1 + I$ is the identity of R/I. (iii) Observe that $(u + I)(u^{-1} + I) = 1 + I = (u^{-1} + I)(u + I)$; thus, $(u + I)^{-1} = u^{-1} + I$. ■

Theorem 3.2.14. *Let R be a ring with ideals I and J, such that $I \subseteq J$. Then J/I is an ideal of R/I.*

Proof. We see that I is a subring of J, and since I enjoys the absorption property in R, it enjoys it in J as well. Thus, I is an ideal of J, and so J/I makes sense. Now, $0 \in J$, and therefore $0 + I \in J/I$. If $j_1, j_2 \in J$, then $(j_1 + I) - (j_2 + I) = (j_1 - j_2) + I \in J/I$. Also,if $j \in J$ and $r \in R$, then $(r+I)(j+I) = rj + I \in J/I$, since $rj \in J$. Similarly, $(j+I)(r+I) = jr + I \in J/I$. The proof is complete. ■

Definition 3.2.15. A prime ideal A of a commutative ring R is a proper ideal of R such that $a, b \in R$ and $ab \in A$ imply $a \in A$ or $b \in A$. A maximal ideal of a commutative ring R is a proper ideal of R such that, whenever B is an ideal of R and $A \subseteq B \subseteq R$, then $B = A$ or $B = R$.

Theorem 3.2.16. *Let R be a commutative ring with unity and let A be an ideal of R. Then R/A is an integral domain if and only if A is prime.*

Proof. Suppose that R/A is an integral domain and $ab \in A$. Then $(a+A)(b+A) = ab + A = A$, the zero element of the ring R/A. So, either $a + A = A$ or $b + A = A$; that is, either $a \in A$ or $b \in A$. Hence, A is prime. To prove the other half of the theorem, we first observe that R/A is a commutative ring with unity for any proper ideal A. Thus, our task is simply to show that when A is prime, R/A has no zero-divisors. So, suppose that A is prime and $(a+A)(b+A) = 0 + A = A$. Then $ab \in A$ and, therefore, $a \in A$ or $b \in A$. Thus, one of $a + A$ or $b + A$ is the zero coset in R/A. ■

Theorem 3.2.17. *Let R be a commutative ring with unity and let A be an ideal of R. Then R/A is a field if and only if A is maximal.*

Proof. Suppose that R/A is a field and B is an ideal of R that properly contains A. Let $b \in B$ but $b \notin A$. Then $b + A$ is a nonzero element of R/A and, therefore, there exists an element $c + A$ such that $(b+A)? \cdot (c+A) = 1 + A$, the multiplicative identity of R/A. Since $b \in B$, we have $bc \in B$. Because

$$1 + A = (b+A)(c+A) = bc + A,$$

we have $1 - bc \in A, \subset B$. So, $1 = (1 - bc) + bc \in B$. Thus, $B = R$. This proves that A is maximal. Now suppose that A is maximal and let $b \in R$ but $b \notin A$. It suffices to show that $b + A$ has a multiplicative inverse. (All other properties for a field follow trivially.) Consider $B = \{br + a \,:\, r \in R, a \in A\}$. This is an ideal of R that properly contains A. Since A is maximal, we must have $B = R$. Thus, $1 \in B$, say, $1 = bc + a'$, where $a' \in A$. Then

$$1 + A = bc + a' + A = bc + A = (b+A)(c+A).$$

■

When a commutative ring has a unity, it follows from Theorems 3.2.16 and 3.2.17 that a maximal ideal is a prime ideal. The next example shows that a prime ideal need not be maximal.

Example 3.2.18. The ideal $\langle x \rangle$ is a prime ideal in $\mathbb{Z}[x]$ but not a maximal ideal in $\mathbb{Z}[x]$. To verify this, we begin with the observation that $\langle x \rangle = \{f(x) \in \mathbb{Z}[x] \,:\, f(0) = 0\}$. Thus, if $g(x)h(x) \in \langle x \rangle$, then $g(0)h(0) = 0$. And since $g(0)$ and $h(0)$ are integers, we have $g(0) = 0$ or $h(0) = 0$. To see that $\langle x \rangle$ is not maximal, we simply note that $\langle x \rangle \subset \langle x, 2 \rangle \subset \mathbb{Z}[x]$.

3.3 Ring Homomorphisms and Embedding

Definition 3.3.1. If R and S are rings, then a ring homomorphism is a map $\phi : R \to S$ satisfying

$$\phi(a+b) = \phi(a) + \phi(b) \text{ and } \phi(ab) = \phi(a)\phi(b)$$

for all $a, b \in R$. If a ring homomorphism $\phi : R \to S$ is a bijection, then ϕ is called an isomorphism of rings. When such an isomorphism exists, we say that R and S are isomorphic rings.

Example 3.3.2. Consider the function $\phi : \mathbb{Z}_{15} \to \mathbb{Z}_3 \times \mathbb{Z}_5$ given by $\phi(x) = (x, x)$ for any $x \in \mathbb{Z}_{15}$. Then $\phi(x+y) = (x+y, x+y) = (x, x) + (y, y) = \phi(x) + \phi(y)$ and $\phi(xy) = (xy, xy) = (x, x)(y, y) = \phi(x)\phi(y)$. Thus, $\mathbb{Z}_{15}$ and $\mathbb{Z}_3 \times \mathbb{Z}_5$ are isomorphic rings.

We must not, however, make the mistake of thinking that rings that are isomorphic as additive groups are necessarily isomorphic as rings!

Example 3.3.3. Let $R = \mathbb{Z}$ and $S = n\mathbb{Z}, n \notin \{1, -1\}$, then R and S are isomorphic as groups but not isomorphic as rings.

The kernel and image of a homomorphism.

Definition 3.3.4. Let ϕ be a ring homomorphism from a ring R to a ring S. Then $Ker\phi = \{r \in R \ : \ f(r) = 0\}$ is an ideal of R. The image of ϕ is the subset $Im(\phi) = \{s \in S \ : \ \exists\, r \in R$ s.t. $\phi(r) = s\} \subseteq S$ is a subring of S.

Theorem 3.3.5. *Let $\phi : R \to S$ be a ring homomorphism. Then*

(i) $ker(\phi)$ is an ideal of R;

(ii) ϕ is one-to-one if and only if $ker(\phi) = \{0\}$;

(iii) $\phi(0) = 0$; and

(iv) if R is a ring with identity, then so is $\phi(R)$, and $\phi(1)$ is the identity of $\phi(R)$.

3.3.1 Ring Isomorphism Theorems

Theorem 3.3.6 (First Isomorphism Theorem for Rings). *Let ϕ be a ring homomorphism from R to S. Then the mapping from $R/Ker\phi$ to $\phi(R)$, given by $r + Ker\phi \to \phi(r)$, is an isomorphism. In symbols, $R/Ker\phi \cong \phi(R)$.*

Proof. Let $K = ker(\phi)$. The map $\psi : R/K \to \phi(R)$ via $\psi(a + K) = \phi(a)$ is a ring homomorphism as follows. $\psi(a+b+K) = \phi(a+b) = \phi(a)+\phi(b) = \psi(a+K)+\psi(b+K)$. Similarly, $\psi((a+K)(b+K)) = \psi(ab + K) = \phi(ab) = \phi(a)\phi(b) = \psi(a + K)\psi)b + K)$, as required. ■

Theorem 3.3.7 (Second Isomorphism Theorem for Rings). *Let A and B be two ideals of a ring R, then $\dfrac{A+B}{A} \cong \dfrac{B}{A \cap B}$.*

Proof. Define $\phi : B \to (A + B)/A$ via $\phi(b) = b + A$, for all $b \in B$. Then $\phi(b_1 + b_2) = b_1 + b_2 + A = (b_1 + A) + (b_2 + A) = \phi(b_1) + \phi(b_2)$, so that ϕ is an onto homomorphism of additive groups with kernel $ker(\phi) = \{b \in B \ : \ \phi(b) = A \text{ i.e. } b \in A\} = A \cap B$. In view of the First Isomorphism Theorem for Rings, it suffices to show that ϕ respects multiplication. But for any $b_1, b_2 \in B$, we have $\phi(b_1b_2) = b_1b_2 + A = (b_1 + A)(b_2 + A) = \phi(b_1)\phi(b_2)$. The proof is complete. ■

Theorem 3.3.8 (Third Isomorphism Theorem for Rings). *Let $A \subseteq B$ be two ideals of a ring R then $R/B \cong (R/A)(B/A)$.*

Proof. Define $\phi : R/A \to R/B$ via $\phi(x + A) = x + B$, for all $a \in R$. Then ϕ is an onto additive group homomorphism with kernel B/A. It remains only to show that ϕ respects multiplication, for then the First Isomorphism Theorem completes the proof. Take $x, y \in R$. Then

$$\phi((x + A)(y + A)) = \phi(xy + A) = xy + B = (x + B)(y + B) = \phi(x + A)\phi(y + A).$$

We are done. ■

Given a ring R, we might well ask if it is a subring of a field. For example, $\mathbb{Z}$ is a subring of $\mathbb{Q}$. But not every ring can be a subring of a field. Indeed, a noncommutative ring or a ring containing zero divisors cannot exist inside a field. So, it seems that integral domains are a good place to start. In fact, if R is an integral domain, then we can construct a field F containing an isomorphic copy of R. The method we will use may seem somewhat familiar; actually, it is exactly the way in which $\mathbb{Q}$ is constructed from $\mathbb{Z}$. We will need something comparable to numerators and denominators. Also, the denominator must not be zero. Furthermore, we need some way of recognizing that $10/25 = 8/20$, for instance. This gives us an idea of how to proceed.
Let R be an integral domain. Then let S be the Cartesian product $R \times (R - \{0\})$. Note that S is only a set, not a ring. (If $R = \mathbb{Z}$, for instance, when we look at $(10, 25) \in S$, we are thinking of the fraction

10/25.) Let us define a relation $\sim$ on S via $(a, b) \sim (c, d)$ if and only if $ad = bc$. (Continuing our parenthetical thought, $(10, 25) \sim (8, 20)$.) Then $\sim$ is an equivalence relation. Let us write F for the set of all equivalence classes of S. The addition and multiplication operations on F work precisely as we would expect with fractions. Specifically,

$$[a, b] + [c, d] = [ad + bc, bd] \text{ and } [a, b][c, d] = [ac, bd].$$

We must verify that these operations are well-defined. Suppose that $[a_1, b_1] = [a, b]$ and $[c_1, d_1] = [c, d]$. Then $[a_1, b_1] + [c_1, d_1] = [a_1d_1 + b_1c_1, b_1d_1]$. But $(ad + bc)(b_1d_1) - (a_1d_1 + b_1c_1)bd = dd_1(ab_1 - a_1b) + bb_1(cd_1 - c_1d) = dd_1(0) + bb_1(0) = 0$; thus, $[a, b] + [c, d] = [a_1, b_1] + [c_1, d_1]$. Similarly, $[a_1, b_1][c_1, d_1] = [a_1c_1, b_1d_1]$, and

$$\begin{aligned} acb_1d_1 - a_1c_1bd &= (acb_1d_1 - a_1cbd_1) + (a_1cbd_1 - a_1c_1bd) \\ &= cd_1(ab_1 - a_1b) + a_1b(cd_1 - c_1d) \\ &= cd_1(0) + a_1b(0) \\ &= 0; \end{aligned}$$

thus, $[a, b][c, d] = [a_1, b_1][c_1, d_1]$. Also, we must note that if b and d are nonzero, then so is bd, since R is an integral domain.

Definition 3.3.9. Let R be an integral domain. Then the field of fractions (or field of quotients) of R is the field F constructed above.

Example 3.3.10 (Equivalence of Fractions)**.** Let $D = \mathbb{Z}$. In the construction of the field of quotients $F_D = \mathbb{Q}$, show that the fraction $\frac{2}{3}$ is equivalent to $\frac{4}{6}$, but not equivalent to $\frac{3}{5}$.

The set of formal fractions is $S = \mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$. The equivalence relation is $(a, b) \sim (c, d)$ if $ad = bc$. We represent $\frac{a}{b}$ by the equivalence class of (a, b).

1. To show $\frac{2}{3}$ is equivalent to $\frac{4}{6}$: We check if $(2, 3) \sim (4, 6)$. This requires $a \cdot d = b \cdot c$, so $2 \cdot 6 = 3 \cdot 4$. $2 \cdot 6 = 12$. $3 \cdot 4 = 12$. Since $12 = 12$, we have $(2, 3) \sim (4, 6)$. Thus, $\frac{2}{3} = \frac{4}{6}$ in $\mathbb{Q}$.

2. To show $\frac{2}{3}$ is not equivalent to $\frac{3}{5}$: We check if $(2, 3) \sim (3, 5)$. This would require $2 \cdot 5 = 3 \cdot 3$. $2 \cdot 5 = 10$. $3 \cdot 3 = 9$. Since $10 \neq 9$, $(2, 3)$ is not equivalent to $(3, 5)$. Thus, $\frac{2}{3} \neq \frac{3}{5}$ in $\mathbb{Q}$.

Example 3.3.11 (Operations in the Field of Quotients)**.** Let $D = \mathbb{Z}$, so $F_D = \mathbb{Q}$. Compute $\frac{2}{5} + \frac{3}{4}$ and $\frac{2}{5} \cdot \frac{3}{4}$ using the definitions for operations in the field of quotients.

1. Addition: $\frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd}$. Here $a = 2, b = 5, c = 3, d = 4$. $\frac{2}{5} + \frac{3}{4} = \frac{(2)(4)+(5)(3)}{(5)(4)} = \frac{8+15}{20} = \frac{23}{20}$.

2. Multiplication: $\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}$. Here $a = 2, b = 5, c = 3, d = 4$. $\frac{2}{5} \cdot \frac{3}{4} = \frac{(2)(3)}{(5)(4)} = \frac{6}{20}$. We can simplify $\frac{6}{20}$. In $\mathbb{Q}$, $\frac{6}{20} = \frac{3}{10}$ because $(6, 20) \sim (3, 10)$ as $6 \cdot 10 = 60$ and $20 \cdot 3 = 60$.

Example 3.3.12 (Field of Quotients of $F[x]$). Let F be a field. The ring of polynomials $D = F[x]$ is an integral domain. Describe its field of quotients F_D.

Let $D = F[x]$. D is an integral domain (since F is a field, the product of two non-zero polynomials is non-zero). The field of quotients F_D consists of equivalence classes of pairs $(p(x), q(x))$ where $p(x), q(x) \in F[x]$ and $q(x) \neq 0$ (the zero polynomial). The equivalence relation is $(p_1(x), q_1(x)) \sim (p_2(x), q_2(x))$ if $p_1(x)q_2(x) = p_2(x)q_1(x)$ in $F[x]$. We denote the equivalence class of $(p(x), q(x))$ as $\dfrac{p(x)}{q(x)}$. The set of these equivalence classes is denoted by $F(x)$, called the **field of rational functions** in x with coefficients in F.

Addition is defined as: $\dfrac{p_1(x)}{q_1(x)} + \dfrac{p_2(x)}{q_2(x)} = \dfrac{p_1(x)q_2(x) + p_2(x)q_1(x)}{q_1(x)q_2(x)}$.

Multiplication is defined as: $\dfrac{p_1(x)}{q_1(x)} \cdot \dfrac{p_2(x)}{q_2(x)} = \dfrac{p_1(x)p_2(x)}{q_1(x)q_2(x)}$.

For example, if $F = \mathbb{R}$, then elements of $\mathbb{R}(x)$ are like $\frac{x^2+1}{x-3}$ or $\dfrac{5}{x^3+x+1}$. The zero element is $\dfrac{0}{1}$ (where 0 is the zero polynomial and 1 is the constant polynomial 1). The unity element is $\dfrac{1}{1}$. The inverse of a non-zero element $\frac{p(x)}{q(x)}$ (where $p(x) \neq 0$) is $\dfrac{q(x)}{p(x)}$.

Theorem 3.3.13. *Let R be an integral domain. Then the field of fractions, F, is a field.*

Proof. The proof of this theorem is not difficult. However, there are many steps to complete, as we must verify that F is an abelian group under addition, then that it has all of the remaining properties of a ring, and finally that it is a field. We will prove a few selected properties, and leave the rest to the reader.
Take any $a, b, c, d, e, f \in R$ with b, d and f all non-zero. Let us show that the addition operation on F is associative. We have

$$\begin{aligned}([a, b] + [c, d]) + [e, f] &= [ad + bc, bd] + [e, f] \\ &= [adf + bcf + bde, bdf] \\ &= [a, b] + [cf + de, df] \\ &= [a, b] + ([c, d] + [e, f]),\end{aligned}$$

as required.
Next, let us prove a distributive law. Observe that
$[a, b]([c, d] + [e, f]) = [a, b][cf + de, df] = [acf + ade, bdf]$
whereas
$[a, b][c, d]+[a, b][e, f] = [ac, bd]+[ae, bf] = [abcf +abde, b^2df]$. But as $(acf +ade)b^2df = (abcf +abde)bdf$, these are equal.
Notice that $[0, 1]$ is the additive identity of F and $[1, 1]$ is the multiplicative identity. Let us show that every nonzero element of F has an inverse. Take a nonzero $[a, b] \in F$. Note that $[0, b] = [0, 1]$, so $a \neq 0$. But then $[b, a] \in F$ as well, and $[a, b][b, a] = [ab, ab] = [1, 1]$. Thus, $[b, a] = [a, b] - 1$. ■

Theorem 3.3.14. *Let D be an integral domain. Then there exists a field F (called the field of quotients of D) that contains a subring isomorphic to D.*

Proof. Let $S = \{(a, b) \ :\ a, b \in D,\ b \neq 0\}$. We define an equivalence relation on S by $(a, b) \equiv (c, d)$ if $ad = bc$. Now, let F be the set of equivalence classes of S under the relation $\equiv$ and denote the equivalence class that contains (x, y) by x/y. We define addition and multiplication on F by

$$a/b + c/d = (ad + bc)/(bd) \text{ and } a/b \cdot c/d = (ac)/(bd).$$

Since there are many representations of any particular element of F (just as in the rationals, we have 1/2 5 3/6 5 4/8), we must show that these two operations are well defined. To do this, suppose that $a/b = a'/b'$ and $c/d = c'/d'$, so that $ab' = a'b$ and $cd' = c'd$. It then follows that

$$\begin{aligned}(ad + bc)b'd' = adb'd' + bcb'd' &= (ab')dd' + (cd')bb' = (a'b)dd' + (c'd)bb' \\ = a'd'bd + b'c'bd &= (a'd' + b'c')bd.\end{aligned}$$

Thus, by definition, we have

$$(ad + bc)/(bd) = (a'd' + b'c')/(b'd'),$$

and, therefore, addition is well defined. We leave the verification that multiplication is well defined as an exercise. That F is a field is straightforward. Let 1 denote the unity of D. Then $0/1$ is the additive identity of F. The additive inverse of a/b is $-a/b$; the multiplicative inverse of a nonzero element a/b is b/a. The remaining field properties can be checked easily. Finally, the mapping $\phi : D \to F$ given by $x \to x/1$ is a ring isomorphism from D to $\phi(D)$. ■

Example 3.3.15. Let $R = \{a + b\sqrt{2} : a, b \in \mathbb{Z}\}$. Then R is an integral domain, and its field of fractions is isomorphic to $\{a + b\sqrt{2} : a, b \in \mathbb{Q}\}$.

Definition 3.3.16. Embedding of a ring R into a ring S means existence of a subring S' in S such that $R \cong S'$.

Theorem 3.3.17. *Every ring can be embedded into a ring with unity.*

Proof. Let R be a ring. Consider, $R \times \mathbb{Z} = \{(r, z) : r \in R, z \in \mathbb{Z}\}$. Then $R \times \mathbb{Z}$ is a ring with unity $(0, 1)$ under the operations

$$(r, n) + (s, m) = (r + s, n + m) \text{ and } (r, n)(s, m) = (rs + ns + mr, nm)$$

The mapping $\phi : R \to R \times \mathbb{Z}$ defined by $\phi(r) = (r, 0)$ is a one-one homomorphism. Thus, $R \cong \phi(R) \subseteq R \times \mathbb{Z}$. ■

Theorem 3.3.18. *An integral domain can be embedded into a field.*

Proof. Proved in Theorem 3.3.14. ■

Example 3.3.19 (Embedding an Integral Domain)**.** Let $D = \mathbb{Z}$. Illustrate the embedding map $\phi : \mathbb{Z} \to \mathbb{Q}$. Show that $\phi(2 + 3) = \phi(2) + \phi(3)$ and $\phi(2 \cdot 3) = \phi(2)\phi(3)$.

The embedding map is $\phi : D \to F_D$ defined by $\phi(a) = \frac{a}{1}$. For $D = \mathbb{Z}$ and $F_D = \mathbb{Q}$, the map is $\phi(n) = \frac{n}{1}$ for $n \in \mathbb{Z}$.

1. Check $\phi(2+3) = \phi(2)+\phi(3)$: LHS: $2+3 = 5$. So $\phi(2+3) = \phi(5) = \frac{5}{1}$. RHS: $\phi(2) = \frac{2}{1}$ and $\phi(3) = \frac{3}{1}$. $\phi(2)+\phi(3) = \frac{2}{1}+\frac{3}{1}$. Using the addition rule for fractions $\frac{a}{b}+\frac{c}{d} = \frac{ad+bc}{bd}$: $\frac{2}{1}+\frac{3}{1} = \frac{2\cdot1+1\cdot3}{1\cdot1} = \frac{2+3}{1} = \frac{5}{1}$. Since LHS = RHS, the additive homomorphism property holds for these elements.
2. Check $\phi(2 \cdot 3) = \phi(2)\phi(3)$: LHS: $2 \cdot 3 = 6$. So $\phi(2 \cdot 3) = \phi(6) = \frac{6}{1}$. RHS: $\phi(2) = \frac{2}{1}$ and $\phi(3) = \frac{3}{1}$. $\phi(2)\phi(3) = \frac{2}{1} \cdot \frac{3}{1}$. Using the multiplication rule for fractions $\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}$: $\frac{2}{1} \cdot \frac{3}{1} = \frac{2\cdot3}{1\cdot1} = \frac{6}{1}$. Since LHS = RHS, the multiplicative homomorphism property holds for these elements.

The map ϕ identifies an integer n with the rational number $\frac{n}{1}$. For instance, the integer 5 is identified with the rational number usually written as 5. This shows how $\mathbb{Z}$ is "contained within" $\mathbb{Q}$ as a subring (more precisely, $\phi(\mathbb{Z})$ is a subring of $\mathbb{Q}$ isomorphic to $\mathbb{Z}$).

Example 3.3.20 (Why a field is its own field of quotients)**.** Let $F = \mathbb{Q}$ (which is already a field). Describe its field of quotients F_F.

Since $F = \mathbb{Q}$ is an integral domain (in fact, a field), we can construct its field of quotients, let's call it $K = F_{\mathbb{Q}}$. Elements of K are of the form $\frac{a}{b}$ where $a, b \in \mathbb{Q}$ and $b \neq 0_{\mathbb{Q}}$. The operations are: $\frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd}$ $\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}$

Consider an element $\frac{a}{b} \in K$. Since $a, b \in \mathbb{Q}$ and $b \neq 0_{\mathbb{Q}}$, b has a multiplicative inverse $b^{-1} \in \mathbb{Q}$. The element $a \cdot b^{-1}$ is an element of $\mathbb{Q}$. Let's show that the fraction $\frac{a}{b}$ in K corresponds to the element $a \cdot b^{-1}$ in $\mathbb{Q}$. The embedding map $\phi : \mathbb{Q} \to K$ is given by $\phi(x) = \frac{x}{1_{\mathbb{Q}}}$. We can define a map $\psi : K \to \mathbb{Q}$ by $\psi\left(\frac{a}{b}\right) = a \cdot b^{-1}$.

- Is ψ well-defined? Suppose $\frac{a}{b} = \frac{c}{d}$ in K. This means $ad = bc$ (as elements of $\mathbb{Q}$). We want to show $a \cdot b^{-1} = c \cdot d^{-1}$. Since $ad = bc$, multiply by $b^{-1}d^{-1}$ (which exist in $\mathbb{Q}$): $ad(b^{-1}d^{-1}) = bc(b^{-1}d^{-1})$ $a(dd^{-1})b^{-1} = c(bb^{-1})d^{-1}$ $a \cdot 1_{\mathbb{Q}} \cdot b^{-1} = c \cdot 1_{\mathbb{Q}} \cdot d^{-1}$ $ab^{-1} = cd^{-1}$. So ψ is well-defined.
- Is ψ a homomorphism? (This is routine to check).
- Is ψ injective? If $\psi\left(\frac{a}{b}\right) = 0_{\mathbb{Q}}$, then $ab^{-1} = 0_{\mathbb{Q}}$. Since $b^{-1} \neq 0_{\mathbb{Q}}$, this implies $a = 0_{\mathbb{Q}}$. So $\frac{a}{b} = \frac{0_{\mathbb{Q}}}{b} = \frac{0_{\mathbb{Q}}}{1_{\mathbb{Q}}}$, which is the zero element in K.
- Is ψ surjective? For any $x \in \mathbb{Q}$, we can write $x = \frac{x}{1_{\mathbb{Q}}}$. Then $\psi\left(\frac{x}{1_{\mathbb{Q}}}\right) = x \cdot (1_{\mathbb{Q}})^{-1} = x \cdot 1_{\mathbb{Q}} = x$. So ψ is surjective.

Thus, ψ is an isomorphism from $K = F_{\mathbb{Q}}$ to $\mathbb{Q}$. This demonstrates that the field of quotients of a field F is isomorphic to F itself. The embedding $\phi(a) = \frac{a}{1}$ is, in this case, an isomorphism.

3.4 Special Types of Domains

Recall the Definition 3.2.12 of a polynomial ring.

Definition 3.4.1. Let R be a ring and let $f(x) = a_0 + a_1x + \cdots + a_nx_n \in R[x]$. Further suppose that $a_m \neq 0$ but $a_k = 0$ for all $k > m$. Then the degree of $f(x)$ is m, and we write $\deg(f(x)) = m$. The leading term of $f(x)$ is a_mx_m, and the leading coefficient is a_m . Note that the zero polynomial, 0, has no degree, leading term or leading coefficient. A constant polynomial has degree 0. If R has an identity, then $f(x)$ is monic if its leading coefficient is 1.

We wish to make $R[x]$ into a ring, and so we need addition and multiplication operations. These will be exactly the same as for real polynomials. Let $f(x) = a_0 + a_1x + \cdots + a_nx_n$ and $g(x) = b_0 + b_1x + \cdots + b_mx_m$. Adding in terms with zero coefficients if necessary, we may assume that $m = n$. Then

$$f(x) + g(x) = (a_0 + b_0) + (a_1 + b_1)x + \cdots + (a_n + b_n)x_n.$$

Similarly,

$$f(x)g(x) = c_0 + c_1x + c_2x^2 + \cdots + c_{m+n}x^{m+n},$$

where

$$c_i = a_0b_i + a_1b_{i-1} + a_2b_{i-2} + \cdots + a_ib_0.$$

Here, we take $a_j = 0$ if $j > n$ and $b_j = 0$ if $j > m$.

Theorem 3.4.2. *Let R be an integral domain, and let $f(x)$ and $g(x)$ be nonzero polynomials in $R[x]$, of degrees m and n respectively. Then*

(i) $\deg(f(x) + g(x)) \leq \max\{m, n\}$ *(or $f(x) + g(x) = 0$);and*

(ii) $\deg(f(x)g(x)) = m + n$.

Theorem 3.4.3. *Let R be a ring. Then*

(i) $R[x]$ is a ring.

(ii) if R has an identity, then so does $R[x]$; and

(iii) if R is commutative, then so is $R[x]$.

(iv) If R is an integral domain, then so is $R[x]$.

Proof. (i). Let us show that $R[x]$ is an abelian group under addition. Clearly the sum of two polynomials is a polynomial. Let $f(x) = a_0 + \cdots + a_nx^n$, $g(x) = b_0 + \cdots + b_mx^m$ and $h(x) = c_0 + \cdots + c_kx^k$. Then the coefficient of x^i in $f(x) + g(x)$ is $a_i + b_i$, and similarly for $g(x) + f(x)$ (adding in terms with zero coefficients if necessary). Thus, addition is commutative. In the same way, because the addition of coefficients is associative, addition in $R[x]$ is associative. The zero polynomial is the additive identity, and $-f(x) = -a_0 - \cdots - a_nx^n$. Therefore, $R[x]$ is an abelian group under addition.
Evidently, the product of two polynomials is a polynomial. Let us check a distributive law. The coefficient of x^i in $f(x)(g(x) + h(x))$ is $a_0(b_i + c_i) + a_1(b_{i-1} + c_{i-1}) + \cdots + a_i(b_0 + c_0)$.
But this is $(a_0b_i + \cdots + a_ib_0) + (a_0c_i + \cdots + a_ic_0)$, which is the coefficient of x^i in $f(x)g(x) + f(x)h(x)$.
The other distributive law is proved similarly. Finally, we must check that multiplication is associative. But by repeated application of the distributive laws, we see that we may reduce to proving that $(a_ux^ub_vx^v)c_wx^w = a_ux^u(b_vx^vc_wx^w)$, for all $a_u, b_v, c_w \in R$ and all $u, v, w \geq 0$. However, both sides of this equation are equal to aubvcwxu+v+w, and the proof is complete.
(ii). The constant polynomial 1 is the identity.
(iii). Repeatedly applying the distributive laws, we see that we need only check that a_ix^i and b_jx^j commute, where $a_i, b_j \in R$ and $i, j \geq 0$. But $a_ix^ib_jx^j = a_ib_jx^{i+j}$ and $b_jx^ja_ix^i = b_ja_ix^{i+j}$. Since R is commutative, these are equal.
(iv). The product of nonzero polynomials cannot be the zero polynomial in an integral domain. ∎

3.4.1 Division Algorithm and Associated Results

Theorem 3.4.4. (Division Algorithm for Polynomials). *Let F be a field, and let $f(x), g(x) \in F[x]$, with $g(x) \neq 0$. Then there exist unique $q(x), r(x) \in F[x]$ such that*

$$f(x) = g(x)q(x) + r(x)$$

with either $r(x) = 0$ or $\deg[r(x)] < \deg[g(x)]$.

Proof. Let us verify the existence of $q(x)$ and $r(x)$. If $f(x) = 0$, there is nothing to do; indeed, we let $q(x) = r(x) = 0$. Therefore, assume that $f(x)$ is not the zero polynomial. We proceed by strong induction on $\deg(f(x))$. Suppose that $\deg(f(x)) = 0$. If $\deg(g(x)) > 0$, then use $q(x) = 0$ and $r(x) = f(x)$. On the other hand, if $\deg(g(x)) = 0$, then $g(x) = b$ is a non-zero constant in F. As F is a field, we have $b^{-1} \in F$, and we can use $q(x) = b^{-1}f(x)$ and $r(x) = 0$.

Thus, suppose that $\deg(f(x)) = n > 0$ and that our result holds for polynomials of smaller degree. Let us write $f(x) = a_0 + a_1x + \cdots + a_nx^n$. Also suppose that $\deg(g(x)) = m$, and write $g(x) = b_0 + b_1x + \cdots + b_mx^m$. If $n < m$, then we can use $q(x) = 0$ and $r(x) = f(x)$. Otherwise, notice that in $f(x) - g(x)b_m^{-1}a_nx^{n-m}$, no term of degree greater than n appears, and the coefficient of x^n is $a_nb_mb_m^{-1}a_n = 0$; thus, either $f(x) - g(x)b_m^{-1}a_nx^{n-m}$ is the zero polynomial, or it has degree strictly smaller than $f(x)$. By our inductive hypothesis, there exist $q(x), r(x) \in F[x]$ such that $f(x) - g(x)b^{-1}a_nx^{n-m} = g(x)q(x) + r(x)$, with $r(x) = 0$ or $\deg(r(x)) < \deg(g(x))$. But then $f(x) = g(x)(q(x) + b^{-1}a_nx^{n-m}) + r(x)$, as required.

Now for uniqueness. Suppose that $f(x) = g(x)q(x) + r(x) = g(x)q_1(x) + r_1(x)$, with $q(x), q_1(x)$, $r(x), r_1(x) \in F[x]$ and each of $r(x)$ and $r_1(x)$ either is 0 or has degree smaller than that of $g(x)$. Then $g(x)(q(x) - q_1(x)) = r_1(x) - r(x)$. Suppose that $q(x) \neq q_1(x)$. Then $\deg(g(x)(q(x) - q_1(x))) \geq \deg(g(x))$, but $r_1(x) - r(x)$ cannot possibly have a degree that large. Thus, $q(x) = q_1(x)$ and hence $r(x) = r_1(x)$. ■

Example 3.4.5 (Polynomial Division in $\mathbb{Z}_5[x]$)**.** Divide $f(x) = 3x^4+2x^3+4x^2+x+2$ by $g(x) = 2x^2+x+3$ in $\mathbb{Z}_5[x]$.

We perform polynomial long division in $\mathbb{Z}_5[x]$. All coefficient arithmetic is modulo 5. We need $2^{-1} \equiv 3 \pmod 5$.

$$
\begin{array}{r l l}
 & \mathbf{4x^2} \quad +\mathbf{4x} \quad +\mathbf{4} & \\
2x^2+x+3 & \overline{3x^4+2x^3+4x^2+x+2} & \\
 & \underline{-(3x^4 \;\; +4x^3 \;\; +2x^2)} & (4x^2 \cdot (2x^2+x+3) \equiv 3x^4+4x^3+2x^2) \\
 & \quad 0x^4-2x^3 \;\; +2x^2 \;\; +x & \\
 & \quad \equiv \mathbf{3x^3} \;\; +\mathbf{2x^2} \;\; +\mathbf{x} & \\
 & \quad \underline{-(3x^3 \;\; +4x^2 \;\; +2x)} & (4x \cdot (2x^2+x+3) \equiv 3x^3+4x^2+2x) \\
 & \qquad 0x^3-2x^2 \; -x \; +2 & \\
 & \qquad \equiv \mathbf{3x^2} \;\; +\mathbf{4x} \; +\mathbf{2} & \\
 & \qquad \underline{-(3x^2 \;\; +4x \; +2)} & (4 \cdot (2x^2+x+3) \equiv 3x^2+4x+2) \\
 & \qquad\qquad \mathbf{0} &
\end{array}
$$

The quotient is $q(x) = 4x^2 + 4x + 4$. The remainder is $r(x) = 0$. Since $r(x) = 0$, the condition $\deg(r(x)) < \deg(g(x))$ (or $r(x) = 0$) is satisfied. Thus, in $\mathbb{Z}_5[x]$: $3x^4 + 2x^3 + 4x^2 + x + 2 = (2x^2 + x + 3)(4x^2 + 4x + 4)$.

Exercise 3.4.6. In $\mathbb{Z}_{11}[x]$, let $f(x) = 2x^3 + 4x^2 + 2x + 5$ and $g(x) = 2x^4 + 5x^3 + 7x + 1$. Find $f(x) - g(x)$ and $f(x)g(x)$. Also apply division alogorithm to find $q(x)$ and $r(x)$.

Definition 3.4.7 (Euclidean function). Let R be an integral domain. Then a Euclidean function is a function δ from the set of non-zero elements of R to the non-negative integers such that, for all nonzero $a, b \in R$, we have
1. $\delta(a) \leq \delta(ab)$;and
2. there exist $q, r \in R$ such that $a = bq + r$, and either $r = 0$ or $\delta(r) < \delta(b)$.

Definition 3.4.8. A Euclidean domain is an integral domain having a Euclidean function.

Or we can define a Euclidean domain as follows:

Definition 3.4.9 (**Euclidean Domain).** An integral domain D is called a **Euclidean Domain** (ED) if there exists a function $d : D \setminus \{0\} \to \mathbb{N}_0 = \{0, 1, 2, \dots\}$ (called a Euclidean function or norm) satisfying:

1. For all non-zero $a, b \in D$, $d(a) \leq d(ab)$.

2. For any $a, b \in D$ with $b \neq 0$, there exist $q, r \in D$ (quotient and remainder) such that $a = bq + r$, where either $r = 0$ or $d(r) < d(b)$. (Division Algorithm property)

Example 3.4.10. The integers form a Euclidean domain. We already know that $\mathbb{Z}$ is an integral domain. Define $\delta(a) = |a|$. If a and b are nonzero integers, then $|ab| = |a||b| \geq |a|$. Furthermore, by the division algorithm, there exist $q, r \in \mathbb{Z}$ such that $a = |b|q + r$, with $0 \leq r < |b|$. If $b > 0$, we are done. Otherwise, simply note that $a = b(-q) + r$.

Example 3.4.11 (The Integers $\mathbb{Z}$ as an ED)**.** Show that the ring of integers $\mathbb{Z}$ is a Euclidean Domain with the Euclidean function $d(n) = |n|$ for $n \neq 0$.

We know $\mathbb{Z}$ is an integral domain. Let $d(n) = |n|$ for $n \in \mathbb{Z} \setminus \{0\}$.

1. **Condition 1:** $d(a) \leq d(ab)$ **for non-zero** $a, b \in \mathbb{Z}$**.** Let $a, b \in \mathbb{Z}$ with $a \neq 0, b \neq 0$. Then $d(a) = |a|$ and $d(ab) = |ab| = |a||b|$. Since $b \neq 0$, b is an integer, so $|b| \geq 1$. Therefore, $|a||b| \geq |a| \cdot 1 = |a|$. So, $d(ab) \geq d(a)$. This condition holds.

2. **Condition 2: Division Algorithm.** Let $a, b \in \mathbb{Z}$ with $b \neq 0$. By the standard Division Algorithm for integers, there exist unique integers q (quotient) and r (remainder) such that $a = bq + r$, where $0 \leq r < |b|$. If $r = 0$, we are done. If $r \neq 0$, then $d(r) = |r| = r$. Since $0 < r < |b|$, we have $d(r) < |b| = d(b)$. Thus, the division algorithm property holds.

Since both conditions are satisfied, $\mathbb{Z}$ is a Euclidean Domain with the Euclidean function $d(n) = |n|$.

Example 3.4.12 (A Field F as an ED)**.** Show that any field F is a Euclidean Domain.

Let F be a field. A field is an integral domain. Define the Euclidean function $d : F \setminus \{0\} \to \mathbb{N}_0$ by $d(a) = 0$ for all $a \in F \setminus \{0\}$. (Other constant positive values like $d(a) = 1$ would also work).

1. **Condition 1:** $d(a) \leq d(ab)$ **for non-zero** $a, b \in F$**.** Let $a, b \in F$ with $a \neq 0, b \neq 0$. Then $d(a) = 0$. Since F is a field and $a, b \neq 0$, $ab \neq 0$. So $d(ab) = 0$. Thus, $d(a) = 0 \leq 0 = d(ab)$. This condition holds.

2. **Condition 2: Division Algorithm.** Let $a, b \in F$ with $b \neq 0$. Since F is a field and $b \neq 0$, b has a multiplicative inverse $b^{-1} \in F$. We can write $a = b(ab^{-1}) + 0$. Let $q = ab^{-1}$. Then $q \in F$ because $a, b^{-1} \in F$ and F is closed under multiplication. Let $r = 0$. So we have $a = bq + r$ with $r = 0$. The condition for the remainder is "either $r = 0$ or $d(r) < d(b)$". Since $r = 0$, the condition is satisfied.

Since both conditions are satisfied, any field F is a Euclidean Domain with the Euclidean function $d(a) = 0$ (or any constant positive integer) for $a \neq 0$.

Example 3.4.13 (Applying the Division Algorithm in $\mathbb{Z}[i]$)**.** In the Euclidean Domain $\mathbb{Z}[i]$ (Gaussian integers), find the quotient q and remainder r when $\alpha = 7 + 5i$ is divided by $\beta = 2 + i$. Verify that $N(r) < N(\beta)$ if $r \neq 0$. Use the norm $N(a + bi) = a^2 + b^2$.

We are given $\alpha = 7 + 5i$ and $\beta = 2 + i$.

1. **Calculate $\frac{\alpha}{\beta}$ in $\mathbb{C}$:**

$$
\begin{aligned}
\frac{\alpha}{\beta} = \frac{7+5i}{2+i} &= \frac{(7+5i)(2-i)}{(2+i)(2-i)} \\
&= \frac{14-7i+10i-5i^2}{2^2-i^2} \\
&= \frac{14+3i+5}{4-(-1)} \\
&= \frac{19+3i}{5} \\
&= \frac{19}{5} + \frac{3}{5}i \\
&= 3.8 + 0.6i
\end{aligned}
$$

2. **Find the closest Gaussian integer $q = m + ni$:** For $x = 3.8$, the closest integer is $m = 4$. For $y = 0.6$, the closest integer is $n = 1$. So, we choose the quotient $q = 4 + i$.

3. **Calculate the remainder $r = \alpha - \beta q$:**

$$
\begin{aligned}
r &= (7+5i) - (2+i)(4+i) \\
&= (7+5i) - (8+2i+4i+i^2) \\
&= (7+5i) - (8+6i-1) \\
&= (7+5i) - (7+6i) \\
&= (7-7) + (5-6)i \\
&= 0 - 1i \\
&= -i
\end{aligned}
$$

4. **Verify the condition on the remainder:** The remainder is $r = -i$. The norm of the divisor $\beta = 2 + i$ is $N(\beta) = N(2+i) = 2^2 + 1^2 = 4 + 1 = 5$. The norm of the remainder $r = -i$ is $N(r) = N(0-i) = 0^2 + (-1)^2 = 1$. We check if $N(r) < N(\beta)$. $1 < 5$. The condition is satisfied.

So, when $7 + 5i$ is divided by $2 + i$, the quotient is $q = 4 + i$ and the remainder is $r = -i$. We have $7 + 5i = (2+i)(4+i) + (-i)$.

Example 3.4.14 (Applying the Division Algorithm in $F[x]$)**.** Let $F = \mathbb{R}$. In the Euclidean Domain $\mathbb{R}[x]$, find the quotient $q(x)$ and remainder $r(x)$ when $a(x) = x^3 - 2x^2 + 4x - 1$ is divided by $b(x) = x - 1$. Verify that $r(x) = 0$ or $\deg(r(x)) < \deg(b(x))$.

We perform polynomial long division.

$$
\begin{array}{r|llll}
 & x^2 & -x & +3 & \\
\hline
x-1 & x^3 & -2x^2 & +4x & -1 \\
 & -(x^3 & -x^2) & & \\
\cline{2-3}
 & 0 & -x^2 & +4x & \\
 & & -(-x^2 & +x) & \\
\cline{3-4}
 & & 0 & 3x & -1 \\
 & & & -(3x & -3) \\
\cline{4-5}
 & & & 0 & 2
\end{array}
$$

The quotient is $q(x) = x^2 - x + 3$. The remainder is $r(x) = 2$.

We verify the condition on the remainder: The divisor is $b(x) = x - 1$, so $\deg(b(x)) = 1$. The remainder is $r(x) = 2$. Since $r(x) \neq 0$, we check its degree. $\deg(r(x)) = \deg(2) = 0$. We need to check if $\deg(r(x)) < \deg(b(x))$. $0 < 1$. The condition is satisfied.

So, $a(x) = b(x)q(x) + r(x)$ becomes: $x^3 - 2x^2 + 4x - 1 = (x-1)(x^2 - x + 3) + 2$.

Example 3.4.15. [$\mathbb{Z}[\sqrt{-5}]$ is NOT an ED] The ring $\mathbb{Z}[\sqrt{-5}] = \{a + b\sqrt{-5} \mid a, b \in \mathbb{Z}\}$ is an integral domain. We know from a previous example (related to UFDs) that $6 = 2 \cdot 3 = (1+\sqrt{-5})(1-\sqrt{-5})$, where $2, 3, 1+\sqrt{-5}, 1-\sqrt{-5}$ are all irreducible and 2 is not an associate of $1+\sqrt{-5}$ or $1-\sqrt{-5}$. Explain why this implies $\mathbb{Z}[\sqrt{-5}]$ cannot be a Euclidean Domain. (See Example 3.4.51).

Example 3.4.16. If F is a field, then $F[x]$ is a Euclidean domain. Indeed, $F[x]$ is an integral domain. For any $0 \neq f(x) \in F[x]$, let $\delta(f(x)) = \deg(f(x))$. If $0 \neq g(x) \in F[x]$, then by Theorem 3.4.2, $\deg(f(x)g(x)) = \deg(f(x)) + \deg(g(x)) \geq \deg(f(x))$. The division algorithm (Theorem 3.4.4) for polynomials completes the proof.

Example 3.4.17. Let $R = \{a + bi : a, b \in Z\}$. We call this the ring of Gaussian integers. Then R is a subring of $F = \{a + bi : a, b \in \mathbb{Q}\}$ which, in turn, is a subfield of $\mathbb{C}$. We claim that R is a Euclidean domain. It is surely an integral domain, since it is a unital subring of a field and therefore has no zero divisors. It remains to construct a Euclidean function. Define $\delta : F \to \mathbb{Q}$ via $\delta(a + bi) = a^2 + b^2$. In particular, if $0 \neq a + bi \in R$, then $\delta(a + bi) \in \mathbb{N}$. If $a, b, c, d \in Q$, then

$$\begin{aligned}\delta((a + bi)(c + di)) &= \delta((ac - bd) + (ad + bc)i) = (ac - bd)^2 + (ad + bc)^2 \\ &= a^2c^2 + b^2d^2 + a^2d^2 + b^2c^2 = (a^2 + b^2)(c^2 + d^2) \\ &= \delta(a + bi)\delta(c + di).\end{aligned}$$

In particular, if $a + bi$ and $c + di$ are nonzero elements of R, then

$$\delta(a + bi) \leq \delta((a + bi)(c + di)).$$

Take any non-zero $u, v \in R$. Then as F is a field, $uv^{-1} \in F$. Let us write $uv^{-1} = s + ti$, with $s, t \in \mathbb{Q}$. Choose integers m and n such that $|s - m| \leq 1/2$ and $|t - n| \leq 1/2$.
Then

$$\begin{aligned}u - v(m + ni) &= u - v((s + ti) + ((m - s) + (n - t)i)) \\ &= u - v(uv^{-1}) + v((s - m) + (t - n)i) \\ &= v((s - m) + (t - n)i).\end{aligned}$$

Now,

$$\delta((s - m) + (t - n)i) = (s - m)2 + (t - n)2 \leq 1/2.$$

Therefore,

$$\delta(u - v(m + n)i) = \delta(v)\delta((s - m) + (t - n)i) < \delta(v).$$

Letting $q = m + ni$ and $r = u - v(m + ni)$, we have $u = vq + r$ and we are done.

What is so special about Euclidean domains? Let us begin with some definitions.

Definition 3.4.18. Let R be a commutative ring with identity. If $a, b \in R$, then we say that a divides b, and write $a \mid b$, if there exists $c \in R$ such that $b = ac$.

Of course, this agrees with our definition of divisibility in $\mathbb{Z}$. We are very much interested in extending the notion of a greatest common divisor as well. For an arbitrary ring, this is problematic, as there is no particular notion of ordering. But for a Euclidean domain, we have δ!.

Definition 3.4.19. Let R be a Euclidean domain, and let $a, b \in R$, not both zero. Then a nonzero element d of R is said to be a greatest common divisor (or gcd) if

(i) $d \mid a$ and $d \mid b$; and

(ii) whenever c is an element of R satisfying $c \mid a$ and $c \mid b$, we have $\delta(c) \leq \delta(d)$.

Certainly a gcd must exist. Indeed, 1 is a common divisor of any two elements, so the set of common divisors is not empty. Furthermore, by definition of a Euclidean function, if $c \mid a$ and $a \neq 0$, then $\delta(c) \leq \delta(a)$. Thus, there is an upper bound on the δ values of the common divisors, so we can select one having the largest possible value.
Notice that we called d "a gcd", not "the gcd". Indeed, this definition does not produce a unique gcd. In particular, in $\mathbb{Z}$, we see that both 5 and -5 would meet the description of "a gcd" of 10 and 35. However, when we say "the gcd", we will still mean the positive one; that is, $(10, 35) = 5$, not -5.
Similarly, if F is a field, suppose that $d(x)$ is a gcd of $f(x)$ and $g(x)$. If u is a non-zero element of F, we see immediately that $ud(x)$ also divides both $f(x)$ and $g(x)$, and that $\deg(ud(x)) = \deg(d(x))$. Thus, $ud(x)$ is also a gcd. But again, we can choose a specific gcd here.

Definition 3.4.20. Let F be a field and let $f(x)$ and $g(x)$ be polynomials in $F[x]$, not both the zero polynomial. By the gcd of $f(x)$ and $g(x)$ we mean a monic gcd. When we write $(f(x), g(x))$, we mean specifically this monic gcd.

For more general Euclidean domains, we cannot easily out a particular gcd in this manner. But we will see that, in fact, the gcd's are all related to each other in a nice way. While proving this, we can produce some other interesting results. For instance, the Euclidean domain is so named because there is a Euclidean algorithm just like in $\mathbb{Z}$.

Theorem 3.4.21. (Euclidean Algorithm for Euclidean Domains). *Let R be a Euclidean domain. Take $a, b \in R$ with $b \neq 0$. If $b \mid a$, then b is a* gcd *of a and b. Otherwise, apply the division algorithm repeatedly. To wit, write*

$$\begin{aligned} a &= bq_1 + r_1 \\ b &= r_1 q_2 + r_2 \\ r_1 &= r_2 q_3 + r_3 \\ &\vdots \\ r_{k-2} &= r_{k-1} q_k + r_k \\ r_{k-1} &= r_k q_{k+1} + 0, \end{aligned}$$

where all $q_i, r_i \in R$ and $ri \neq 0$, with $\delta(r_1) < \delta(b)$ and $\delta(r_j) < \delta(r_{j-1})$ for all $j \geq 2$. Then r_k is a gcd *of a and b.*

Proof. If $b \mid a$, then b is a common divisor of a and b. Also, if $c \mid b$, then $\delta(c) \leq \delta(b)$, and so b is a gcd. Assume that b does not divide a, and we perform the division algorithm repeatedly, as indicated. Note, first of all, that this process must end, as the $\delta(r_i)$ are strictly decreasing integers and cannot be negative. Suppose that $c \mid a$ and $c \mid b$, say $a = ca_1$ and $b = ca_2$, with $a_i \in R$. Then $r_1 = c(a_1 - a_2 q_1)$, and hence $c \mid r_1$. Similarly, any common divisor of b and r_1 must also divide a. Thus, the set of common divisors of a and b is precisely the same as the set of common divisors of b and r_1. In particular, they have the same set of gcds.
By the same argument, the gcds of b and r_1 are the same as those of r_1 and r_2. We then repeat this and find that the gcds of a and b are the same as the gcds of r_k and 0. But as everything divides 0, we are looking only for the largest value of δ among the divisors of r_k . However, as if $u \mid v$ and $v \neq 0$, then $\delta(u) \leq \delta(v)$, we see that r_k is a gcd of r_k and 0, as required. ■

Example 3.4.22. Let us apply the Euclidean algorithm and its corollary in $\mathbb{Z}_7[x]$, starting with $f(x) = 2x^3 + 4x^2 + x + 1$ and $g(x) = 6x^3 + 4x^2 + 4x + 5$. We write

$$\begin{aligned} 2x^3 + 4x^2 + x + 1 &= (6x^3 + 4x^2 + 4x + 5)(5) + (5x^2 + 2x + 4) \\ 6x^3 + 4x^2 + 4x + 5 &= (5x^2 + 2x + 4)(4x + 2) + (5x + 4) \\ 5x^2 + 2x + 4 &= (5x + 4)(x + 1) + 0. \end{aligned}$$

Thus, $5x + 4$ is a gcd of $f(x)$ and $g(x)$. Now let us apply the method discussed in the proof of the preceding corollary. We have

$$\begin{aligned} 5x + 4 &= g(x) - (5x^2 + 2x + 4)(4x + 2) \\ &= g(x) - (f(x) - g(x)(5))(4x + 2) \\ &= f(x)(3x + 5) + g(x)(6x + 4). \end{aligned}$$

If we want to use $(f(x), g(x))$, we must make it monic. Now, $5^{-1} = 3$, and therefore $(f(x), g(x)) = 3(5x + 4) = x + 5$. Then we get

$$x + 5 = 3(5x + 4) = f(x)(2x + 1) + g(x)(4x + 5).$$

When F is a field, $a \in F$, and $f(x) \in F[x]$, we say that a is a zero of multiplicity $k (k \geq 1)$ if $(x - a)^k$ is a factor of $f(x)$ but $(x - a)^{k+1}$ is not a factor of $f(x)$. With these definitions, we may now give several important corollaries of the division algorithm. No doubt you have seen these for the special case where F is the field of real numbers.

Lemma 3.4.23. *Let R be a commutative ring and fix $r \in R$. Then the function $\alpha : R[x] \to R$ given by $\alpha(f(x)) = f(r)$ is a homomorphism.*

Proof. Let $f(x) = a_0 + \cdots + a_n x^n$ and $g(x) = b_0 + \cdots + b_n x^n$ be arbitrary polynomials in $R[x]$ (adding in terms with coefficient zero if necessary). Then

$$\begin{aligned}\alpha(f(x) + g(x)) &= a_0 + b_0 + a_1 r + b_1 r + \cdots + a_n r^n + b_n r^n \\ &= (a_0 + \cdots + a_n r^n) + (b_0 + \cdots + b_n r^n) \\ &= \alpha(f(x)) + \alpha(g(x)).\end{aligned}$$

Also,writing $f(x)g(x) = c_0 + \cdots + c_{2n}x^{2n}$, where $c_i = a_0 b_i + \cdots + a_i b_0$, we have

$$\alpha(f(x)g(x)) = c_0 + \cdots + c_{2n} r^{2n},$$

whereas

$$\alpha(f(x))\alpha(g(x)) = (a_0 + \cdots + a_n r^n)(b_0 + \cdots + b_n r^n).$$

But for any i,

$$a_0 b_i r^i + a_1 r b_{i-1} r^{i-1} + \cdots + a_i r^i b_0 = c_i r^i,$$

and so $\alpha(f(x)g(x)) = \alpha(f(x))\alpha(g(x))$. ∎

Theorem 3.4.24. (The Remainder Theorem.) *Let F be a field, $a \in F$, and $f(x) \in F[x]$. Then $f(a)$ is the remainder in the division of $f(x)$ by $x - a$.*

Proof. By the division algorithm for polynomials, $f(x) = (x - a)q(x) + r(x)$, where $q(x), r(x) \in F[x]$, and either $r(x)$ is the zero polynomial, or $\deg(r(x)) < \deg(x - a) = 1$. That is, $r(x)$ is some constant, $b \in F$. By the Lemma 3.4.23,

$$f(a) = (a - a)q(a) + b = b$$

∎

Theorem 3.4.25. (The Factor Theorem.) *Let F be a field, $a \in F$, and $f(x) \in F[x]$. Then a is a zero of $f(x)$ if and only if $x - a$ is a factor of $f(x)$.*

Proof. Suppose that a is a root of $f(x)$. By the Remainder Theorem, we have $f(x) = (x - a)q(x)$, and hence $(x - a) \mid f(x)$. Conversely, suppose that $(x - a) \mid f(x)$. Then $f(x) = (x - a)g(x)$, for some $g(x) \in F[x]$. In this case, $f(a) = (a - a)g(a) = 0$, and hence a is a root. ∎

Theorem 3.4.26. (Polynomials of Degree n Have at Most n Zeros). *A polynomial of degree n over a field has at most n zeros, counting multiplicity.*

Proof. We proceed by induction on n. Clearly, a polynomial of degree 0 over a field has no zeros. Now suppose that $f(x)$ is a polynomial of degree n over a field and a is a zero of $f(x)$ of multiplicity k. Then, $f(x) = (x - a)^k q(x)$ and $q(a) \neq 0$; and, since $n = \deg f(x) = \deg(x - a)^k q(x) = k + \deg q(x)$, we have $k \leq n$. If f(x) has no zeros other than a, we are done. On the other hand, if $b \neq a$ and b is a zero of $f(x)$, then $0 = f(b) = (b - a)^k q(b)$, so that b is also a zero of $q(x)$ with the same multiplicity as it has for $f(x)$. By the Principle of Mathematical Induction, we know that $q(x)$ has at most $\deg q(x) = n - k$ zeros, counting multiplicity. Thus, $f(x)$ has at most $k + n - k = n$ zeros, counting multiplicity. ∎

Definition 3.4.27. An ideal A of a ring R that is generated by a single element of A, that is, $A = \langle a \rangle = \{ra : r \in R\}$ is called a principal ideal. A principal ideal domain is an integral domain R in which every ideal has the form $\langle a \rangle = \{ra : r \in R\}$ for some a in R.

Theorem 3.4.28. *Let F be a field. Then $F[x]$ is a principal ideal domain.*

Proof. we know that $F[x]$ is an integral domain. Now, let I be an ideal in $F[x]$. If $I = \{0\}$, then $I = \langle 0 \rangle$. If $I \neq \{0\}$, then among all the elements of I, let $g(x)$ be one of minimum degree. We will show that $I = \langle g(x) \rangle$. Since $g(x) \in I$, we have $\langle g(x) \rangle \subseteq I$. Now let $f(x) \in I$. Then, by the division algorithm, we may write $f(x) = g(x)q(x) + r(x)$, where $r(x) = 0$ or $\deg r(x) < \deg g(x)$. Since $r(x) = f(x) - g(x)q(x) \in I$, the minimality of $\deg g(x)$ implies that the latter condition cannot hold. So, $r(x) = 0$ and, therefore, $f(x) \in \langle g(x) \rangle$. This shows that $I \subseteq \langle g(x) \rangle$. ∎

Theorem 3.4.29. *Every Euclidean domain is a principal ideal domain.*

Proof. Let I be an ideal of R. If I is the zero ideal, there is nothing to prove. Otherwise let d be any non-zero element of I such that $\delta(d)$ is minimum. (Since δ takes on values that are nonnegative integers, there must be a smallest such value.) Clearly $\langle d\rangle \subseteq I$ since d is an element of I . To show the reverse inclusion let a be any element of I and use the Division Algorithm to write $a = qd + r$ with $r = 0$ or $\delta(r) < \delta\langle d\rangle$. Then $r = a - qd$ and both a and qd are in I, so r is also an element of I. By the minimality of $\delta(d)$, we see that r must be 0. Thus $a = qd \in \langle d\rangle$ showing $I = \langle d\rangle$. ∎

Example 3.4.30. Since $\mathbb{Z}$ isa Euclidean domain, it is a PID.

Example 3.4.31. Let F be a field. Since $F[x]$ is a Euclidean domain, it is a PID.

Example 3.4.32. We claim that $\mathbb{Z}[x]$ is not a PID, and hence not a Euclidean domain. To prove this, consider the set I of all $f(x) \in \mathbb{Z}[x]$ whose constant terms are divisible by 5. Then I is an ideal (Verify!). But it is not a principal ideal. Indeed, suppose that $I = (f(x))$.Then as the constant polynomial 5 is in I, we see that $f(x) \mid 5$. Therefore, $f(x)$ is a constant polynomial. As it divides 5, the constant must be in $\{\pm1, \pm5\}$. However, $(1) = (-1) = \mathbb{Z}[x]$, whereas $(5) = (-5) = 5\mathbb{Z}[x]$, which does not include x. But $x \in I$, and therefore $5\mathbb{Z}[x] \neq I$.

Theorem 3.4.33. *Let R be a PID. Suppose that R has ideals $I_k, k \in \mathbb{N}$, such that $I_1 \subseteq I_2 \subseteq I_3 \subseteq \cdots$. Then there exists a positive integer n such that $I_k = I_n$ for all $k \geq n$.*

Proof. Let $I = \bigcup_{k=1}^{\infty} I_k$. We claim that I is an ideal. Certainly $0 \in I_1 \subseteq I$. If $a, b \in I$, then there exist positive integers k and l such that $a \in I_k$ and $b \in Il$. Let m be the larger of k and l. Then $a, b \in I_m$, and hence $a - b \in I_m \subseteq I$. Similarly, if $a \in I$, say $a \in I_k$, and $r \in R$, then $ra \in Ik \subseteq I$. Thus, I is an ideal. As R is a PID, we must have $I = (c)$ for some $c \in I$. But then $c \in I_n$, for some positive integer n. It now follows that $I = (c) \subseteq I_n$. That is, $I = I_n$, and hence $I_k = I_n$ for all $k \geq n$. ∎

Theorem 3.4.34. *Let $p > 1$ be a positive integer. Then the following are equivalent:*

(i) p is prime; and

(ii) if a and b are integers such that $p \mid ab$, then $p \mid a$ or $p \mid b$.

Proof. Proof is left as an exercise. ∎

Definition 3.4.35. Let R be an integral domain. Then an element p of R is prime if it is not zero, not a unit, and if $p \mid ab$, with $a, b \in R$, then $p \mid a$ or $p \mid b$.

Definition 3.4.36. Let R be a commutative ring with identity. If $a, b \in R$, then we say that a and b are associates if there exists a unit u of R such that $b = au$.

Lemma 3.4.37. *Let R be an integral domain. Then a and b are associates in R if and only if $a \mid b$ and $b \mid a$.*

Proof. If a and b are associates, the fact that $a \mid b$ and $b \mid a$ follows from the definition. Suppose that $a \mid b$ and $b \mid a$, say $b = ar$ and $a = bs$, with $r, s \in R$. Then $a = ars$. If $a = 0$, then $b = 0$, so $a = b \cdot 1$. Otherwise, by cancellation, $rs = 1$, and hence r is a unit. ∎

Theorem 3.4.38. *Let R be an integral domain, and take $0 \neq p \in R$. Then p is prime if and only if $\langle p\rangle$ is a prime ideal.*

Proof. Let p be prime. If $\langle p\rangle = R$, then there exists an $r \in R$ such that $rp = 1$; hence, p is a unit. But primes cannot be units, so this is impossible. If $ab \in \langle p\rangle$, then $p \mid ab$, and hence $p \mid a$ or $p \mid b$. Thus, $a \in \langle p\rangle$ or $b \in \langle p\rangle$, and $\langle p\rangle$ is a prime ideal. Conversely, suppose that $\langle p\rangle$ is a prime ideal and $p \mid ab$. Then $ab \in \langle p\rangle$, and hence $a \in \langle p\rangle$ or $b \in \langle p\rangle$. That is, $p \mid a$ or $p \mid b$. Furthermore, if p is a unit, then $\langle p\rangle = R$, which contradicts the assumption that $\langle p\rangle$ is a prime ideal. Thus, p is prime. ∎

Definition 3.4.39. Let R be an integral domain, and take $p \in R$. We say that p is irreducible if it is not zero, not a unit, and if $p = ab$, with $a, b \in R$, then either a or b must be a unit.

Theorem 3.4.40. *$\mathbb{Z}$ is a Principal Ideal Domain (PID).*

Proof. Let I be a non-zero ideal of $\mathbb{Z}$. Let d be the smallest positive integer in I. We claim $I = \langle d \rangle$. Clearly, $\langle d \rangle \subseteq I$. For any $a \in I$, by the Division Algorithm, $a = qd + r$ where $0 \leq r < d$. Since $a \in I$ and $qd \in I$, we have $r = a - qd \in I$. By minimality of d, r must be 0. Thus, $a = qd \in \langle d \rangle$, implying $I \subseteq \langle d \rangle$. Hence, $I = \langle d \rangle$, and $\mathbb{Z}$ is a PID. ■

Theorem 3.4.41. *If F is a field, then $F[x]$ is a Principal Ideal Domain (PID).*

Proof. Let I be a non-zero ideal of $F[x]$. Let $p(x)$ be a non-zero polynomial of least degree in I. We claim $I = \langle p(x) \rangle$. Clearly, $\langle p(x) \rangle \subseteq I$. For any $f(x) \in I$, by the Division Algorithm, $f(x) = q(x)p(x) + r(x)$ where $r(x) = 0$ or $\deg(r(x)) < \deg(p(x))$. Since $f(x) \in I$ and $q(x)p(x) \in I$, we have $r(x) = f(x) - q(x)p(x) \in I$. By minimality of $\deg(p(x))$, $r(x)$ must be 0. Thus, $f(x) = q(x)p(x) \in \langle p(x) \rangle$, implying $I \subseteq \langle p(x) \rangle$. Hence, $I = \langle p(x) \rangle$, and $F[x]$ is a PID. ■

Theorem 3.4.42. *Let R be an integral domain. Then every prime in R is irreducible.*

Proof. Let p be a prime, and suppose that $p = ab$, with $a, b \in R$. Then $p \mid ab$, so $p \mid a$ or $p \mid b$. Without loss of generality, say $p \mid a$. But $a \mid p$ as well. By Lemma 3.4.37, p and a are associates. Thus, b is a unit, as required. ■

Theorem 3.4.43. *Let R be a PID and $p \in R$. Then p is prime if and only if p is irreducible.*

Proof. In view of Theorem 3.4.43, we only need to show the converse. Let p be irreducible, and let $I = (p)$. We claim that I is a maximal ideal of R. If not, suppose that J is an ideal of R with $IsubsetJ \subsetneq R$. Since R is a PID, we have $J = (a)$, for some $a \in J$. Now, $p \in I \subseteq J$, so $p = ab$, for some $b \in R$. As p is irreducible, either a or b is a unit. If a is a unit, then J = R, which is not permitted. Therefore, b is a unit. But then $a = pb^{-1} \in I$. Thus, $J \subseteq I$, which is also not allowed. On the other hand, if $I = R$, then p is a unit, which is impossible. Our claim is proved. But a maximal ideal is necessarily prime. Lemma 3.4.38 completes the proof. ■

2nd Proof. Let R be a Principal Ideal Domain and $p \in R$ be an irreducible element. We aim to show that p is prime. Assume $p \mid ab$ for some $a, b \in R$. We need to show that $p \mid a$ or $p \mid b$.

Consider the ideal generated by p and a: $I = \langle p, a \rangle$. Since R is a PID, I must be a principal ideal, so $I = \langle d \rangle$ for some $d \in R$.

As $p \in \langle p, a \rangle = \langle d \rangle$, it follows that $d \mid p$. Since p is irreducible, its only divisors are units or associates of p. Thus, we have two cases for d:

1. d **is an associate of** p**:** In this case, $\langle d \rangle = \langle p \rangle$. Since $a \in I = \langle d \rangle = \langle p \rangle$, this implies that a is a multiple of p, i.e., $p \mid a$.

2. d **is a unit in** R**:** If d is a unit, then $\langle d \rangle = R$. Therefore, $I = \langle p, a \rangle = R$. This means that 1 can be expressed as a linear combination of p and a: $1 = xp + ya$ for some $x, y \in R$.

   Multiply this equation by b: $b = xpb + yab$.

   We are given that $p \mid ab$. So, $ab = pk$ for some $k \in R$. Substitute $ab = pk$ into the equation for b: $b = xpb + y(pk) = p(xb + yk)$.

   This last expression shows that b is a multiple of p, i.e., $p \mid b$.

In both possible cases, we have shown that $p \mid a$ or $p \mid b$. Therefore, by definition, p is a prime element. ■

Definition 3.4.44. A Unique factorisation Domain (U.F.D.) is an integral domain R in which every non-zero element $r \in R$ which is not a unit has the following two properties:

(i) r can be written as a finite product of irreducibles p_i of R (not necessarily distinct): $r = p_1 p_2 \dots p_n$ and

(ii) the decomposition in (i) is unique up to associates: namely, if $r = q_1 q_2 \dots q_m$ is another factorisation of r into irrreducibles, then $m = n$ and there is some renumbering of the factors so that p_i is associate to q_i for $i = 1, 2, \dots, n$.

Example 3.4.45. 1. A field F is trivially a Unique factorisation Domain since every nonzero element is a unit, so there are no elements for which properties (i) and (ii) must be verified.

2. For any field F, $F[x]$ is a UFD.

3. The subring of the Gaussian integers $R = \mathbb{Z}[2i] = \{a + 2bi \mid a,\ b \in \mathbb{Z}\}$, where $i^2 = -1$, is an integral domain but not a Unique factorisation Domain. The elements 2 and $2i$ are irreducibles which are not associates in R since $i \notin R$, and $4 = 2{\cdot}2 = (-2i){\cdot}(2i)$ has two distinct factorisations in R.

4. The quadratic integer ring $\mathbb{Z}[\sqrt{-5}]$ is another example of an integral domain that is not a Unique factorisation Domain, since $6 = 2 \cdot 3 = (1 + \sqrt{-5})(1 - \sqrt{-5})$ gives two distinct factorisations of 6 into irreducibles.

Theorem 3.4.46. *In a Unique factorisation Domain a nonzero element is a prime if and only if it is irreducible.*

Proof. Let R be a Unique factorisation Domain. Since by Theorem 3.4.42, primes of R are irreducible it remains to prove that each irreducible element is a prime. Let p be an irreducible in R and assume $p \mid ab$ for some $a, b \in R$; we must show that p divides either a or b. To say that p divides ab is to say $ab = pc$ for some c in R. Writing a and b as a product of irreducibles,we see from this last equation and from the uniqueness of the decomposition into irreducibles of ab that the irreducible element p must be associate to one of the irreducibles occurring either in the factorisation of a or in the factorisation of b. We may assume that p is associate to one of the irreducibles in the factorisation of a, i.e., that a can be written as a product $a = (up)p_2 \cdots p_n$ for u a unit and some (possibly empty set of) irreducibles $p_2, \dots, p_n$. But then p divides a, since $a = pd$ with $d = up_2 \cdots p_n$, completing the proof. ■

3.4.2 Relations between ED, PID and UFD

Theorem 3.4.47. *Every Euclidean Domain is a Principal Ideal Domain (PID).*

Proof. Let R be a Euclidean Domain with a Euclidean function $\delta : R \setminus \{0\} \to \mathbb{N} \cup \{0\}$. Let I be any ideal of R. If $I = \{0\}$, then $I = \langle 0 \rangle$, which is principal. Assume $I \neq \{0\}$. Then I contains non-zero elements. Let $d \in I$ be a non-zero element such that $\delta(d)$ is minimal among all non-zero elements in I. Such a d exists because the range of δ is the set of non-negative integers.

We claim $I = \langle d \rangle$. Clearly, since $d \in I$ and I is an ideal, all multiples of d are in I. Thus, $\langle d \rangle \subseteq I$.

Now, let a be any element in I. By the definition of a Euclidean Domain, for $a, d \in R$ (with $d \neq 0$), there exist elements $q, r \in R$ such that $a = qd + r$, where either $r = 0$ or $\delta(r) < \delta(d)$.

Since $a \in I$ and $d \in I \implies qd \in I$, we have $r = a - qd \in I$. If $r \neq 0$, then $\delta(r) < \delta(d)$. However, d was chosen as a non-zero element in I with the *minimum* Euclidean value. This means r cannot be a non-zero element in I. Therefore, r must be 0.

If $r = 0$, then $a = qd$. This means a is a multiple of d, so $a \in \langle d \rangle$. Thus, $I \subseteq \langle d \rangle$.

Since $I \subseteq \langle d \rangle$ and $\langle d \rangle \subseteq I$, we conclude that $I = \langle d \rangle$. Therefore, every ideal in R is principal, and thus R is a PID. ■

Theorem 3.4.48. *Every Principal Ideal Domain is a Unique Factorisation Domain.*

Proof. Let R be a PID. We prove the existence and uniqueness of factorization into irreducibles.

Existence of Factorization: Suppose, for contradiction, there exists a non-zero, non-unit element $r \in R$ that is not a product of irreducibles. Then r is reducible, so $r = r_1 s_1$ for non-units r_1, s_1. At least one of r_1, s_1 must also not be a product of irreducibles; assume it's r_1. This gives a proper ideal inclusion $\langle r \rangle \subset \langle r_1 \rangle$. Repeating this, we construct an infinite ascending chain of proper ideals: $\langle r \rangle \subset \langle r_1 \rangle \subset \langle r_2 \rangle \subset \cdots$.

However, in a PID, every ascending chain of ideals $I_1 \subseteq I_2 \subseteq \dots$ stabilizes as follows: Let $I = \cup_{i=1}^{\infty} I_i$. Then I is an ideal. Since R is a Principal Ideal Domain it is principally generated, say $I = \langle a \rangle$. Since I is the union of the ideals above, a must be an element of one of the ideals in the chain, say $a \in I_n$. But then we have $I_n \subseteq I = \langle a \rangle \subseteq I_n$ and so $I = I_n$ and the chain becomes stationary at I_n. Let $I = \bigcup_{k=1}^{\infty} I_k$. I is an ideal, so $I = \langle a \rangle$ for some $a \in R$. Since $a \in I$, $a \in I_n$ for some n. Thus, $\langle a \rangle \subseteq I_n$, implying $I \subseteq I_n$. As $I_n \subseteq I$, we have $I_n = I$, so the chain stabilizes. This contradiction proves every non-zero, non-unit element has a factorization into irreducibles.

Uniqueness of Factorization: We proceed by induction on the number of irreducible factors n. Base case ($n = 1$): Let $r = p_1$ be irreducible. If $r = q_1 \cdots q_m$ is another factorization, then p_1 divides

$q_1 \cdots q_m$. Since p_1 is irreducible, it must divide some q_j. Since q_j is irreducible, p_1 and q_j are associates. If $m > 1$, then $p_1 = uq_1 \cdots q_m$ for some unit u if p_1 is an associate of q_1. This leads to a contradiction ($1 = uq_2 \cdots q_m$, implying $q_2, \ldots, q_m$ are units). Thus $m = 1$, and p_1 is an associate of q_1.

Inductive step: Assume uniqueness holds for elements with $n-1$ irreducible factors. Let $r = p_1p_2 \cdots p_n = q_1q_2 \cdots q_m$ be two factorizations into irreducibles. Since p_1 divides $q_1 \cdots q_m$, p_1 must divide some q_j. Without loss of generality, let p_1 divide q_1. Since q_1 is irreducible, p_1 and q_1 are associates, so $q_1 = p_1u$ for some unit u. Cancelling p_1 (since R is an integral domain), we get $p_2 \cdots p_n = uq_2 \cdots q_m = (uq_2)q_3 \cdots q_m$. Let $q_2' = uq_2$, which is irreducible and an associate of q_2. By the induction hypothesis, $n-1 = m-1$ (so $n = m$), and $p_2, \ldots, p_n$ are associates of $q_2', \ldots, q_m$ in some order. Since p_1 is an associate of q_1, the uniqueness of factorization is proven. ■

Corollary 3.4.49. (Fundamental Theorem of Arithmetic) *The integers $\mathbb{Z}$ are a Unique factorisation Domain.*

Proof. The integers $\mathbb{Z}$ are a Euclidean Domain, hence are a Unique factorisation Domain. ■

Theorem 3.4.50. *In a UFD, every pair of non-zero elements have an HCF and LCM.*

Proof. Let R be a UFD, and let a, b be non-zero in R. If a is a unit then $b = aa^{-1}b$ and so $HCF(a, b) = a$ and $LCM[a, b] = b$. If neither of a and b is a unit, then R being a UFD gives

$$a = up_1^{\alpha_1}p_2^{\alpha_2} \ldots p_k^{\alpha_k} \text{ and } b = vq_2^{\beta_1}q_1^{\beta_2} \ldots q_k^{\beta_k}$$

where u, v are units. (We can take $p^0 \mid r$ for any $0 \neq r \in R$ to make it k in both factorizations). (For example in $\mathbb{Q}[x]$, we can write $6x + 3 = 3(2x+1)^1(x+1)^0$ and $5x + 5 = 5(x+1)^1(2x+1)^0$).
If $\mu_i = \max(\alpha_{\rm i}, \beta_{\rm i})$ and $\lambda_i = \min(\alpha_{\rm i}, \beta_{\rm i})$, then

$$HCF[a, b] = p_1^{\lambda_1}p_2^{\lambda_2} \ldots p_k^{\lambda_k} \text{ and } LCM[a, b] = p_1^{\mu_1}p_2^{\mu_2} \ldots p_k^{\mu_k}$$

■

Example 3.4.51. We have the important theorem: **Every Euclidean Domain is a Principal Ideal Domain (PID)**, and **Every PID is a Unique factorisation Domain (UFD)**. So, ED $\implies$ PID $\implies$ UFD.

In $\mathbb{Z}[\sqrt{-5}]$, we have shown that 6 has two distinct factorisations into irreducibles: $6 = 2 \cdot 3$ $6 = (1+\sqrt{-5})(1-\sqrt{-5})$ The elements $2, 3, 1+\sqrt{-5}, 1-\sqrt{-5}$ are irreducible, and 2 is not an associate of $1+\sqrt{-5}$ (nor $1-\sqrt{-5}$), and 3 is not an associate of $1+\sqrt{-5}$ (nor $1-\sqrt{-5}$). This means the factorisation of 6 into irreducibles is not unique up to order and associates. Therefore, $\mathbb{Z}[\sqrt{-5}]$ is **not a UFD**.

Since ED $\implies$ UFD, if a ring is not a UFD, it cannot be an ED. (Contrapositive: Not UFD $\implies$ Not ED). Because $\mathbb{Z}[\sqrt{-5}]$ is not a UFD, it cannot be a Euclidean Domain.

Alternatively, one could try to show it's not a PID directly. For example, the ideal $I = (2, 1+\sqrt{-5})$ in $\mathbb{Z}[\sqrt{-5}]$ is not principal. If it were principal, say $I = (\alpha)$, then α must divide 2 and α must divide $1+\sqrt{-5}$. This would involve checking norms. Since not all ideals are principal, it's not a PID, and therefore not an ED. The UFD argument is often more straightforward once non-unique factorisation is established.

3.5 Examples of PIDs and Non-PIDs, UFD's and Non-UFDs

Example 3.5.1 (The Integers $\mathbb{Z}$ as a PID)**.** Show that the ring of integers $\mathbb{Z}$ is a Principal Ideal Domain. Find a generator for the ideal $I = (6, 10)$ in $\mathbb{Z}$.

Part 1: Show $\mathbb{Z}$ is a PID. We know that $\mathbb{Z}$ is an integral domain. Let I be an arbitrary ideal in $\mathbb{Z}$. If $I = \{0\}$, then $I = (0)$, which is generated by 0, so it is a principal ideal. Suppose $I \neq \{0\}$. Then I contains non-zero integers. Since I is an ideal, if $x \in I$, then $-x \in I$. Thus, I must contain positive integers. Let $S = \{k \in I \mid k > 0\}$. Since $I \neq \{0\}$, S is a non-empty set of positive integers. By the Well-Ordering Principle, S has a least element, say d. So $d \in I$ and $d > 0$. We claim that $I = (d)$.

- $(d) \subseteq I$: Since $d \in I$ and I is an ideal, any multiple zd (for $z \in \mathbb{Z}$) must also be in I. Thus, $(d) \subseteq I$.
- $I \subseteq (d)$: Let a be any element in I. By the Division Algorithm in $\mathbb{Z}$, there exist integers q (quotient) and r (remainder) such that $a = qd + r$, where $0 \le r < d$. Since $a \in I$ and $d \in I$, $qd \in I$ (as I is an ideal). Therefore, $r = a - qd \in I$ (as I is closed under subtraction). If $r > 0$, then r would be a positive integer in I such that $r < d$. This contradicts the choice of d as the *least* positive integer in I. Thus, we must have $r = 0$. So, $a = qd$, which means a is a multiple of d. Hence $a \in (d)$.

Since $I \subseteq (d)$ and $(d) \subseteq I$, we have $I = (d)$. As I was an arbitrary ideal, every ideal in $\mathbb{Z}$ is principal. Therefore, $\mathbb{Z}$ is a PID.

Part 2: Find a generator for $I = (6, 10)$ in $\mathbb{Z}$. The ideal $(6, 10)$ is defined as $\{6x + 10y \mid x, y \in \mathbb{Z}\}$. We know that any ideal in $\mathbb{Z}$ is principal, so $(6, 10) = (d)$ for some $d \in \mathbb{Z}$. It is a known result that for integers a, b, the ideal (a, b) is generated by their greatest common divisor, $\gcd(a, b)$. We find $\gcd(6, 10)$. $6 = 2 \cdot 3$ $10 = 2 \cdot 5$ So, $\gcd(6, 10) = 2$. Thus, the ideal $I = (6, 10)$ is generated by 2. That is, $(6, 10) = (2) = 2\mathbb{Z}$.

To verify:

- $(2) \subseteq (6, 10)$: We need to show 2 can be written as $6x + 10y$. $2 = 6(2) + 10(-1) = 12 - 10 = 2$. So $2 \in (6, 10)$. Since $2 \in (6, 10)$, all multiples of 2 are in $(6, 10)$, so $(2) \subseteq (6, 10)$.
- $(6, 10) \subseteq (2)$: Any element $6x + 10y$ is $2(3x) + 2(5y) = 2(3x + 5y)$, which is a multiple of 2. So $(6, 10) \subseteq (2)$.

Therefore, $(6, 10) = (2)$. A generator for the ideal is 2.

Example 3.5.2 (A Field F as a PID)**.** Show that any field F is a Principal Ideal Domain.

Let F be a field. A field is an integral domain (it's a commutative ring with unity $1 \neq 0$, and every non-zero element is a unit, implying no zero divisors). Let I be an ideal in F. There are two cases for I:

1. **Case 1: $I = \{0\}$ (the zero ideal).** Then $I = (0)$, which is generated by the element 0. So I is a principal ideal.
2. **Case 2: $I \neq \{0\}$.** If $I \neq \{0\}$, then I must contain at least one non-zero element. Let $a \in I$ such that $a \neq 0$. Since F is a field and $a \neq 0$, a is a unit. This means a has a multiplicative inverse $a^{-1} \in F$. We have a theorem stating: "If R is a ring with identity and an ideal I of R contains a unit, then $I = R$." Since $a \in I$ is a unit, it follows that $I = F$. The ideal F can be generated by the unity element $1 \in F$, i.e., $F = (1)$. (Any element $x \in F$ can be written as $x \cdot 1$, so $x \in (1)$). Thus, if $I \neq \{0\}$, then $I = F = (1)$, which is a principal ideal.

In both cases, any ideal I in F is principal. Therefore, any field F is a PID.

Example 3.5.3 (Polynomial Ring $F[x]$ over a Field F as a PID)**.** Show that if F is a field, then the polynomial ring $F[x]$ is a Principal Ideal Domain.

We know that $F[x]$ is an integral domain when F is a field. We also know that $F[x]$ is a Euclidean Domain with the Euclidean function $d(f(x)) = \deg(f(x))$ for any non-zero polynomial $f(x) \in F[x]$. Since **every Euclidean Domain is a Principal Ideal Domain**, it follows directly that $F[x]$ is a PID.

To illustrate the generator for an ideal: Let I be a non-zero ideal in $F[x]$. Let $g(x)$ be a non-zero polynomial in I of minimal degree. Then $I = (g(x))$. For instance, consider $F = \mathbb{R}$ and the ideal $I = (x^2 - 1, x^2 + x - 2)$ in $\mathbb{R}[x]$. We know $I = (d(x))$ where $d(x) = \gcd(x^2 - 1, x^2 + x - 2)$. $x^2 - 1 = (x - 1)(x + 1)$ $x^2 + x - 2 = (x - 1)(x + 2)$ So, $\gcd(x^2 - 1, x^2 + x - 2) = x - 1$. Thus, $I = (x^2 - 1, x^2 + x - 2) = (x - 1)$. This means any polynomial of the form $A(x)(x^2 - 1) + B(x)(x^2 + x - 2)$ is a multiple of $(x - 1)$, and $(x - 1)$ itself can be expressed in this form (e.g., by using the extended Euclidean algorithm for polynomials, or in this simple case: $(x^2 + x - 2) - (x^2 - 1) = x - 1$).

Example 3.5.4 (Gaussian Integers $\mathbb{Z}[i]$ as a PID)**.** Show that the ring of Gaussian integers $\mathbb{Z}[i]$ is a Principal Ideal Domain. (This is also verified in Example 3.4.17)

$\mathbb{Z}[i]$ is an integral domain. We have shown previously that $\mathbb{Z}[i]$ is a Euclidean Domain with the norm $N(a+bi) = a^2+b^2$ as the Euclidean function. Since every Euclidean Domain is a Principal Ideal Domain, $\mathbb{Z}[i]$ is a PID.

For example, consider the ideal $I = (2+i, 5)$ in $\mathbb{Z}[i]$. We can find a generator by using the Euclidean algorithm for Gaussian integers. $N(2+i) = 2^2+1^2 = 5$. $N(5) = 5^2 = 25$. Divide 5 by $2+i$: $\frac{5}{2+i} = \frac{5(2-i)}{(2+i)(2-i)} = \frac{10-5i}{4-i^2} = \frac{10-5i}{5} = 2-i$. So $5 = (2+i)(2-i)+0$. The remainder is 0. The gcd is the last non-zero remainder, which in this context is $2+i$. (More precisely, the Euclidean algorithm process for gcd means the generator of (a,b) is the last non-zero remainder.) So, $(2+i, 5) = (2+i)$. This means that 5 is a multiple of $2+i$ (which we saw: $5 = (2+i)(2-i)$), and $2+i$ is obviously a multiple of $2+i$. Any element $A(2+i)+B(5)$ can be written as $A(2+i)+B(2+i)(2-i) = (2+i)(A+B(2-i))$, which is a multiple of $2+i$.

Example 3.5.5 (A Ring that is an Integral Domain but NOT a PID: $\mathbb{Z}[x]$)**.** We showed in a previous example (related to UFDs or general PID properties) that $\mathbb{Z}[x]$ (the ring of polynomials with integer coefficients) is an integral domain. Show again that the ideal $I = (2, x)$ in $\mathbb{Z}[x]$ is not principal, thus $\mathbb{Z}[x]$ is not a PID.

The ideal $I = (2, x) = \{2f(x) + xg(x) \mid f(x), g(x) \in \mathbb{Z}[x]\}$. Assume, for the sake of contradiction, that I is a principal ideal, so $I = (d(x))$ for some $d(x) \in \mathbb{Z}[x]$. Since $2 \in I$, 2 must be a multiple of $d(x)$. So $2 = d(x)\cdot h(x)$ for some $h(x) \in \mathbb{Z}[x]$. Taking degrees, $\deg(2) = \deg(d(x)) + \deg(h(x))$. $0 = \deg(d(x)) + \deg(h(x))$. Since degrees are non-negative, this implies $\deg(d(x)) = 0$ and $\deg(h(x)) = 0$. Thus, $d(x)$ must be a constant integer, say $d(x) = d \in \mathbb{Z}$. From $2 = d\cdot h(x)$, d must be a divisor of 2 in $\mathbb{Z}$. So $d \in \{\pm1, \pm2\}$.

Also, since $x \in I$, x must be a multiple of $d(x) = d$. So $x = d\cdot k(x)$ for some $k(x) \in \mathbb{Z}[x]$. If $d = \pm2$, then $x = (\pm2)k(x)$. Let $k(x) = c_m x^m + \cdots + c_1 x + c_0$. Then $x = (\pm2)(c_1x + c_0 + \text{higher order terms...})$. Comparing the coefficient of x on both sides: $1 = (\pm2)c_1$. Since $c_1 \in \mathbb{Z}$, there is no integer c_1 such that $1 = \pm2c_1$. This is a contradiction. So $d \neq \pm2$.

The only remaining possibility is $d = \pm1$. If $d = \pm1$, then $I = (d(x)) = (\pm1) = \mathbb{Z}[x]$ (the entire ring). However, $I = (2, x)$ is not the entire ring $\mathbb{Z}[x]$. For example, the constant polynomial $1 \notin I$. To see this, suppose $1 \in I$. Then $1 = 2f(x) + xg(x)$ for some $f(x), g(x) \in \mathbb{Z}[x]$. Substitute $x = 0$: $1 = 2f(0) + 0\cdot g(0) = 2f(0)$. Since $f(0)$ is an integer (the constant term of $f(x)$), $2f(0)$ must be an even integer. But 1 is odd. So $1 = 2f(0)$ is impossible for $f(0) \in \mathbb{Z}$. Thus, $1 \notin I$, which means $I \neq \mathbb{Z}[x]$. This contradicts our finding that if I were principal, it would have to be $\mathbb{Z}[x]$.

Therefore, the assumption that $I = (2, x)$ is a principal ideal is false. Since $\mathbb{Z}[x]$ has an ideal that is not principal, $\mathbb{Z}[x]$ is not a PID.

Example 3.5.6 (Gaussian Integers $\mathbb{Z}[i]$ as a UFD)**.** Explain why $\mathbb{Z}[i]$ is a UFD. Factor 30 into irreducibles in $\mathbb{Z}[i]$.

The ring of Gaussian integers $\mathbb{Z}[i]$ is an integral domain. We know that $\mathbb{Z}[i]$ is a Euclidean Domain (with norm $N(a+bi) = a^2+b^2$). Since every ED is a PID, and every PID is a UFD, it follows that $\mathbb{Z}[i]$ is a UFD.

Factorizing 30 in $\mathbb{Z}[i]$: First, factor 30 in $\mathbb{Z}$: $30 = 2\cdot3\cdot5$. Now, we need to see how these integer primes factor in $\mathbb{Z}[i]$. The units in $\mathbb{Z}[i]$ are $\{\pm1, \pm i\}$. An integer prime p factors in $\mathbb{Z}[i]$ as follows:

- If $p = 2$, then $2 = -i(1+i)^2 = (1-i)(1+i)$. $N(1+i) = 1^2+1^2 = 2$ (prime), so $1+i$ is irreducible. $N(1-i) = 2$, so $1-i$ is irreducible. Note that $1-i = (-i)(1+i)$, so $1-i$ and $1+i$ are associates. A common factorisation is $2 = (1+i)(1-i)$.

- If $p \equiv 1 \pmod 4$, then p factors as $p = \pi\bar{\pi}$, where π and $\bar{\pi}$ are non-associate irreducibles in $\mathbb{Z}[i]$ with $N(\pi) = N(\bar{\pi}) = p$. For $p = 5$: $5 \equiv 1 \pmod 4$. We found $5 = (1+2i)(1-2i)$. $N(1+2i) = 1^2+2^2 = 5$ (prime), so $1+2i$ is irreducible. $N(1-2i) = 1^2+(-2)^2 = 5$ (prime), so $1-2i$ is irreducible. Are $1+2i$ and $1-2i$ associates? If $1+2i = u(1-2i)$ for $u \in \{\pm1, \pm i\}$: If $u = 1$, $1+2i = 1-2i \implies 4i = 0$, false. If $u = -1$, $1+2i = -1+2i \implies 1 = -1$, false. If $u = i$, $1+2i = i(1-2i) = i-2i^2 = i+2 = 2+i$. Not equal. If $u = -i$, $1+2i = -i(1-2i) = -i+2i^2 = -i-2 = -2-i$. Not equal. So $1+2i$ and $1-2i$ are not associates. (Alternatively, $1+2i = a+bi$ and $1-2i = a-bi$. They are conjugates). Other choices for factors of 5 are $2+i$ and $2-i$. $N(2+i) = 5, N(2-i) = 5$. $(2+i)(2-i) = 4-i^2 = 5$.

- If $p \equiv 3 \pmod 4$, then p remains irreducible in $\mathbb{Z}[i]$. For $p = 3$: $3 \equiv 3 \pmod 4$. So 3 is irreducible in $\mathbb{Z}[i]$. (Its norm is $N(3) = 9 = 3^2$).

Now, combine these for $30 = 2{\cdot}3{\cdot}5$: $30 = [(1+i)(1-i)]{\cdot}[3]{\cdot}[(2+i)(2-i)]$ $30 = (1+i)(1-i){\cdot}3{\cdot}(2+i)(2-i)$. All factors $(1+i), (1-i), 3, (2+i), (2-i)$ are irreducible in $\mathbb{Z}[i]$. This is a factorisation of 30 into irreducibles. Any other factorisation will consist of associates of these elements, in some order. For example, $1-i = -i(1+i)$, so we could write $2 = (1+i)(-i(1+i)) = -i(1+i)^2$. $30 = (-i(1+i)^2) \cdot 3 \cdot (1+2i)(1-2i)$. (Using the other factorisation of 5, for variety).

Example 3.5.7 ($\mathbb{Z}[x]$ as a UFD)**.** Explain why $\mathbb{Z}[x]$, the ring of polynomials with integer coefficients, is a Unique factorisation Domain. Give an example of factoring a polynomial.

1. We know that $\mathbb{Z}$ is an integral domain.
2. We have shown that $\mathbb{Z}$ is a UFD (this is the Fundamental Theorem of Arithmetic).
3. A fundamental theorem by Gauss states: **If D is a UFD, then the polynomial ring $D[x]$ is also a UFD.**

Since $\mathbb{Z}$ is a UFD, by Gauss's Lemma (the theorem stated above), $\mathbb{Z}[x]$ is also a UFD.

It is important to note that $\mathbb{Z}[x]$ is **not** a PID. For example, the ideal $(2, x)$ is not principal. This example shows that UFD $\not\Rightarrow$ PID.

Example of factoring a polynomial in $\mathbb{Z}[x]$: Consider the polynomial $p(x) = 6x^2 - 6$.

1. **Factor out the greatest common divisor (gcd) of the coefficients (content):** The coefficients are 6 and -6. The gcd is 6. $p(x) = 6(x^2 - 1)$.
2. **Factor the integer part (content) into primes in $\mathbb{Z}$:** $6 = 2 \cdot 3$. Both 2 and 3 are irreducible in $\mathbb{Z}$ (and also in $\mathbb{Z}[x]$ as constant polynomials).
3. **Factor the primitive polynomial part into irreducibles in $\mathbb{Z}[x]$:** The primitive part is $x^2 - 1$. (A polynomial in $\mathbb{Z}[x]$ is primitive if the gcd of its coefficients is 1). $x^2 - 1 = (x-1)(x+1)$. Are $x - 1$ and $x + 1$ irreducible in $\mathbb{Z}[x]$? A non-constant polynomial in $\mathbb{Z}[x]$ that is irreducible over $\mathbb{Q}$ and is primitive is irreducible over $\mathbb{Z}$. $x - 1$ is primitive. If $x - 1 = f(x)g(x)$ in $\mathbb{Z}[x]$, then $\deg(f) + \deg(g) = 1$. So one has degree 1, the other degree 0 (a constant). If $x - 1 = c(ax + b)$ where $c \in \mathbb{Z}$ and $ax + b \in \mathbb{Z}[x]$ is primitive. Then c must be ± 1 for $ax + b$ to be $x - 1$. So $x - 1$ is irreducible. Similarly, $x + 1$ is irreducible.

Combining these, the factorisation of $6x^2 - 6$ into irreducibles in $\mathbb{Z}[x]$ is: $6x^2 - 6 = 2 \cdot 3 \cdot (x-1) \cdot (x+1)$. The irreducibles are $2, 3, (x-1), (x+1)$. Any other factorisation would involve associates. The units in $\mathbb{Z}[x]$ are just ± 1 (the units of $\mathbb{Z}$). So, another valid factorisation might be $(-2) \cdot (-3) \cdot (x-1) \cdot (x+1)$, or $2 \cdot 3 \cdot (1-x) \cdot (-(x+1))$.

Example 3.5.8 ($\mathbb{Z}[\sqrt{-5}]$ is NOT a UFD – Recap)**.** We have shown previously that in $\mathbb{Z}[\sqrt{-5}] = \{a + b\sqrt{-5} \mid a, b \in \mathbb{Z}\}$, the element 6 has two distinct factorisations into irreducibles: $6 = 2 \cdot 3$ $6 = (1+\sqrt{-5})(1-\sqrt{-5})$ where $2, 3, 1+\sqrt{-5}, 1-\sqrt{-5}$ are all irreducible, and 2 is not an associate of $1+\sqrt{-5}$ or $1-\sqrt{-5}$ (and 3 is not an associate of $1+\sqrt{-5}$ or $1-\sqrt{-5}$).

Since the element 6 has two factorisations into irreducible elements that are not unique up to order and associates, the second condition for a UFD is violated. Therefore, $\mathbb{Z}[\sqrt{-5}]$ is an example of an integral domain that is **not** a Unique factorisation Domain. Since it's not a UFD, it also cannot be a PID or an ED.

Example 3.5.9 (Field F as a UFD)**.** Explain why any field F is a Unique factorisation Domain.

Let F be a field. Then F is an integral domain. In a field F, every non-zero element is a unit. The definition of a UFD concerns the factorisation of *non-zero, non-unit* elements. In a field F, there are no non-zero, non-unit elements. The set of non-zero, non-unit elements is empty. Therefore, the conditions for being a UFD are vacuously satisfied.

1. **Existence of factorisation:** The statement "Every non-zero, non-unit element of F can be written as a product of irreducible elements" is true because there are no such elements that need to be factored. (An empty product is sometimes defined as the multiplicative identity, which is a unit, or the condition is simply considered met if no such elements exist).

2. **Uniqueness of factorisation:** Similarly, this condition is vacuously true as there are no non-zero, non-unit elements to have multiple factorisations.

Thus, any field F is a UFD. (Alternatively, since any field is an ED, and ED $\implies$ PID $\implies$ UFD, a field is a UFD).

3.6 Problems for Practice

1. If R is a ring with unity 1 and 1=0, prove that R is a zero ring.
2. Show that the ring of integers modulo n is a field if and only if n is prime.
3. If for any x, y in a ring R, $(xy)^2 = (yx)^2$, prove that R is commutative.
4. Prove that the ring $\mathbb{Z}[i] = \{a + ib : a, b \in \mathbb{Z}\}$ is a ring under usual addition and multpliication of complex numbers. This ring is called the ring of Gaussian integers.
5. Show that a ring R is commutative if it has the property that $ab = ca$ implies $b = c$ when $a \neq 0$.
6. Show that a ring that is cyclic under addition is commutative.
7. Show that every ring with a prime number of elements is commutative.
8. Let R be a ring and $a \in R$. Let $S = \{x \in R \,:\, ax = 0\}$. Show that S is a subring of R.
9. Give an example of an integral domain with nonzero elements a, b such that $a^2 + b^2 = 0$.
10. An element a in a ring R is called idempotent if $a^2 = a$. A ring in which each element is idempotent is called a Boolean ring. Show that a Boolean ring has exactly one unit.
11. Give an example to show that a unity of a ring and its proper subring need not be same.
12. Give an example of a subset of a ring that is a subgroup under addition but not a subring.
13. List all of the elements in each of the principal ideals $((0, 5))$ and $((2, 3))$ of $\mathbb{Z}_4 \oplus \mathbb{Z}_6$.
14. Let I be the set of all polynomials in $\mathbb{Z}[x]$ whose constant term is a multiple of 5. Show that I is an ideal of $\mathbb{Z}[x]$.
15. Show that the intersection of ideals I and J in a ring R is also an ideal. Extend this to the intersection of an arbitrary collection of ideals.
16. Let m and n be positive integers. Show that $m\mathbb{Z} + n\mathbb{Z} = (m, n)\mathbb{Z}$.
17. Prove that $\mathbb{Z}[x]$ and $\mathbb{R}[x]$ are not isomorphic.
18. Find ideals I and J in a ring R such that $IJ \neq I \cap J$.
19. Prove that a ring R is a field if and only if R has exactly two ideals.
20. Show that the union of two ideals need not be an ideal. Find a condition that union of two ideals A and B is also an ideal.
21. Prove that for any two ideals A and B of a ring R, $A + B = \langle A \cup B \rangle$.
22. Find all maximal ideals in **(a).** $\mathbb{Z}_8$, **(b).** $\mathbb{Z}_{10}$, **(c).** $\mathbb{Z}_{12}$, **(d).** $\mathbb{Z}_n$.
23. Show that the intersection of maximal (prime) ideals need not be maximal (prime).
24. If R is a finite commutative ring with unity, prove that every prime ideal of R is a maximal ideal of R.
25. Show that the characteristic of $\mathbb{Z}[i]/\langle a + bi \rangle$ divides $a^2 + b^2$.
26. If p is a prime number, prove that $\langle p, x, y \rangle$ is a maximal ideal in $\mathbb{Z}[x, y]$.
27. If x is a nilpotent element in a commutative ring R, prove that rx is nilpotent for all r in R.

28. Show that the correspondence $x \mapsto 5x$ from $\mathbb{Z}_5$ to $\mathbb{Z}_{10}$ does not preserve addition.
29. Show that the correspondence $x \mapsto 3x$ from $\mathbb{Z}_4$ to $\mathbb{Z}_{12}$ does not preserve multiplication.
30. Determine all ring isomorphisms from $\mathbb{Z}_n$ to itself.
31. Determine all ring homomorphisms from $\mathbb{Z}$ to $\mathbb{Z}$.
32. Prove that every ring homomorphism ϕ from $\mathbb{Z}_n$ to itself has the form $\phi(x) = ax$, where a is an idempotent.
33. For any ring R, show that R and $R[x]$ have the same characteristic.
34. If F is a field, is $F[x]$ a field?
35. If R and S are isomorphic rings, show that $R[x]$ and $S[x]$ are also isomorphic.
36. Let R be a commutative ring with identity and P a prime ideal of R. Show that $P[x]$ is a prime ideal of $R[x]$.
37. Let R be a commutative ring. Show that $R[x]$ has a subring isomorphic to R.
38. Show that the polynomial $2x + 1$ in $\mathbb{Z}_4[x]$ has a multiplicative inverse in $\mathbb{Z}_4[x]$.
39. Prove that the ideal $\langle x \rangle$ in $\mathbb{Q}[x]$ is maximal.

Chapter 4

Polynomial Rings

4.1 Irreducibility of Polynomials

Definition 4.1.1 (Irreducible and reducible polynomial). Let D be an integral domain. The units in $D[x]$ are precisley the units in D. A polynomial $f(x)$ from $D[x]$ that is neither the zero polynomial nor a unit in $D[x]$ is said to be irreducible over D if, whenever $f(x)$ is expressed as a product $f(x) = g(x)h(x)$, with $g(x)$ and $h(x)$ from $D[x]$, then $g(x)$ or $h(x)$ is a unit in $D[x]$, that is, either $g(x)$ or $h(x)$ is an element of D. A nonzero, nonunit element of $D[x]$ that is not irreducible over D is called reducible over D. Note that irreducibility depends very much upon the particular field.

Example 4.1.2.

1. A linear polynomial over $\mathbb{R}$ is irreducible over $\mathbb{R}$.
2. All polynomials over $\mathbb{R}$ of degree ≥ 2 are reducible over $\mathbb{C}$.
3. All polynomials over $\mathbb{R}$ of degree ≥ 3 are reducible over $\mathbb{R}$ (Prove it.).
4. The polynomial $f(x) = 2x^2+4$ is irreducible over $\mathbb{Q}$ but reducible over $\mathbb{Z}$, since $2x^2+4 = 2(x^2+2)$ and neither 2 nor x^2+2 is a unit in $\mathbb{Z}[x]$.
5. The polynomial $x^2 - 2 \in \mathbb{Q}[x]$ is irreducible since it cannot be factored any further over the rational numbers. Similarly, $x^2 + 1$ is irreducible over the real numbers.

Example 4.1.3. The polynomial $x^2 - 2$ is irreducible over $\mathbb{Q}$, but reducible over $\mathbb{R}$, since $x^2 - 2 = (x - \sqrt{2})(x + \sqrt{2})$.

Theorem 4.1.4. (Rational Roots Theorem). *Let $f(x) = a_0 + a_1x + \cdots + a_nx_n \in \mathbb{Z}[x]$, with $a_n \neq 0$. Suppose that $q \in \mathbb{Q}$is a root of $f(x)$. If $q = rs$, with $r, s \in \mathbb{Z}$ and $(r, s) = 1$, then $r \mid a_0$ and $s \mid a_n$.*

Proof. We have

$$0 = f\left(\frac{r}{s}\right) = a_0 + \frac{a_1r}{s} + \cdots + \frac{a_{n-1}r^{n-1}}{s^{n-1}} + \frac{a_nr^n}{s^n}.$$

Multiplying through by s^n, we obtain

$$a_0s^n + a_1rs^{n-1} + \cdots + a_{n-1}r^{n-1}s + a_nr^n = 0.$$

As s divides $a_0s^n + a_1rs^{n-1} + \cdots + a_{n-1}r^{n-1}s$, it must also divide a_nr^n. Since $(r, s) = 1$, therefore, $s \mid a_n$. Similarly, r divides every term except a_0s_n, so it also divides a_0s_n. Since $(r, s) = 1$, we see that $r \mid a_0$. ■

Example 4.1.5. Let $f(x) = 3x^3 + 2x^2 - 2x - 8$. In view of the Rational Roots Theorem, the only possible rational roots of $f(x)$ are $\pm1, \pm2, \pm4, \pm8, \pm1/3, \pm2/3, \pm4/3$ and $\pm8/3$. Trying them all, we see that the only rational root of $f(x)$ is $4/3$.

Theorem 4.1.6. *Let F be a field. If $f(x) \in F[x]$ and $\deg f(x)$ is 2 or 3, then $f(x)$ is reducible over F if and only if $f(x)$ has a zero in F.*

Proof. Suppose that $f(x) = g(x)h(x)$, where both $g(x)$ and $h(x)$ belong to $F[x]$ and have degrees less than that of $f(x)$. Since $\deg f(x) = \deg g(x) + \deg h(x)$ and $\deg f(x)$ is 2 or 3, at least one of $g(x)$ and $h(x)$ has degree 1. Say $g(x) = ax + b$. Then, clearly, $\frac{-b}{a}$ is a zero of $g(x)$ and therefore a zero of $f(x)$ as well.

Conversely, suppose that $f(a) = 0$, where $a \in F$. Then, by the Factor Theorem (Theorem 3.4.25), we know that $x - a$ is a factor of $f(x)$ and, therefore, $f(x)$ is reducible over F. ■

4.1.1 The Gauss Lemma

Definition 4.1.7. If $f(x)$ is a nonzero polynomial in $\mathbb{Z}[x]$, then the **content** of $f(x)$ is the largest positive integer that divides every coefficient of $f(x)$. We say that $f(x)$ is primitive if its content is 1.

Example 4.1.8. The polynomial $6x^3 - 15x^2 + 81x - 12$ has content 3, whereas $5x^2 + 14x - 2$ is primitive.

We can now present a famous result due to Carl F. Gauss.

Theorem 4.1.9. (Gauss's Lemma). *The product of two primitive polynomials in $\mathbb{Z}[x]$ is also a primitive polynomial.*

Proof. Let $f(x) = a_0 + \cdots + a_n x^n$ and $g(x) = b_0 + \cdots + b_m x^m$ be primitive. Suppose that $f(x)g(x)$ is not primitive. Let p be a prime dividing the content of $f(x)g(x)$. As p cannot divide all of the coefficients of $f(x)$, let i be the smallest nonnegative integer such that p does not divide a_i . Similarly, let j be the smallest nonnegative integer such that b_j is not divisible by p. Then the coefficient of x^{i+j} in $f(x)g(x)$ is

$$a_0 b_{i+j} + a_1 b_{i+j-1} + \cdots + a_{i-1} b_{j+1} + a_i b_j + a_{i+1} b_{j-1} + \cdots + a_{i+j-1} b_1 + a_{i+j} b_0,$$

where we add terms with coefficient zero if necessary. Now, this coefficient must be divisible by p. Also, p divides a_k, $0 \leq k < i$, and p divides b_l, $0 \leq l < j$. Thus, every term in the sum is divisible by p except $a_i b_j$, which means that $p \mid a_i b_j$ as well. But this contradicts Euclid's Lemma (Theorem 3.4.34). ■

Theorem 4.1.10. *Let $f(x)$ be a polynomial in $\mathbb{Z}[x]$, and suppose that $f(x) = g(x)h(x)$, where $g(x), h(x) \in \mathbb{Q}[x]$. Then there is a positive rational number q such that $qg(x)$ and $\frac{1}{q}h(x)$ lie in $\mathbb{Z}[x]$.*

Proof. Assume, first of all, that $f(x)$ is primitive. Choose positive integers a and b such that $ag(x), bh(x) \in \mathbb{Z}[x]$. Then $abf(x) = (ag(x))(bh(x))$.

Let c be the content of $ag(x)$ and d the content of $bh(x)$. Then $acg(x), dbh(x) \in \mathbb{Z}[x]$, and both are primitive polynomials. By Gauss's lemma, their product, $\frac{ab}{cd}f(x)$, is also primitive. Thus, the content of $abf(x)$ is cd. But as $f(x)$ is primitive, the content of $abf(x)$ is also ab. Thus, $ab = cd$, and hence letting $q = a/c$, we see that $b/d = 1/q$.

Suppose that $f(x)$ is not primitive. If it is the zero polynomial, then either $g(x)$ or $h(x)$ must be as well. Without loss of generality, say that $h(x)$ is the zero polynomial. Then let q be a positive integer such that $qg(x) \in \mathbb{Z}[x]$. On the other hand, if $f(x)$ is not the zero polynomial, then let k be its content. Writing $f(x) = kf(x)$, with $f_1(x) \in \mathbb{Z}[x]$, we have $f_1(x) = \left(\frac{1}{k}g(x)\right) h(x)$. By the argument above, there exists a positive rational number q such that $\frac{q}{k}g(x), \frac{1}{q}h(x) \in \mathbb{Z}[x]$. But then $qg(x), \frac{1}{q}h(x) \in \mathbb{Z}[x]$ as well. ■

Example 4.1.11. The polynomial $f(x) = 3x^3 + 2x^2 - 2x - 8$ has $\frac{4}{3}$ as a rational root. Thus, by Theorem 3.4.25 (Factor Theorem), $g(x) = x - \frac{4}{3}$ is a divisor of $f(x)$ in $\mathbb{Q}[x]$. Performing polynomial long division, we see that $f(x) = g(x)h(x)$, where $h(x) = 3x^2 + 6x + 6$. Using $q = 3$ in the above theorem, we find that $f(x) = (3x - 4)(x^2 + 2x + 2)$, and we have a factorisation in $\mathbb{Z}[x]$.

4.1.2 Eisenstein's Criterion

Theorem 4.1.12. (Eisenstein's Criterion). *Let $f(x) = a_0 + a_1 + \cdots + a_n x^n \in \mathbb{Z}[x]$, with $n \geq 1$ and $a_n \neq 0$. Suppose that there exists a prime p such that $p \mid a_i$, $0 \leq i < n$, but $p \nmid a_n$ and $p^2 \nmid a_0$. Then $f(x)$ is irreducible in $\mathbb{Q}[x]$.*

Proof. If $f(x)$ is reducible, then there exist nonconstant polynomials $g(x) = b_0 + \cdots + b_l x^l$ and $h(x) = c_0 + \cdots + c_m x^m$ in $\mathbb{Z}[x]$, with $b_l \neq 0 \neq c_m$ and $f(x) = g(x)h(x)$. Now, p divides $a_0 = b_0c_0$, but p^2 does not. Thus, p divides exactly one of $\{b_0, c_0\}$. Without loss of generality, say $p \mid b_0$. But p does not divide $a_n = b_l c_m$. Thus, p divides neither b_l nor c_m. Let i be the smallest positive integer such that $p \nmid b_i$. Then

$$a_i = b_0c_i + b_1c_{i-1} + \cdots + b_{i-1}c_1 + b_ic_0.$$

Now, $p \mid b_j$, $0 \leq j < i$. Furthermore, as $i \leq l < n$, we know that $p \mid a_i$. Thus, $p \mid b_ic_0$. But p divides neither b_i nor c_0, and we have a contradiction. ■

4.1.3 Solved Examples using Eisenstein's Criterion

Example 4.1.13. Show that the polynomial $f(x) = x^5 + 6x^4 - 9x^2 + 12x - 15$ is irreducible over $\mathbb{Q}$.

We apply Eisenstein's Criterion with $D = \mathbb{Z}$ and $F = \mathbb{Q}$. The polynomial is $f(x) = a_5x^5 + a_4x^4 + a_3x^3 + a_2x^2 + a_1x + a_0$, where: $a_5 = 1$, $a_4 = 6$, $a_3 = 0$, $a_2 = -9$, $a_1 = 12$, $a_0 = -15$

We need to find a prime number p in $\mathbb{Z}$ satisfying the conditions. Let's try the prime $p = 3$.

1. $p \nmid a_n$: Does $3 \nmid a_5$? $a_5 = 1$. $3 \nmid 1$. This condition is satisfied.
2. $p \mid a_i$ for $i = 0, 1, 2, 3, 4$:
   - $a_4 = 6$. $3 \mid 6$. (True)
   - $a_3 = 0$. $3 \mid 0$. (True, since $0 = 3 \cdot 0$)
   - $a_2 = -9$. $3 \mid -9$. (True, since $-9 = 3 \cdot (-3)$)
   - $a_1 = 12$. $3 \mid 12$. (True, since $12 = 3 \cdot 4$)
   - $a_0 = -15$. $3 \mid -15$. (True, since $-15 = 3 \cdot (-5)$)

   This condition is satisfied.
3. $p^2 \nmid a_0$: We need $3^2 \nmid a_0$. So, $9 \nmid -15$. -15 is not divisible by 9 (since $-15 = 9 \cdot (-1) - 6$ or $-15 = 9 \cdot (-2) + 3$). This condition is satisfied.

Since all three conditions of Eisenstein's Criterion are met with $p = 3$, the polynomial $f(x) = x^5 + 6x^4 - 9x^2 + 12x - 15$ is irreducible over $\mathbb{Q}$.

Furthermore, the coefficients are $1, 6, 0, -9, 12, -15$. The gcd of these coefficients is $\gcd(1, 6, 0, -9, 12, -15)$ 1. So $f(x)$ is primitive. Thus, $f(x)$ is also irreducible over $\mathbb{Z}$.

Example 4.1.14. Show that for any prime number p, the p-th cyclotomic polynomial $\Phi_p(x) = \frac{x^p-1}{x-1} = x^{p-1} + x^{p-2} + \cdots + x + 1$ is irreducible over $\mathbb{Q}$.

Eisenstein's Criterion cannot be applied directly to $\Phi_p(x)$ as it is. However, a polynomial $f(x)$ is irreducible if and only if $f(x+c)$ is irreducible for any constant c. Let's consider $\Phi_p(x+1)$.

$$\begin{aligned}
\Phi_p(x+1) &= \frac{(x+1)^p - 1}{(x+1) - 1} = \frac{(x+1)^p - 1}{x} \\
&= \frac{1}{x}\left[\sum_{k=0}^{p}\binom{p}{k}x^k(1)^{p-k} - 1\right] \\
&= \frac{1}{x}\left[\binom{p}{0}x^0 + \binom{p}{1}x^1 + \binom{p}{2}x^2 + \cdots + \binom{p}{p-1}x^{p-1} + \binom{p}{p}x^p - 1\right] \\
&= \frac{1}{x}\left[1 + px + \binom{p}{2}x^2 + \cdots + \binom{p}{p-1}x^{p-1} + x^p - 1\right] \\
&= \frac{1}{x}\left[px + \binom{p}{2}x^2 + \cdots + \binom{p}{p-1}x^{p-1} + x^p\right] \\
&= x^{p-1} + \binom{p}{p-1}x^{p-2} + \cdots + \binom{p}{2}x + \binom{p}{1} \\
&= x^{p-1} + px^{p-2} + \cdots + \frac{p(p-1)}{2}x + p
\end{aligned}$$

Let $g(x) = \Phi_p(x+1)$. The coefficients of $g(x)$ are: $a_{p-1} = 1$ (leading coefficient) $a_{p-2} = \binom{p}{p-1} = p$ ... $a_1 = \binom{p}{2} = \frac{p(p-1)}{2}$ $a_0 = \binom{p}{1} = p$ (constant term)

We apply Eisenstein's Criterion to $g(x)$ with the prime p (the same prime as in $\Phi_p(x)$).

1. $p \nmid a_{p-1}$: The leading coefficient is $a_{p-1} = 1$. $p \nmid 1$. This is satisfied.
2. $p \mid a_i$ for $i = 0, 1, \ldots, p-2$: The coefficients are binomial coefficients $\binom{p}{k}$ for $k = 1, 2, \ldots, p-1$. We know that for a prime p and $1 \le k \le p-1$, p divides $\binom{p}{k} = \frac{p!}{k!(p-k)!}$. This is because p appears in the numerator $p!$, but since $k < p$ and $p-k < p$, p does not appear in $k!$ or $(p-k)!$. Thus, p cannot be cancelled out. So, p divides all coefficients $a_0, a_1, \ldots, a_{p-2}$. This condition is satisfied.
3. $p^2 \nmid a_0$: The constant term is $a_0 = \binom{p}{1} = p$. Since p is prime, $p^2 \nmid p$. This condition is satisfied.

Since all conditions of Eisenstein's Criterion are met for $g(x) = \Phi_p(x+1)$ with the prime p, $g(x)$ is irreducible over $\mathbb{Q}$. Since $\Phi_p(x+1)$ is irreducible over $\mathbb{Q}$, it follows that $\Phi_p(x)$ is also irreducible over $\mathbb{Q}$. (Note: $\Phi_p(x)$ is primitive since its leading coefficient is 1, so it's also irreducible over $\mathbb{Z}$.)

Example 4.1.15. Determine if $f(x) = x^4 + x^2 + 1$ is irreducible over $\mathbb{Q}$ using Eisenstein's Criterion.

The polynomial is $f(x) = x^4 + 0x^3 + x^2 + 0x + 1$. Coefficients: $a_4 = 1, a_3 = 0, a_2 = 1, a_1 = 0, a_0 = 1$. Let's try to find a prime p:

- $p \nmid a_4 = 1$. This is true for any prime p.
- $p \mid a_3, a_2, a_1, a_0$. We need $p \mid 0, p \mid 1, p \mid 0, p \mid 1$. For $p \mid 1$ to be true, p would have to be ± 1, which are not primes.

Eisenstein's Criterion does not apply directly.

Let's try a substitution $x \mapsto x + c$. If $x \mapsto x+1$: $f(x+1) = (x+1)^4 + (x+1)^2 + 1$ $(x+1)^4 = x^4 + 4x^3 + 6x^2 + 4x + 1$ $(x+1)^2 = x^2 + 2x + 1$ So $f(x+1) = (x^4 + 4x^3 + 6x^2 + 4x + 1) + (x^2 + 2x + 1) + 1$ $f(x+1) = x^4 + 4x^3 + 7x^2 + 6x + 3$. Coefficients: $a_4 = 1, a_3 = 4, a_2 = 7, a_1 = 6, a_0 = 3$. Let's try prime $p = 3$:

1. $p \nmid a_4$: $3 \nmid 1$. (True)
2. $p \mid a_3, a_2, a_1, a_0$: $3 \mid 4$? No.

So $p = 3$ does not work for $f(x+1)$. Let's try prime $p = 2$ for $f(x+1)$:

1. $p \nmid a_4$: $2 \nmid 1$. (True)
2. $p \mid a_3, a_2, a_1, a_0$: $2 \mid 4$? Yes. $2 \mid 7$? No.

So $p = 2$ does not work for $f(x+1)$.

Eisenstein's criterion (even with shift) is not immediately helpful. This does not mean the polynomial is reducible. Eisenstein's criterion is a sufficient condition, not a necessary one. In fact, $x^4 + x^2 + 1$ can be factored: $x^4 + x^2 + 1 = (x^4 + 2x^2 + 1) - x^2 = (x^2+1)^2 - x^2 = (x^2 + 1 - x)(x^2 + 1 + x)$. So $f(x) = (x^2 - x + 1)(x^2 + x + 1)$. The factors $x^2 - x + 1$ and $x^2 + x + 1$ are irreducible over $\mathbb{Q}$ (their discriminants $b^2 - 4ac = (-1)^2 - 4(1)(1) = -3 < 0$ and $1^2 - 4(1)(1) = -3 < 0$, so no real roots, hence no rational roots for quadratics). Thus, $f(x) = x^4 + x^2 + 1$ is reducible over $\mathbb{Q}$. This example illustrates that if Eisenstein's criterion fails to apply, no conclusion can be drawn about reducibility.

Example 4.1.16. Show that $f(x) = 2x^{10} - 25x^3 + 10x^2 - 30$ is irreducible over $\mathbb{Q}$.

The polynomial is $f(x) = 2x^{10} + 0x^9 + \cdots - 25x^3 + 10x^2 + 0x - 30$. Coefficients: $a_{10} = 2, a_3 = -25, a_2 = 10, a_0 = -30$. Other $a_i = 0$ for $i \notin \{0, 2, 3, 10\}$. Let's try the prime $p = 5$.

1. $p \nmid a_{10}$: We need $5 \nmid 2$. This is true.
2. $p \mid a_i$ for $i = 0, 1, \ldots, 9$:
   - $a_9 = 0$, $5 \mid 0$. (True)

- ...
- $a_4 = 0, 5 \mid 0$. (True)
- $a_3 = -25, 5 \mid -25$. (True)
- $a_2 = 10, 5 \mid 10$. (True)
- $a_1 = 0, 5 \mid 0$. (True)
- $a_0 = -30, 5 \mid -30$. (True)

This condition is satisfied.

3. $p^2 \nmid a_0$: We need $5^2 \nmid -30$, i.e., $25 \nmid -30$. -30 is not divisible by 25. This is true.

Since all three conditions of Eisenstein's Criterion are met with $p = 5$, the polynomial $f(x) = 2x^{10} - 25x^3 + 10x^2 - 30$ is irreducible over $\mathbb{Q}$.

Is $f(x)$ primitive? The coefficients are $2, 0, \ldots, -25, 10, 0, -30$. $\gcd(2, -25, 10, -30, \text{zeros}) = \gcd(2, 25, 10, 3$
1. Since $f(x)$ is primitive and irreducible over $\mathbb{Q}$, it is also irreducible over $\mathbb{Z}$.

Example 4.1.17. The polynomial $13x^3 - 42x^2 + 81x - 15$ is irreducible over $\mathbb{Q}$, using Eisenstein's criterion with $p = 3$.

Example 4.1.18. For any positive integer n and any prime p, we observe that $x^n - p$ is irreducible over $\mathbb{Q}$.

Example 4.1.19. Let $f(x) = 2x - 6$. Then $f(x)$ is irreducible in $\mathbb{Q}[x]$. But $f(x)$ is reducible in $\mathbb{Z}[x]$; indeed, $f(x) = 2(x - 3)$, and neither 2 nor $x - 3$ is a unit in $\mathbb{Z}[x]$.

The problem, then, is that the nonzero constants are not necessarily units in $\mathbb{Z}[x]$, and this affects irreducibility.

Theorem 4.1.20. *Let $f(x) \in \mathbb{Z}[x]$. If $f(x)$ is reducible over $\mathbb{Q}$, then it is reducible over $\mathbb{Z}$.*

Proof. Suppose that $f(x) = g(x)h(x)$, where $g(x)$ and $h(x) \in \mathbb{Q}[x]$. Clearly, we may assume that $f(x)$ is primitive because we can divide both $f(x)$ and $g(x)$ by the content of $f(x)$. Let a be the least common multiple of the denominators of the coefficients of $g(x)$, and b the least common multiple of the denominators of the coefficients of $h(x)$. Then $abf(x) = ag(x) \cdot bh(x)$, where $ag(x)$ and $bh(x) \in \mathbb{Z}[x]$. Let c_1 be the content of $ag(x)$ and let c_2 be the content of $bh(x)$. Then $ag(x) = c_1 g_1(x)$ and $bh(x) = c_2 h_1(x)$, where both $g_1(x)$ and $h_1(x)$ are primitive and $abf(x) = c_1 c_2 g_1(x) h_1(x)$. Since $f(x)$ is primitive, the content of $abf(x)$ is ab. Also, since the product of two primitive polynomials is primitive (Theorem 4.1.9), it follows that the content of $c_1 c_2 g_1(x) h_1(x)$ is $c_1 c_2$. Thus, $ab = c_1 c_2$ and $f(x) = g_1(x) h_1(x)$, where $g_1(x)$ and $h_1(x) \in \mathbb{Z}[x]$ and $\deg g_1(x) = \deg g(x)$ and $\deg h_1(x) = \deg h(x)$. ■

2nd Proof: Theorem 4.1.10 proves as required.

Example 4.1.21. We illustrate the proof of Theorem 4.1.20 by tracing through it for the polynomial $f(x) = 6x^2 + x - 2 = (3x - 3/2)(2x + 4/3) = g(x)h(x)$. In this case we have $a = 2$, $b = 3$, $c_1 = 3$, $c_2 = 2$, $g_1(x) = 2x - 1$, and $h_1(x) = 3x + 2$, so that $2 \cdot 3(6x^2 + x - 2) = 3 \cdot 2(2x - 1)(3x + 2)$ or $6x^2 + x - 2 = (2x - 1)(3x + 2)$.

Theorem 4.1.22. *Let $f(x) \in \mathbb{Z}[x]$. Then $f(x)$ is irreducible in $\mathbb{Z}[x]$ if and only if either*

(i) $f(x)$ is a (positive or negative) prime in $\mathbb{Z}$; or

(ii) $f(x)$ is a primitive polynomial that is irreducible in $\mathbb{Q}[x]$.

Proof. Note that a unit in $\mathbb{Z}[x]$ is also a unit in $\mathbb{Q}[x]$, and hence a constant. But the only constants having inverses in $\mathbb{Z}[x]$ are ± 1, so those are the only units.

Suppose that $f(x)$ is a constant $c \in \mathbb{Z}$. If c is prime, then its only factorisations are $1 \cdot c$ and $(-1)(-c)$, so $f(x)$ is irreducible. Otherwise, c has some other factorisation, and $f(x)$ is not irreducible.

So, let $\deg(f(x)) \geq 1$. Suppose that $f(x)$ is irreducible in $\mathbb{Z}[x]$. If $f(x)$ has content $d > 1$, then $f(x)$ has a factorisation $d \cdot \left(\frac{1}{d} f(x)\right)$, and therefore $f(x)$ is reducible. So, we may assume that $f(x)$ is primitive.

If it is reducible in $\mathbb{Q}[x]$, then, it is reducible in $\mathbb{Z}[x]$ as well (Theorem 4.1.20). Conversely, assume that $f(x)$ is irreducible in $\mathbb{Q}[x]$ and primitive. If $f(x) = g(x)h(x)$, with $g(x), h(x) \in \mathbb{Z}[x]$, then we have a factorisation in $\mathbb{Q}[x]$ as well, which would make $f(x)$ reducible over $\mathbb{Q}$, unless either $g(x)$ or $h(x)$ is a constant. Without loss of generality, let $g(x) = e \neq 0$. If $e = \pm 1$, then $g(x)$ is a unit in $\mathbb{Z}[x]$. If not, then $f(x)$ has content $|e|$ times the content of $h(x)$, contradicting the assumption that $f(x)$ is primitive. Thus, $f(x)$ is irreducible in $\mathbb{Z}[x]$ in this case. ■

We already know that $\mathbb{Z}[x]$ is not a PID. But we have the following.

Theorem 4.1.23. *The ring $\mathbb{Z}[x]$ is a UFD.*

Proof. Let $f(x) \in \mathbb{Z}[x]$ be a nonzero nonunit. We will show that $f(x)$ is a product of irreducibles. First, suppose that $\deg(f(x)) = n \geq 1$. We claim that $f(x)$ is a product of polynomials in $\mathbb{Z}[x]$ that are irreducible in $\mathbb{Q}[x]$. Our proof is by strong induction on n. If $n = 1$, then $f(x)$ is irreducible in $\mathbb{Q}[x]$ and there is nothing to do. Let $n \geq 2$, and suppose that our claim holds for polynomials of smaller degree. If $f(x)$ is irreducible in $\mathbb{Q}[x]$, then again, there is nothing to do. Otherwise, we know that $f(x) = g(x)h(x)$, where $g(x)$ and $h(x)$ are polynomials of degree less than n in $\mathbb{Q}[x]$. By Theorem 4.1.20, we may choose $g(x)$ and $h(x)$ to be in $\mathbb{Z}[x]$. Then our inductive hypothesis tells us that $g(x)$ and $h(x)$ are products of polynomials in $\mathbb{Z}[x]$ that are irreducible in $\mathbb{Q}[x]$, and hence, so is $f(x)$, proving the claim.

If $f(x) = f_1(x) \cdots f_k(x)$, where each $f_i(x)$ is irreducible in $\mathbb{Q}[x]$, then let c_i be the content of $f_i(x)$. We now have

$$f(x) = (c1 \cdots ck)\left(\frac{1}{c_1} f_1(x)\right) \cdots \left(\frac{1}{c_k} f_k(x)\right),$$

where each $\frac{1}{c_i} f_i(x)$ is irreducible in $\mathbb{Z}[x]$, by the preceding Theorem.

Thus, bringing the $\deg(f(x)) = 0$ possibility back into consideration, we see that $f(x)$ is either an integer not in $\{0, \pm 1\}$, or a nonzero integer multiplied by a product of irreducibles in $\mathbb{Z}[x]$.

It remains only to consider the case of an integer. But the Fundamental Theorem of Arithmetic tells us that any integer not in $\{0, \pm 1\}$ is a product of (positive or negative) primes, which are certainly irreducible in $\mathbb{Z}[x]$. We do still have to deal with $f(x) = (-1)g_1(x) \cdots g_k(x)$, where each $g_i(x)$ is irreducible, but then this is $(-g_1(x))g_2(x) \cdots g_k(x)$, and $-g_1(x)$ is irreducible as well.

Let us verify the uniqueness. Suppose that

$$f(x) = p_1 \cdots p_k g_1(x) \cdots g_l(x) = q_1 \cdots q_m h_1(x) \cdots h_n(x),$$

where each p_i and q_i is a (positive or negative) prime in $\mathbb{Z}$, and each $g_i(x)$ and $h_i(x)$ is a primitive polynomial which is irreducible in $\mathbb{Q}[x]$. (We allow the possibility that k, l, m or n may be zero.) By Gauss's lemma (Lemma 4.1.9, the product of primitive polynomials is primitive.Thus, the content of $f(x)$ is $|p_1 \cdots p_k| = |q_1 \cdots q_l|$. By the Fundamental Theorem of Arithmetic, $k = l$ and after rearranging, each $p_i = \pm q_i$. Cancelling, we have $g_1(x) \cdots g_m(x) = \pm h_1(x) \cdots h_n(x)$. But these are products of irreducible polynomials in $\mathbb{Q}[x]$. As $\mathbb{Q}[x]$ is a UFD, $m = n$ and, after rearranging, each $g_i(x) = q_i h_i(x)$, for some $qi \in \mathbb{Q}$. Write $q_i = \frac{r_i}{s_i}$, with $r_i, s_i \in \mathbb{Z}$ and $s_i \neq 0$. Then $s_i g_i(x) = r_i h_i(x)$. As $g_i(x)$ and $h_i(x)$ are primitive, looking at the content of each side of the equation, we have $|s_i| = |r_i|$, and hence $q_i \in \{1, -1\}$. We are done. ■

While the real numbers may seem like a more natural field with which to work, the complex numbers have a more attractive algebraic structure. Indeed, we wish to consider the complex numbers, because there are nonconstant real polynomials, such as $x^2 + 1$, having no real roots. The complex numbers do not have this problem. Indeed, a a famous result known as the Fundamental Theorem of Algebra. There are many different proofs of this theorem. Curiously, to the best of the author's knowledge, all of these proofs require results from outside of algebra. A proof that is mostly algebraic can be found in the advanced textbook of Dummit and Foote [**?**]. (Sadly, the algebra involved is somewhat beyond the scope of this course.)

Theorem 4.1.24. *Let $f(x)$ be a nonconstant polynomial in $\mathbb{C}[x]$. Then $f(x)$ has a root in $\mathbb{C}$.*

Lemma 4.1.25. *Let $f(x) \in \mathbb{R}[x]$. If $c, d \in \mathbb{R}$, and $c + di$ is a complex root of $f(x)$, then so is $c - di$.*

Proof. Hint. $f(\overline{z}) = \overline{f(z)}$, where $z = c + di$. ■

Theorem 4.1.26. *Let $f(x) \in \mathbb{R}[x]$. Then $f(x)$ is irreducible over $\mathbb{R}$ if and only if either*

(i) $\deg(f(x)) = 1$*; or*

(ii) $f(x) = ax^2 + bx + c$*, where* $a \neq 0$ *and* $b^2 < 4ac$*.*

Proof. Since R is a field, constant polynomials are either 0 or a unit, and therefore need not be considered. If $\deg(f(x)) = 1$, then $f(x)$ is indeed irreducible. Therefore, let $f(x)$ have degree at least 2. Suppose that $f(x)$ is irreducible. By Theorem 4.1.24, $f(x)$ has a root $z = a + bi \in \mathbb{C}$. If $z \in \mathbb{R}$, then $f(x)$ is reducible, a contradiction. Assume otherwise. By Lemma 4.1.25, $a - bi$ is also a root. then $(x - (a + bi))(x - (a - bi)) \mid f(x)$ in $\mathbb{C}[x]$. But $(x - (a + bi))(x - (a - bi)) = x^2 - 2ax + (a^2 + b^2) \in \mathbb{R}[x]$. Thus, applying the division algorithm, we see that there exist $q(x), r(x) \in \mathbb{R}[x]$ such that $f(x) = (x^2 - 2ax + (a^2 + b^2))q(x) + r(x)$, and $r(x) = 0$ or $\deg(r(x)) < 2$. By the uniqueness of the division algorithm in $\mathbb{C}[x]$, we must have $r(x) = 0$ and $x^2 - 2ax + (a^2 + b^2)$ divides $f(x)$ in $\mathbb{R}[x]$. In particular, if $\deg(f(x)) > 2$, then $f(x)$ must be reducible.

Thus, we may assume that $f(x) = ax^2 + bx + c$, with $a \neq 0$. By Corollary 11.2, such a polynomial is irreducible over $\mathbb{R}$ if and only if it has no roots in $\mathbb{R}$. But the quadratic formula tells us that this happens if and only if $b^2 - 4ac < 0$. We are done ■

Corollary 4.1.27. *Let $f(x) \in \mathbb{R}[x]$ be a polynomial of odd degree. Then $f(x)$ has a real root.*

Proof. We know that $\mathbb{R}[x]$ is a UFD. Thus, write $f(x)$ as a product of irreducible polynomials. By Theorem 4.1.26, each such irreducible has degree 1 or 2. Since $f(x)$ has odd degree, at least one of these irreducible polynomials has degree 1. Therefore, there exist $a, b \in \mathbb{R}$, with $a \neq 0$, such that $ax + b$ divides $f(x)$. But then $-a^{-1}b$ is a root of $f(x)$. ■

Example 4.1.28. Let $f(x) = x^4 + x^3 + x^2 + x + 1 \in \mathbb{Z}_2[x]$. We claim that $f(x)$ is irreducible over $\mathbb{Z}_2$. If not, there are two possibilities. First, $f(x)$ could be a product of a degree 1 polynomial and a degree 3 polynomial. But if it has a degree 1 polynomial as a factor, then it has a root. There are only two possible roots in $\mathbb{Z}_2$, namely, 0 and 1, and neither works. Second, $f(x)$ could be a product of two polynomials of degree 2. Now, the only possible coefficients are 0 and 1. Furthermore, the leading coefficients and the constant terms must multiply to give 1. Thus, the only possible factors are $x^2 + 1$ and $x^2 + x + 1$. But $x^2 + 1$ has 1 as a root, and $f(x)$ does not, so only $x^2 + x + 1$ remains. However, $(x^2 + x + 1)^2 = x^4 + x^2 + 1 \neq f(x)$. Thus, $f(x)$ is indeed irreducible.

Example 4.1.29. Let $f(x) = 3x^5 + x^4 + 4x^3 + 4x^2 + 3x + 2 \in \mathbb{Z}_5[x]$. We would like to write $f(x)$ as a product of irreducibles. The first thing we should do is check for roots. We run through the five elements of $\mathbb{Z}_5$ and find that 3 is a root. Thus, $x - 3$ (or, equivalently, $x + 2$) divides $f(x)$. Performing polynomial long division, we find that $f(x) = (x+2)(3x^4 + 4x^2 + x + 1)$. Let $g(x) = 3x^4 + 4x^2 + x + 1$. Evaluating $g(x)$ at each element of $\mathbb{Z}_5$, we see that $g(x)$ has no roots. Thus, if it is to be factored, it must be as a product of two polynomials of degree 2. Up to a unit in $\mathbb{Z}_5$, these factors would have to be $x^2 + ax + b$ and $3x^2 + cx + d$, for some $a, b, c, d \in \mathbb{Z}_5$. Furthermore, looking at the constant terms, we have $bd = 1$. Thus, once b is decided, $d = b - 1$. Looking at the coefficients of x^3, we have $3a + c = 0$. Thus, once a is decided, we have $c = 2a$. Trying the various possibilities for a and b, we have $g(x) = (x^2 + 2x + 3)(3x^2 + 4x + 2)$. Since $g(x)$ has no roots, this cannot be factored any further. Thus, $f(x) = (x + 2)(x^2 + 2x + 3)(3x^2 + 4x + 2)$ is a product of irreducibles in $\mathbb{Z}_5[x]$.

Theorem 4.1.30. *Let $f(x) = a_0 + a_1x + \cdots + a_nx^n \in \mathbb{Z}[x]$. Let p be a prime such that $p \nmid a_n$. Reducing all of the coefficients modulo p, if $[a_0] + [a_1]x + \cdots + [a_n]x^n$ is irreducible in $\mathbb{Z}_p[x]$, then $f(x)$ is irreducible in $\mathbb{Q}[x]$.*

Proof. Suppose $f(x)$ is reducible in $\mathbb{Q}[x]$. Then, we must have $f(x) = g(x)h(x)$, where $g(x) = b_0 + \cdots + b_kx^k$ and $h(x) = c_0 + \cdots + c_mx^m$ are polynomials in $\mathbb{Z}[x]$, with $k, m > 0$ and $b_k \neq 0 \neq c_m$. Now, we have $a_i = b_0c_i + b_1c_{i-1} + \cdots + b_ic_0$, for each i. The function from $\mathbb{Z}$ to $\mathbb{Z}_p$ sending d to $[d]$, where $[d]$ is an equivalence induced by modulo p, is a ring homomorphism. Thus,

$$[a_i] = [b_0][c_i] + [b_1][c_{i-1}] + \cdots + [b_i][c_0].$$

It now follows that

$$[a_0] + \cdots + [a_n]x^n = ([b_0] + \cdots + [b_k]x^k)([c_0] + \cdots + [c_m]x^m).$$

That is, $[a_0] + \cdots + [a_n]x^n$ is reducible, unless one of the factors is a constant polynomial. But as $p \nmid a_n$,we see that the degree of $[a_0] + \cdots + [a_n]x^n$ is $n = k + m$. The only way the product will have the correct degree is if $[b_k] \neq [0] \neq [c_m]$. This contradiction completes the proof. ■

Note that the condition that p does not divide the leading coefficient is important. Indeed, $3x^2 + x - 4$ is reducible in $\mathbb{Q}[x]$, as it is $(x - 1)(3x + 4)$. But if we tried to use $p = 3$, we would obtain $x + 2 \in \mathbb{Z}_3[x]$, which is certainly irreducible. Also, the converse of the theorem is not true. For instance, x^2+1 is irreducible over $\mathbb{Q}$ (or, for that matter, $\mathbb{R}$), but in $\mathbb{Z}_5[x]$, we have $x^2+1 = (x+2)(x+3)$.

Example 4.1.31. We claim that $15x^4-29x^3+13x^2+33x-201$ is irreducible over $\mathbb{Q}[x]$. Use Eisenstein's Criterion (Theorem 4.1.12) with $p = 2$. In $\mathbb{Z}_2[x]$, we obtain $x^4 + x^3 + x^2 + x + 1$ which, by Example 4.1.28, is irreducible.

Sometimes, we might have to try more than one prime.

Example 4.1.32. Let $f(x) = 5x^3 + 3x^2 + x + 1$. If we use $p = 2$, we obtain $x^3 + x^2 + x + 1 \in \mathbb{Z}_2[x]$. But this polynomial has 1 as a root, so it is reducible. No help here! Let us try $p = 3$. Then we get $2x^3 + x + 1 \in \mathbb{Z}_3[x]$. By Theorem 4.1.6, it is irreducible if it has no roots. But trying 0, 1 and 2, we see that it has no roots in $\mathbb{Z}_3$. Thus, $f(x)$ is irreducible in $\mathbb{Q}[x]$.

Recall that the complex zeros of $x^n - 1$ are $1, \omega = \cos\left(\frac{2\pi}{n}\right) + i\sin\left(\frac{2\pi}{n}\right), \omega^2, \ldots, \omega^{n-1}$. Since $\omega = \cos\left(\frac{2\pi}{n}\right) + i\sin\left(\frac{2\pi}{n}\right)$ generates a cyclic group of order n under multiplication and the generators of $\langle\omega\rangle$ are elements of the form ω^k, where, $a \leq k \leq n$ and $\gcd(n, k) = 1$. These generators are called the *primitive* n^{th} roots of unity. Recalling that we use $\phi(n)$ to denote the number of positive integers less than or equal to n and relatively prime to n, we see that for each positive integer n there are precisely $\phi(n)$ primitive n^{th} roots of unity. The polynomials whose zeros are the $\phi(n)$ primitive n^{th} roots of unity have a special name.

Solved Examples using Mod p Irreducibility Test

Example 4.1.33. Show that the polynomial $f(x) = x^3 + x + 1$ is irreducible over $\mathbb{Q}$.

Let $f(x) = x^3 + x + 1 \in \mathbb{Z}[x]$. The coefficients are $a_3 = 1, a_2 = 0, a_1 = 1, a_0 = 1$. The polynomial is primitive since $\gcd(1, 0, 1, 1) = 1$. Let's try reducing modulo $p = 2$. The leading coefficient is $a_3 = 1$. $2 \nmid 1$. This condition is satisfied. Now, reduce the coefficients of $f(x)$ modulo 2: $\bar{f}(x) = [1]x^3 + [0]x^2 + [1]x + [1] = x^3 + x + 1 \in \mathbb{Z}_2[x]$. The degree of $\bar{f}(x)$ is 3. A cubic polynomial in $\mathbb{Z}_p[x]$ is reducible if and only if it has a root in $\mathbb{Z}_p$. We check for roots of $\bar{f}(x)$ in $\mathbb{Z}_2 = \{[0], [1]\}$:

- $\bar{f}([0]) = [0]^3 + [0] + [1] = [0] + [0] + [1] = [1] \neq [0]$.
- $\bar{f}([1]) = [1]^3 + [1] + [1] = [1] + [1] + [1] = [3] = [1] \neq [0]$ (since we are in $\mathbb{Z}_2$).

Since $\bar{f}(x)$ has no roots in $\mathbb{Z}_2$ and its degree is 3, $\bar{f}(x) = x^3 + x + 1$ is irreducible in $\mathbb{Z}_2[x]$. By the Mod p Irreducibility Test (with $p = 2$), since $\bar{f}(x)$ is irreducible in $\mathbb{Z}_2[x]$ and $2 \nmid a_3$, $f(x) = x^3 + x + 1$ is irreducible over $\mathbb{Q}$. Since $f(x)$ is primitive, it is also irreducible over $\mathbb{Z}$.

Example 4.1.34. Show that the polynomial $f(x) = 7x^3 - 6x^2 + 3x + 9$ is irreducible over $\mathbb{Q}$.

Let $f(x) = 7x^3 - 6x^2 + 3x + 9 \in \mathbb{Z}[x]$. Coefficients: $a_3 = 7, a_2 = -6, a_1 = 3, a_0 = 9$. The polynomial is primitive since $\gcd(7, -6, 3, 9) = 1$. Let's try reducing modulo $p = 2$. Leading coefficient $a_3 = 7$. $2 \nmid 7$. Condition satisfied. Reduce coefficients modulo 2: $a_3 = 7 \equiv 1 \pmod 2$ $a_2 = -6 \equiv 0 \pmod 2$ $a_1 = 3 \equiv 1 \pmod 2$ $a_0 = 9 \equiv 1 \pmod 2$ So, $\bar{f}(x) = [1]x^3 + [0]x^2 + [1]x + [1] = x^3 + x + 1 \in \mathbb{Z}_2[x]$. This is the same polynomial as in the previous example. We found it is irreducible in $\mathbb{Z}_2[x]$ because it has no roots in $\mathbb{Z}_2$. Since $\bar{f}(x)$ is irreducible in $\mathbb{Z}_2[x]$ and $2 \nmid a_3 = 7$, by the Mod p Irreducibility Test, $f(x) = 7x^3 - 6x^2 + 3x + 9$ is irreducible over $\mathbb{Q}$. Since $f(x)$ is primitive, it is also irreducible over $\mathbb{Z}$.

Example 4.1.35. Show that $f(x) = x^4 + 3x^3 + 3x^2 - x + 5$ is irreducible over $\mathbb{Q}$.

Let $f(x) = x^4 + 3x^3 + 3x^2 - x + 5 \in \mathbb{Z}[x]$. Coefficients: $a_4 = 1, a_3 = 3, a_2 = 3, a_1 = -1, a_0 = 5$. This polynomial is primitive. Let's try modulo $p = 2$. Leading coefficient $a_4 = 1$. $2 \nmid 1$. Condition satisfied. Reduce coefficients modulo 2: $a_4 = 1 \equiv 1 \pmod 2$ $a_3 = 3 \equiv 1 \pmod 2$ $a_2 = 3 \equiv 1 \pmod 2$ $a_1 = -1 \equiv 1 \pmod 2$ $a_0 = 5 \equiv 1 \pmod 2$ So, $\bar{f}(x) = x^4 + x^3 + x^2 + x + 1 \in \mathbb{Z}_2[x]$. This is a degree 4 polynomial. We first check for roots in $\mathbb{Z}_2 = \{[0], [1]\}$:

- $\bar{f}([0]) = [0]^4 + [0]^3 + [0]^2 + [0] + [1] = [1] \neq [0]$.
- $\bar{f}([1]) = [1]^4 + [1]^3 + [1]^2 + [1] + [1] = [1] + [1] + [1] + [1] + [1] = [5] = [1] \neq [0]$.

Since $\bar{f}(x)$ has no roots in $\mathbb{Z}_2$, it has no linear factors in $\mathbb{Z}_2[x]$. If $\bar{f}(x)$ is reducible in $\mathbb{Z}_2[x]$, it must be a product of two irreducible quadratic polynomials. The irreducible quadratic polynomials in $\mathbb{Z}_2[x]$ are: Quadratic polynomials in $\mathbb{Z}_2[x]$ are of the form $x^2 + ax + b$. x^2 (reducible, $x \cdot x$) $x^2 + 1 = (x + 1)^2$ (reducible) $x^2 + x$ (reducible, $x(x+1)$) $x^2 + x + 1$ (irreducible, no roots: $0^2 + 0 + 1 = 1$, $1^2 + 1 + 1 = 1$) So the only irreducible quadratic in $\mathbb{Z}_2[x]$ is $x^2 + x + 1$. If $\bar{f}(x)$ factors into two irreducible quadratics, it must be $(x^2+x+1)(x^2+x+1)$. $(x^2+x+1)^2 = (x^2+x+1)(x^2+x+1) = x^4+x^3+x^2+x^3+x^2+x+x^2+x+1$ $= x^4 + (1 + 1)x^3 + (1 + 1 + 1)x^2 + (1 + 1)x + 1 = x^4 + 0x^3 + 1x^2 + 0x + 1 = x^4 + x^2 + 1$. This is not equal to $\bar{f}(x) = x^4 + x^3 + x^2 + x + 1$. Therefore, $\bar{f}(x)$ cannot be factored into two irreducible quadratics. Since $\bar{f}(x)$ has no linear factors and no factorisation into two irreducible quadratics, $\bar{f}(x) = x^4 + x^3 + x^2 + x + 1$ is irreducible in $\mathbb{Z}_2[x]$. By the Mod p Irreducibility Test (with $p = 2$), $f(x) = x^4 + 3x^3 + 3x^2 - x + 5$ is irreducible over $\mathbb{Q}$. Since $f(x)$ is primitive, it is also irreducible over $\mathbb{Z}$.

Example 4.1.36. Consider $f(x) = x^4 - 4$. Can we use the Mod p Irreducibility Test to show it's irreducible over $\mathbb{Q}$?

Let $f(x) = x^4 - 4 \in \mathbb{Z}[x]$. This is primitive. If we try $p = 2$: The leading coefficient $a_4 = 1$. $2 \nmid 1$. $\bar{f}(x) = x^4 - [4] = x^4 - [0] = x^4 \in \mathbb{Z}_2[x]$. x^4 is reducible in $\mathbb{Z}_2[x]$ (e.g., $x^4 = x \cdot x^3$). So, $p = 2$ does not help.

If we try $p = 3$: The leading coefficient $a_4 = 1$. $3 \nmid 1$. $\bar{f}(x) = x^4 - [4] = x^4 - [1] \in \mathbb{Z}_3[x]$. Let's check for roots of $\bar{g}(x) = x^4 - 1$ in $\mathbb{Z}_3 = \{[0], [1], [2] = [-1]\}$. $\bar{g}([0]) = [0]^4 - [1] = -[1] \neq [0]$. $\bar{g}([1]) = [1]^4 - [1] = [1] - [1] = [0]$. So $x - [1]$ (or $x + 2$) is a factor. $\bar{g}([2]) = [2]^4 - [1] = [-1]^4 - [1] = [1] - [1] = [0]$. So $x - [2]$ (or $x + 1$) is a factor. Since $\bar{f}(x)$ has roots in $\mathbb{Z}_3[x]$, it is reducible in $\mathbb{Z}_3[x]$. So $p = 3$ does not help.

If we try $p = 5$: The leading coefficient $a_4 = 1$. $5 \nmid 1$. $\bar{f}(x) = x^4 - [4] \in \mathbb{Z}_5[x]$. Check for roots in $\mathbb{Z}_5 = \{[0], [1], [2], [3], [4]\}$: $\bar{f}([0]) = -[4] \neq [0]$. $\bar{f}([1]) = [1] - [4] = -[3] \neq [0]$. $\bar{f}([2]) = [2]^4 - [4] = [16]-[4] = [1]-[4] = -[3] \neq [0]$. $\bar{f}([3]) = [3]^4-[4] = [81]-[4] = [1]-[4] = -[3] \neq [0]$. $\bar{f}([4]) = [4]^4-[4] = [(-1)]^4 - [4] = [1] - [4] = -[3] \neq [0]$. No roots in $\mathbb{Z}_5$. So no linear factors. Could it be a product of two irreducible quadratics? Irreducible quadratics in $\mathbb{Z}_5[x]$ are those of the form $x^2 + ax + b$ with no roots. $x^4 - 4 = x^4 + 1 \pmod 5$. Consider $(x^2+ax+b)(x^2+cx+d) = x^4+(a+c)x^3+(b+d+ac)x^2+(ad+bc)x+bd$. We want $x^4 + 1$. So $a + c = 0 \implies c = -a$. $ad + bc = 0 \implies ad - ab = 0 \implies a(d - b) = 0$. $bd = 1$. $b + d + ac = b + d - a^2 = 0$. From $bd = 1$, possibilities for (b, d) are $(1, 1), (2, 3), (3, 2), (4, 4)$. Case 1: $b = 1, d = 1$. Then $a(1 - 1) = 0$ is always true. $1 + 1 - a^2 = 0 \implies 2 - a^2 = 0 \implies a^2 = 2$. No integer a has $a^2 \equiv 2 \pmod 5$ ($0^2 = 0, 1^2 = 1, 2^2 = 4, 3^2 = 9 \equiv 4, 4^2 = 16 \equiv 1$). So this case fails. Case 2: $b = 2, d = 3$. Then $a(3 - 2) = 0 \implies a = 0$. So $c = 0$. Then $2 + 3 - 0^2 = 5 \equiv 0 \pmod 5$. This works. So possible factorisation is $(x^2 + 2)(x^2 + 3) = x^4 + 3x^2 + 2x^2 + 6 = x^4 + 5x^2 + 6 \equiv x^4 + 1 \pmod 5$. Are $x^2 + 2$ and $x^2 + 3$ irreducible in $\mathbb{Z}_5[x]$? For $x^2 + 2$: roots? $x^2 \equiv -2 \equiv 3 \pmod 5$. No solution ($0^2 = 0, 1^2 = 1, 2^2 = 4, 3^2 = 4, 4^2 = 1$). So $x^2 + 2$ is irreducible. For $x^2 + 3$: roots? $x^2 \equiv -3 \equiv 2 \pmod 5$. No solution. So $x^2 + 3$ is irreducible. Thus $\bar{f}(x) = x^4 - 4 \equiv (x^2 + 2)(x^2 + 3) \pmod 5$ is reducible in $\mathbb{Z}_5[x]$. So $p = 5$ doesn't help.

In this case, the Mod p test fails to show irreducibility for $p = 2, 3, 5$. This is expected because $f(x) = x^4 - 4 = (x^2 - 2)(x^2 + 2)$. While $x^2 + 2$ is irreducible over $\mathbb{Q}$ (and $\mathbb{Z}$), $x^2 - 2 = (x - \sqrt{2})(x + \sqrt{2})$ is irreducible over $\mathbb{Q}$ but its factors are not in $\mathbb{Q}[x]$. Thus, $x^4 - 4$ is reducible over $\mathbb{R}$ as $(x - \sqrt{2})(x + \sqrt{2})(x^2 + 2)$. Over $\mathbb{Q}$, the factors are $(x^2 - 2)$ and $(x^2 + 2)$. Both are irreducible over $\mathbb{Q}$ (e.g., by Eisenstein with $p = 2$ for $x^2 - 2$, and $x^2 + 2$ has no rational roots). So $f(x) = (x^2 - 2)(x^2 + 2)$ is reducible over $\mathbb{Q}$. The Mod p test is consistent: it did not falsely claim irreducibility.

4.1.4 Cyclotomic Polynomials

Definition 4.1.37. For any positive integer n, let $\omega_1, \omega_2, \ldots, \omega_{\phi(n)}$ denote the primitive n^{th} roots of unity. The n^{th} cyclotomic polynomial over $\mathbb{Q}$ is the polynomial $\Phi_n(x) = (x-\omega_1)(x-\omega_2)\cdots(x-\omega_{\phi(n)})$.

Example 4.1.38. $\Phi_1(x) = x-1$,since 1 is the only zero of $x-1$. $\Phi_2(x) = x+1$, since the zeros of x^2-1 are 1 and -1, and -1 is the only primitive root. $\Phi_3(x) = (x-\omega)(x-\omega^2)$, where ,$\omega = \cos\left(\frac{2\pi}{n}\right) + i\sin\left(\frac{2\pi}{n}\right)$, and direct calculations show that $\Phi_3(x) = x^2 + x + 1$. Since the zeros of $x^4 - 1$ are ± 1 and $\pm i$ and only i and $-i$ are primitive, $\Phi_4(x) = (x - i)(x + i) = x^2 + 1$.

Theorem 4.1.39. *For every positive integer n, $x^n - 1 = \prod_{d|n} \Phi_d(x)$, where the product runs over all positive divisors d of n.*

Proof. Since both polynomials in the statement are monic, it suffices to show that they have the same zeros and that all zeros have multiplicity 1. Let $\omega = \cos\left(\frac{2\pi}{n}\right) + i\sin\left(\frac{2\pi}{n}\right)$. Then $\langle\omega\rangle$ is a cyclic group of order n, and $\langle v\rangle$ contains all the n^{th} roots of unity. By Fundamental Theorem of cyclic groups, we know that for each j, $|\omega^j|$ divides n so that $(x-\omega^j)$ appears as a factor in $\Phi_{|\omega^j|}(x)$. On the other hand, if $x - \alpha$ is a linear factor of $\Phi_d(x)$ for some divisor d of n, then $\alpha^d = 1$,and therefore $\alpha^n = 1$.Thus, $x - a$ is a factor of $x^n - 1$. Finally, since no zero of $x^n - 1$ can be a zero of $\Phi_d(x)$ for two different d's, the result is proved. ■

Example 4.1.40 (Calculate $\Phi_4(x)$)**.** Calculate the 4^{th} cyclotomic polynomial, $\Phi_4(x)$.

We need to find the primitive 4^{th} roots of unity. The 4^{th} roots of unity are the solutions to $x^4 = 1$. These are $1, i, -1, -i$. In exponential form, $e^{2\pi ik/4}$ for $k = 0, 1, 2, 3$:

- $k = 0 : e^0 = 1$
- $k = 1 : e^{2\pi i(1)/4} = e^{\pi i/2} = \cos(\pi/2) + i\sin(\pi/2) = 0 + i(1) = i$
- $k = 2 : e^{2\pi i(2)/4} = e^{\pi i} = \cos(\pi) + i\sin(\pi) = -1 + i(0) = -1$
- $k = 3 : e^{2\pi i(3)/4} = e^{3\pi i/2} = \cos(3\pi/2) + i\sin(3\pi/2) = 0 + i(-1) = -i$

A root $e^{2\pi ik/4}$ is primitive if $\gcd(k, 4) = 1$.

- For $k = 1$, $\gcd(1, 4) = 1$. So $e^{2\pi i(1)/4} = i$ is a primitive 4^{th} root of unity.
- For $k = 3$, $\gcd(3, 4) = 1$. So $e^{2\pi i(3)/4} = -i$ is a primitive 4^{th} root of unity.

For $k = 0$, $\gcd(0, 4) = 4 \neq 1$. For $k = 2$, $\gcd(2, 4) = 2 \neq 1$. The number of primitive 4^{th} roots is $\phi(4) = 4(1 - \dfrac{1}{2}) = 4(\dfrac{1}{2}) = 2$. The primitive 4^{th} roots of unity are $\omega_1 = i$ and $\omega_2 = -i$.

By definition, $\Phi_4(x) = (x - \omega_1)(x - \omega_2) = (x - i)(x - (-i))$.

$$\begin{aligned}\Phi_4(x) &= (x - i)(x + i)\\ &= x^2 - (i)^2\\ &= x^2 - (-1)\\ &= x^2 + 1\end{aligned}$$

So, $\Phi_4(x) = x^2 + 1$. This polynomial has integer coefficients and is irreducible over $\mathbb{Q}$ (it has no rational roots as a quadratic, or apply Eisenstein to $\Phi_4(x+1) = (x+1)^2+1 = x^2+2x+2$ with $p = 2$).

Alternative method using $x^n - 1 = \prod_{d|n}\Phi_d(x)$: For $n = 4$, the divisors are $d = 1, 2, 4$. So, $x^4 - 1 = \Phi_1(x)\Phi_2(x)\Phi_4(x)$. We need $\Phi_1(x)$ and $\Phi_2(x)$:

- $\Phi_1(x)$: Primitive 1^{st} root of unity is $e^{2\pi i(1)/1} = e^{2\pi i} = 1$. $\phi(1) = 1$. So $\Phi_1(x) = x - 1$.
- $\Phi_2(x)$: Primitive 2^{nd} root of unity is $e^{2\pi i(1)/2} = e^{\pi i} = -1$. $\phi(2) = 1$. So $\Phi_2(x) = x - (-1) = x + 1$.

Substitute these into the formula: $x^4 - 1 = (x - 1)(x + 1)\Phi_4(x)$ $x^4 - 1 = (x^2 - 1)\Phi_4(x)$ Therefore, $\Phi_4(x) = \frac{x^4-1}{x^2-1}$. We can factor $x^4 - 1 = (x^2 - 1)(x^2 + 1)$. So, $\Phi_4(x) = \frac{(x^2-1)(x^2+1)}{x^2-1} = x^2 + 1$. This confirms the result.

Example 4.1.41 (Calculate $\Phi_6(x)$)**.** Calculate the 6^{th} cyclotomic polynomial, $\Phi_6(x)$.

We need the primitive 6^{th} roots of unity. These are $e^{2\pi ik/6}$ where $\gcd(k, 6) = 1$. The values of k such that $1 \leq k < 6$ and $\gcd(k, 6) = 1$ are $k = 1$ and $k = 5$. The number of primitive roots is $\phi(6) = 6(1 - \frac{1}{2})(1 - \frac{1}{3}) = 6(\frac{1}{2})(\frac{2}{3}) = 2$. The primitive 6^{th} roots are:

- $\omega_1 = e^{2\pi i(1)/6} = e^{\pi i/3} = \cos(\pi/3) + i\sin(\pi/3) = \frac{1}{2} + i\frac{\sqrt{3}}{2}$.
- $\omega_2 = e^{2\pi i(5)/6} = e^{5\pi i/3} = \cos(5\pi/3) + i\sin(5\pi/3) = \frac{1}{2} - i\frac{\sqrt{3}}{2}$.

Note that $\omega_2 = \bar{\omega}_1$. By definition, $\Phi_6(x) = (x-\omega_1)(x-\omega_2)$.

$$\begin{aligned}
\Phi_6(x) &= \left(x - \left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right)\right)\left(x - \left(\frac{1}{2} - i\frac{\sqrt{3}}{2}\right)\right) \\
&= \left(\left(x - \frac{1}{2}\right) - i\frac{\sqrt{3}}{2}\right)\left(\left(x - \frac{1}{2}\right) + i\frac{\sqrt{3}}{2}\right) \\
&= \left(x - \frac{1}{2}\right)^2 - \left(i\frac{\sqrt{3}}{2}\right)^2 \\
&= x^2 - 2 \cdot x \cdot \frac{1}{2} + \left(\frac{1}{2}\right)^2 - \left(i^2\frac{3}{4}\right) \\
&= x^2 - x + \frac{1}{4} - \left(-\frac{3}{4}\right) \\
&= x^2 - x + \frac{1}{4} + \frac{3}{4} \\
&= x^2 - x + \frac{4}{4} \\
&= x^2 - x + 1
\end{aligned}$$

So, $\Phi_6(x) = x^2 - x + 1$. This polynomial has integer coefficients. To check for irreducibility over $\mathbb{Q}$: for a quadratic, we can check the discriminant $b^2 - 4ac = (-1)^2 - 4(1)(1) = 1 - 4 = -3 < 0$. Since the discriminant is not a perfect square of a rational number (in fact, it's negative), the quadratic has no rational roots and is thus irreducible over $\mathbb{Q}$.

Alternative method using $x^n - 1 = \prod_{d|n} \Phi_d(x)$: For $n = 6$, the divisors are $d = 1, 2, 3, 6$. So, $x^6 - 1 = \Phi_1(x)\Phi_2(x)\Phi_3(x)\Phi_6(x)$. We need $\Phi_1(x), \Phi_2(x), \Phi_3(x)$:

- $\Phi_1(x) = x - 1$.
- $\Phi_2(x) = x + 1$.
- $\Phi_3(x)$: Primitive 3^{rd} roots are $e^{2\pi i(1)/3} = \cos(2\pi/3) + i\sin(2\pi/3) = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$ and $e^{2\pi i(2)/3} = \cos(4\pi/3) + i\sin(4\pi/3) = -\frac{1}{2} - i\frac{\sqrt{3}}{2}$. $\phi(3) = 2$. $\Phi_3(x) = (x - (-\frac{1}{2} + i\frac{\sqrt{3}}{2}))(x - (-\frac{1}{2} - i\frac{\sqrt{3}}{2})) = ((x+\frac{1}{2}) - i\frac{\sqrt{3}}{2})((x+\frac{1}{2}) + i\frac{\sqrt{3}}{2}) = (x+\frac{1}{2})^2 - (i\frac{\sqrt{3}}{2})^2 = x^2 + x + \frac{1}{4} - (-\frac{3}{4}) = x^2 + x + 1$. Alternatively, $x^3 - 1 = \Phi_1(x)\Phi_3(x) = (x-1)\Phi_3(x)$. So $\Phi_3(x) = \frac{x^3-1}{x-1} = x^2 + x + 1$.

Substitute these into the formula for $x^6 - 1$: $x^6 - 1 = (x-1)(x+1)(x^2+x+1)\Phi_6(x)$ $x^6 - 1 = (x^2-1)(x^2+x+1)\Phi_6(x)$ We know $x^3 - 1 = (x-1)(x^2+x+1)$ and $x^3 + 1 = (x+1)(x^2-x+1)$. Also $x^6 - 1 = (x^3-1)(x^3+1) = (x-1)(x^2+x+1)(x+1)(x^2-x+1)$. Comparing $(x^2-1)(x^2+x+1)\Phi_6(x)$ with $(x-1)(x+1)(x^2+x+1)(x^2-x+1)$, we see $(x^2-1) = (x-1)(x+1)$. So, $(x^2-1)(x^2+x+1)\Phi_6(x) = (x^2-1)(x^2+x+1)(x^2-x+1)$. Dividing by $(x^2-1)(x^2+x+1)$ (assuming x is such that this is non-zero), we get: $\Phi_6(x) = x^2 - x + 1$. This confirms the result.

Theorem 4.1.42. *For every positive integer n, $\Phi_n(x)$ has integer coefficients.*

Proof. The case $n - 1$ is trivial. By induction, we may assume that $g(x) = \prod_{d|n, d<n} \Phi_d(x)$ has integer coefficients. By Theorem 4.1.39 we know that $x^n - 1 = \Phi_n(x)g(x)$, and, because $g(x)$ is monic, we may carry out the division in $\mathbb{Z}[x]$. Thus, $\Phi_n(x) \in \mathbb{Z}[x]$. ■

Theorem 4.1.43. *The cyclotomic polynomials $\Phi_n(x)$ are irreducible over $\mathbb{Z}$.*

Proof. Let $f(x) \in \mathbb{Z}[x]$ be a monic irreducible factor of $\Phi_n(x)$. Because $\Phi_n(x)$ is monic and has no multiple zeros, it suffices to show that every zero of $\Phi_n(x)$ is a zero of $f(x)$.

Since $\Phi_n(x)$ divides $x^n - 1$ in $\mathbb{Z}[x]$, we may write $x^n - 1 = f(x)g(x)$, where $g(x) \in \mathbb{Z}[x]$. Let ω

be a primitive n^{th} root of unity that is a zero of $f(x)$. Then $f(x)$ is the minimal polynomial for ω over $\mathbb{Q}$. Let p be any prime that does not divide n. Then, ω^p is also a primitive n^{th} root of unity, and therefore $0 = (\omega^p)^n - 1 = f(\omega^p)g(\omega^p)$, and so $f(\omega^p) = 0$ or $g(\omega^p) = 0$. Suppose $f(\omega^p) \neq 0$. Then $g(\omega^p) = 0$, and so ω is a zero of $g(x^p)$. Thus, by divisibility property, $f(x)$ divides $g(x^p)$ in $\mathbb{Q}[x]$. Since $f(x)$ is monic, $f(x)$ actually divides $g(x^p)$ in $\mathbb{Z}[x]$. Say $g(x^p) = f(x)h(x)$, where $h(x) \in \mathbb{Z}[x]$. Now let $g(x)$, $f(x)$, and $h(x)$ denote the polynomials in $\mathbb{Z}_p[x]$ obtained from $g(x)$, $f(x)$, and $h(x)$, respectively, by reducing each coefficient modulo p. Since this reduction process is a ring homomorphism from $\mathbb{Z}[x] \to \mathbb{Z}_p[x]$, we have $g(x^p) = f(x)h(x)$ in $\mathbb{Z}_p[x]$. By Fermat's Little theorem, we then have $(g(x))^p = g(x^p) = f(x)h(x)$, and since $\mathbb{Z}_p[x]$ is a unique factorisation domain, it follows that $g(x)$ and $f(x)$ have an irreducible factor in $\mathbb{Z}_p[x]$ in common; call it $m(x)$. Thus, we may write $f(x) = k_1(x)m(x)$ and $g(x) = k_2(x)m(x)$, where $k_1(x),\ k_2(x) \in \mathbb{Z}_p[x]$. Then, viewing $x^n - 1$ as a member of $\mathbb{Z}_p[x]$, we have $x^n - 1 = f(x)g(x) = k_1(x)k_2(x)(m(x))^2$. In particular, $x^n - 1$ has a multiple zero in some extension of $\mathbb{Z}_p$. But because p does not divide n, the derivative nx^{n-1} of $x^n - 1$ is not 0, and so nx^{n-1} and $x^n - 1$ do not have a common factor of positive degree in $\mathbb{Z}_p[x]$. This is a contradiction (A polynomial $p(x)$ has a multiple zero if and only if $p(x)$ and its derivative $p'(x)$ have a common factor), we must have $f(\omega^p) = 0$.

We reformulate what we have thus far proved as follows: If β is any primitive n^{th} root of unity that is a zero of $f(x)$ and p is any prime that does not divide n, then β^p is a zero of $f(x)$. Now let k be any integer between 1 and n that is relatively prime to n. Then we can write $k = p_1p_2\dots p_t$, where each p_i is a prime that does not divide n (repetitions are permitted). It follows then that each of $\omega, \omega^{p_1}, (\omega^{p_1})^{p_2}, \dots, (\omega^{p_1p_2\dots p_{t-1}})^{p_t} = \omega^k$ is a zero of $f(x)$. Since every zero of $\Phi_n(x)$ has the form ω^k, where k is between 1 and n and is relatively prime to n, we have proved that every zero of $\Phi_n(x)$ is a zero of $f(x)$. This completes the proof. ■

4.2 PROBLEMS

1. Write each of the following as products of irreducibles in $\mathbb{Z}_5[x]$.
   (a) $x^3 + 3x^2 + 3x + 2$
   (b) $x^3 + 2x^2 + 4x + 2$
   (c) $x^4 + 2x^3 + 4x + 3$
2. Find every irreducible polynomial of degree 3 over $\mathbb{Z}_2$.
3. Let F be an infinite field. If $f(x), g(x) \in F[x]$, and $f(a) = g(a)$ for all $a \in F$, show that $f(x) = g(x)$.
4. Let p be a prime. Find infinitely many polynomials $f_1(x), f_2(x), \dots$ in $\mathbb{Z}_p[x]$ such that $f_i(a) = 0$ for all $a \in \mathbb{Z}_p$ and all positive integers i.
5. Are the following polynomials irreducible over $\mathbb{Q}$?
   (a) $3x^4 + 15x^3 - 25x^2 + 45x + 10$.
   (b) $2x^3 + 5x^2 + x + 7$.
   (c) $x^{14} - 75$.
6. Let F be a field, $a \in F$ and $f(x) \in F[x]$. Show that $f(x)$ is irreducible if and only if $f(x + a)$ is irreducible.
7. Find a non-zero polynomial in $\mathbb{R}[x]$ having $2 - 5i$, $4 + i$ and 6 as roots.
8. Let F be a finite field with n elements. How many monic irreducible polynomials of degree 2 are there in $F[x]$?
9. Let p be an odd prime. Show that $x^4 + 1$ is reducible over $\mathbb{Z}_p$ in each of the following cases:
   (a) there exists an $a \in \mathbb{Z}_p$ such that $a^2 = p - 1$;
   (b) there exists an $a \in \mathbb{Z}_p$ such that $a^2 = p - 2$; or

(c) there exists an $a \in \mathbb{Z}_p$ such that $a^2 = 2$.

10. Is $7x^6 + 21x^5 - 49x^3 + 14x^2 + 7x + 2$ reducible or irreducible over $\mathbb{Q}$?

11. Given that a is a root of $f(x)$, find all complex roots of $f(x)$.

    1. $f(x) = x^3 + x^2 - 5x - 21$, $a = 3$
    2. $f(x) = x^4 - 6x^3 + 33x^2 - 84x + 136$, $a = 1 + 4i$